

FLEXIBLE DEEP NEURAL NETWORKS FOR PARTIALLY LINEAR SURVIVAL DATA: ESTIMATION AND SURVIVAL INFERENCE

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We propose a flexible deep neural network (DNN) framework for modeling survival data within a partially linear regression structure. The approach preserves interpretability through a parametric linear component for covariates of primary interest, while a nonparametric DNN component captures complex time-covariate interactions among nuisance variables. We refer to the method as FLEXI-Haz, a FLEXible Hazard model with a partially linear structure. In contrast to existing DNN approaches for partially linear Cox models, FLEXI-Haz does not rely on the proportional hazards assumption. We establish theoretical guarantees: the neural network component attains minimax-optimal convergence rates over composite Hölder classes, the linear estimator is $\sqrt{n}$ -consistent, asymptotically normal, and semiparametrically efficient, and we develop a cross-fitted one-step estimator of the cumulative hazard and survival function for a new subject, together with pointwise asymptotic confidence intervals. To the best of our knowledge, this is the first frequentist asymptotic pointwise inference result for a survival function in a DNN survival model, with or without a linear component. Simulations and real-data analyses demonstrate the utility of FLEXI-Haz as a principled and interpretable alternative to methods based on proportional hazards.

1. Introduction. Deep neural networks (DNNs) have become a central tool in modern data analysis because of their ability to capture complex nonlinear structure, accommodate high-dimensional inputs, and scale to large data sets. They have achieved strong empirical performance across a wide range of domains, including image recognition, speech processing, and natural language understanding (LeCun, Bengio and Hinton, 2015). These developments have also stimulated substantial interest in using DNNs for survival analysis.

Survival analysis concerns the distribution of event times in the presence of censoring and arises in many fields, including medicine, public health, epidemiology, engineering, and finance (Kalbfleisch and Prentice, 2011; Klein and Moeschberger, 2006). In recent years, DNN-based survival methods have become increasingly popular because they can model complex nonlinear relationships between covariates and event times while remaining computationally feasible in high dimensions. A large part of this literature extends the Cox proportional hazards (PH) model (Cox, 1972) by replacing the linear risk score with a neural network and training by the negative partial likelihood. Early work by Faraggi and Simon (1995) used a single hidden-layer network, and Katzman et al. (2018) extended this idea to deeper architectures in DeepSurv. Related PH-based DNN methods have been developed for genomics (Yousefi et al., 2017; Ching, Zhu and Garmire, 2018), medical imaging (Haarburger et al., 2019; Li et al., 2019), recurrent and discrete-time settings (Giunchiglia, Nemchenko and Van Der Schaar, 2018; Gensheimer and Narasimhan, 2019), and hypothesis testing for covariate effects on survival (Zhong, Mueller and Wang, 2026). While these approaches can

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perform well empirically, they inherit the PH restriction, which may be too rigid when covariate effects vary over time.

To move beyond the PH assumption, several DNN-based methods allow time-varying effects. One approach augments the network input with time while retaining an unspecified baseline hazard, as in Cox-Time (Kvamme, Borgan and Scheel, 2019). Other approaches use baseline-free formulations based on the continuous-time hazard; see, for example, Hu and Nan (2023); Su et al. (2026). These methods naturally accommodate time-varying effects and avoid the PH constraint, but Hu and Nan (2023) do not provide asymptotic theory, and Su et al. (2026) focus on one- and two-sample hypothesis testing.

At the same time, there is strong interest in semiparametric survival models that retain a low-dimensional interpretable component for covariates of primary interest while modeling high-dimensional or nuisance covariates flexibly. This partly linear structure is attractive because it combines interpretability with statistical efficiency. In the non-survival setting, theoretical foundations for DNN estimation over structured nonparametric classes were developed by Schmidt-Hieber (2020). In survival analysis, Zhong, Mueller and Wang (2022) extended this paradigm to the partially linear Cox model $\log h_0(t) + \zeta^\top \mathbf{Z} + f(\mathbf{X})$, where $\mathbf{Z}$ is the low-dimensional covariate vector of primary interest, $\mathbf{X}$ is a nuisance covariate vector, and f is approximated by a DNN. Their approach yields optimal convergence of the nuisance estimator and $\sqrt{n}$ -consistent estimation of ζ , but it remains tied to the PH assumption and to baseline hazard estimation.

In this paper we develop a flexible semiparametric survival model that extends the partially linear framework of Zhong, Mueller and Wang (2022) to a substantially more general non-PH setting. Specifically, we model the log-hazard as $\theta_o^\top \mathbf{Z} + g_o(t, \mathbf{X})$, where $\theta_o \in \mathbb{R}^p$ is the finite-dimensional parameter of interest and $g_o(t, \mathbf{X})$ is an unknown nuisance function approximated by a DNN. This formulation is baseline-free: all time dependence is absorbed into $g_o(t, \mathbf{X})$, rather than decomposed into a baseline hazard and covariate effects. As a result, the model allows arbitrary interactions between time and the nuisance covariates, naturally accommodates time-varying effects, and strictly enlarges the class of partially linear Cox-type models.

This extension is methodologically nontrivial. Once the nuisance component depends on both t and $\mathbf{X}$, the Cox partial likelihood is no longer available as a loss function, and the full likelihood must be used instead. This introduces a predictable cumulative-hazard integral that must be numerically approximated during training and leads to a substantially different asymptotic analysis. In particular, the time-dependent nuisance component requires new empirical-process arguments, a new characterization of the nuisance tangent space, and a new control of the bias induced by DNN approximation. We show, nevertheless, that the estimator of θ_o remains $\sqrt{n}$ -consistent, asymptotically normal, and semiparametrically efficient.

A second and particularly important contribution of the paper is inference for the survival curve of a new subject. For an independent draw $(\mathbf{X}_0, \mathbf{Z}_0)$ from the covariate distribution, we construct a cross-fitted one-step estimator of the cumulative hazard $H_o(t | \mathbf{X}_0, \mathbf{Z}_0)$ and the survival function $S_o(t | \mathbf{X}_0, \mathbf{Z}_0)$, together with Wald-type pointwise confidence intervals. To the best of our knowledge, this is the first frequentist asymptotic pointwise inference result for a survival curve in a DNN survival model, with or without a separate linear component. This result is established under an additional assumption that the nuisance covariate vector $\mathbf{X}$ has discrete finite support, which yields an explicit closed-form Riesz representer and therefore a tractable one-step correction. From a practical perspective, this restriction is often acceptable because nuisance covariates frequently include categorical variables or grouped continuous features, and because the finite-support assumption is required only for the explicit survival-inference theory rather than for estimation and efficiency of θ_o . The broader survival-inference problem can also be studied under general covariate distributions through

a high-level semiparametric characterization of the efficient influence function. Since that formulation is more technical and does not yield a comparably explicit estimator without additional approximation arguments, we present it in the appendix.

Our framework is also relevant beyond the partially linear setting. If the linear component is absorbed into the nuisance function, then the one-step construction and efficient influence function derived here remain valid, with the parametric term dropping out. In this sense, the paper also contributes asymptotic survival-inference tools for baseline-free DNN hazard models more broadly. Compared with [Su et al. \(2026\)](#), who derive functional asymptotic normality for smooth hazard-function functionals and use it for testing, we study a different and sharper target: semiparametric efficiency for an interpretable linear effect and explicit Wald-type pointwise inference for the survival curve of a new subject.

The remainder of the paper is organized as follows. Sections 2 and 3 introduce the model, the DNN class, and the FLEXI-Haz estimation and survival-inference procedures. Sections 4 and 5 develop the asymptotic theory for estimation and pointwise survival inference. Simulation studies and real-data analyses are presented later. Code for FLEXI-Haz and for reproducing the simulations and data analyses is available at <https://github.com/AsafBanana/FLEXI-Haz>.

2. Model Formulation. We consider right-censored survival data, where for each individual the true event time is denoted by U and the censoring time by C . We observe only $T = \min(U, C)$ and $\Delta = \mathbf{1}(U \leq C)$, indicating the observed time and whether the event occurred. Let the covariate vector be $(\mathbf{Z}^\top, \mathbf{X}^\top)^\top$, where $\mathbf{Z} \in \mathbb{R}^p$ contains the variables of primary scientific interest and $\mathbf{X} \in \mathbb{R}^d$ denotes nuisance covariates. Our goal is to characterize the effect of $\mathbf{Z}$ on the event time while appropriately adjusting for $\mathbf{X}$. For example, $\mathbf{X}$ may represent a high-dimensional set of genomic or imaging features, whereas $\mathbf{Z}$ may contain key clinical variables such as treatment assignment or targeted biomarkers. Understanding the effect of $\mathbf{Z}$ while properly adjusting for $\mathbf{X}$ is essential for identifying clinically actionable signals. For simplicity, we focus here on time-independent covariates; the extension to time-dependent covariates will be discussed in Section 7. We assume that U and C are conditionally independent given $(\mathbf{X}, \mathbf{Z})$. The observed data consist of n independent replicates, $\{(T_i, \Delta_i, \mathbf{X}_i, \mathbf{Z}_i)\}_{i=1}^n$.

Under the above setting, the Cox PH model ([Cox, 1972](#)) specifies the instantaneous hazard function at time $t > 0$ as

$$h_1(t \mid \mathbf{X}, \mathbf{Z}) = h_0(t) \exp(\boldsymbol{\alpha}_1^\top \mathbf{X} + \boldsymbol{\alpha}_2^\top \mathbf{Z}),$$

where h_0 is an unspecified baseline hazard function and $\boldsymbol{\alpha}_1 \in \mathbb{R}^d$ and $\boldsymbol{\alpha}_2 \in \mathbb{R}^p$ are unknown parameters. The model remains widely used due to its interpretability, semiparametric structure, and the availability of efficient estimation procedures. DeepSurv ([Katzman et al., 2018](#)) extends the Cox model by replacing the linear risk score $\exp(\boldsymbol{\alpha}_1^\top \mathbf{X} + \boldsymbol{\alpha}_2^\top \mathbf{Z})$ with a flexible nonlinear function $f_1(\mathbf{X}, \mathbf{Z})$ estimated by a DNN. Cox-Time ([Kvamme, Borgan and Scheel, 2019](#)) further generalized this formulation by replacing f_1 with a time-dependent function, $f_2(t, \mathbf{X}, \mathbf{Z})$, thereby relaxing the PH assumption and permitting time-covariate interactions. Because f_2 is time-dependent while the baseline hazard is left unspecified, identifiability becomes nontrivial; [Kvamme, Borgan and Scheel \(2019\)](#) address this challenge through a numerical normalization strategy. More recently, [Hu and Nan \(2023\)](#) proposed a baseline-free hazard model of the form

$$h_3(t \mid \mathbf{X}, \mathbf{Z}) = \exp\{f_3(t, \mathbf{X}, \mathbf{Z})\},$$

which provides full flexibility for modeling non-proportional, time-varying effects via a DNN. Their estimation procedure is evaluated only through simulations, with no accompanying asymptotic guarantees. Although these DNN-based approaches are highly flexible,

they do not provide interpretable estimates for the effect of $\mathbf{Z}$. To address this limitation, [Zhong, Mueller and Wang \(2022\)](#) introduced the deep partially linear Cox model

$$h_4(t | \mathbf{X}, \mathbf{Z}) = h_0(t) \exp\{\zeta^\top \mathbf{Z} + f(\mathbf{X})\},$$

which augments the Cox framework by separating the time component from the covariates and retaining a parametric term $\zeta^\top \mathbf{Z}$ for interpretability, while modeling nuisance effects through a DNN. This structure enables valid inference for ζ while still allowing complex, nonlinear effect of $\mathbf{X}$.

In this work, we extend the deep partially linear framework of [Zhong, Mueller and Wang \(2022\)](#) to the non-PH setting by considering a baseline-free model of the form

$$(2.1) \quad h_o(t | \mathbf{X}, \mathbf{Z}) = \exp\{\theta_o^\top \mathbf{Z} + g_o(t, \mathbf{X})\},$$

where $g_o(t, \mathbf{X})$ is estimated by a DNN. Model (2.1) is baseline-free in the sense that the entire time dependence of the hazard is absorbed into the nuisance function $g_o(t, \mathbf{X})$, rather than being decomposed into a baseline hazard and covariate effects. This is not merely a reparameterization of a Cox-type model with time-varying coefficients: allowing $g_o(t, \mathbf{X})$ to be an unrestricted function of $(t, \mathbf{X})$ strictly enlarges the model class relative to representations of the form $h_0(t) \exp\{\zeta^\top \mathbf{Z} + f(\mathbf{X})\}$. This extension overcomes the PH restriction inherent in [Zhong, Mueller and Wang \(2022\)](#), resolves the non-identifiability issues that arise when time is included in the DNN, and greatly expands the class of survival models that can be handled within a semiparametric, partially linear framework. Identifiability of θ_o in this setting follows from the partially linear structure together with Assumption A5 (see Section 4.2). These conditions ensure that no nontrivial linear function of $\mathbf{Z}$ can be represented almost surely as a function of $(t, \mathbf{X})$, and therefore prevent $g_o(t, \mathbf{X})$ from absorbing components of the linear effect. This is formulated in Lemma 3 (Section 4.3).

As we demonstrate in Section 7, restricting the nuisance function to depend only on $\mathbf{X}$, as in the partially linear Cox model, can induce substantial bias in the estimation of the linear effect whenever the nuisance covariates exhibit time-varying effects. Allowing g_o to depend on both t and $\mathbf{X}$ eliminates this source of bias but introduces significant methodological challenges. First, the negative partial likelihood can no longer be used as the DNN loss function. Instead, the full likelihood is adopted, which includes the predictable cumulative hazard that must be numerically approximated at every training step, increasing the computational burden. Second, the presence of this integral markedly complicates the asymptotic analysis: the time dimension introduces additional stochastic terms that must be controlled, requiring new empirical process arguments beyond those used for time-invariant nuisance functions. Our estimation framework in Section 3 addresses these computational difficulties, and the technical developments leading to the asymptotic results are presented in Sections 4 and 5.

2.1. Composite Hölder Class of g_o . To formalize the complexity assumptions on the nuisance component g_o , we follow the compositional function framework of [Juditsky, Lepski and Tsybakov \(2009\)](#) and [Schmidt-Hieber \(2020\)](#), and use the notation of [Zhong, Mueller and Wang \(2022\)](#). We assume that the nuisance function g_o admits a hierarchical, low-dimensional representation of the form

$$g_o = g_q \circ g_{q-1} \circ \cdots \circ g_1 \circ g_0,$$

with the convention that $(g \circ f)(x) = g(f(x))$. This composition is associated with a dimension vector $\mathbf{d}^* = (d_0^*, \dots, d_{q+1}^*) \in \mathbb{N}_+^{q+2}$ and an intrinsic dimension vector $\tilde{\mathbf{d}} = (\tilde{d}_0, \dots, \tilde{d}_q) \in \mathbb{N}_+^{q+1}$, $\tilde{d}_\ell \leq d_\ell^*$ for each ℓ . The outermost input is $d_0^* = r = 1 + d$, corresponding to observed times and covariates $\mathbf{X}$, and the final output is scalar $d_{q+1}^* = 1$. For each layer

$l, l = 0, \dots, q$, the vector-valued mapping $g_\ell = (g_{\ell 1}, \dots, g_{\ell d_{\ell+1}^*})^\top$ maps the bounded set $[a_\ell, b_\ell]^{d_\ell^*}$ into $[a_{\ell+1}, b_{\ell+1}]^{d_{\ell+1}^*}$, where $a_\ell, b_\ell \in \mathbb{R}$ are fixed finite constants ensuring each intermediate representation remains bounded. Additionally, each component $g_{\ell j}$ depends on at most $\tilde{d}_\ell$ coordinates of its input.

For illustration, consider $f : \mathbb{R}^6 \rightarrow \mathbb{R}$ defined by

$$f(x_1, \dots, x_6) = \sin(x_1 + x_2) + \log(1 + x_3) + x_4^2.$$

which depends on low-dimensional groupings of the input coordinates. A valid three-layer decomposition is $g = g_2 \circ g_1 \circ g_0$ where

$$g_0 : \mathbb{R}^6 \rightarrow \mathbb{R}^3, \quad g_0(x) = (x_1 + x_2, x_3, x_4),$$

$$g_1 : \mathbb{R}^3 \rightarrow \mathbb{R}^3, \quad g_1(y_1, y_2, y_3) = (\sin(y_1), \log(1 + y_2), y_3^2),$$

$$g_2 : \mathbb{R}^3 \rightarrow \mathbb{R}, \quad g_2(z_1, z_2, z_3) = z_1 + z_2 + z_3.$$

For this representation, $\mathbf{d}^* = (d_0^*, d_1^*, d_2^*, d_3^*) = (6, 3, 3, 1)$, and the intrinsic dimension vector is $\tilde{\mathbf{d}} = (\tilde{d}_0, \tilde{d}_1, \tilde{d}_2) = (2, 1, 3)$. That is, components of g_0 depend on at most two inputs, those of g_1 are univariate, and g_2 combines all three outputs. Other valid decompositions may exist. In general, convergence rates depend on the choice of the optimal decomposition used in the asymptotic analysis.

We also assume that each component function satisfies a standard Hölder–smoothness condition. Let $\alpha > 0$ and $M > 0$. A function is said to have Hölder smoothness index α if all mixed partial derivatives up to order $\lfloor \alpha \rfloor$ exist and are uniformly bounded, and if the derivatives of order exactly $\lfloor \alpha \rfloor$ are $(\alpha - \lfloor \alpha \rfloor)$ –Hölder continuous. Here, $\lfloor \alpha \rfloor$ denotes the greatest integer strictly smaller than α . The ball of α –Hölder functions with radius M consists of all such functions whose derivatives and Hölder constants are bounded by M . Formally, for a domain $\mathbb{D} \subset \mathbb{R}^r$, the Hölder ball is defined as

$$\mathcal{H}_r^\alpha(\mathbb{D}, M) = \left\{ g : \mathbb{D} \rightarrow \mathbb{R} : \sum_{\beta: |\beta| < \alpha} \|\partial^\beta g\|_\infty + \sum_{\beta: |\beta| = \lfloor \alpha \rfloor} \sup_{u, v \in \mathbb{D}, u \neq v} \frac{|\partial^\beta g(u) - \partial^\beta g(v)|}{\|u - v\|^{\alpha - \lfloor \alpha \rfloor}} \leq M \right\},$$

where $\beta^\top = (\beta_1, \dots, \beta_r) \in \mathbb{N}^r$ is a multi-index $\partial^\beta = \partial^{\beta_1} \dots \partial^{\beta_r}$, $|\beta| = \sum_{j=1}^r \beta_j$, and $\|g\|_\infty$ is the sup–norm of a function over its domain.

To formalize the function class used for the nuisance component, let $\alpha^\top = (\alpha_0, \dots, \alpha_q) \in \mathbb{R}_+^{q+1}$ denote the layerwise smoothness parameters, $\mathbf{d}$ the dimension vector, and $\tilde{\mathbf{d}}$ the intrinsic dimension vector. The composite Hölder class is defined as

$$\mathcal{H}(q, \alpha, \mathbf{d}^*, \tilde{\mathbf{d}}, M) = \left\{ g = g_q \circ \dots \circ g_0 : g_{\ell j} \in \mathcal{H}_{\tilde{d}_\ell}^{\alpha_\ell}([a_\ell, b_\ell]^{\tilde{d}_\ell}, M), |a_\ell|, |b_\ell| \leq M \right\}.$$

This class accommodates a broad range of structured functions, such as additive models, single- and multiple-index models, and hierarchical interactions, through the intrinsic dimensions $\tilde{d}_\ell$. Finally, we assume that the true nuisance function g_o belongs to the composite class, $g_o \in \mathcal{H}(q, \alpha, \mathbf{d}^*, \tilde{\mathbf{d}}, M)$.

In the classical (non-composite) Hölder class $\mathcal{H}_d^\alpha$, the minimax-optimal rate is $n^{-\alpha/(2\alpha+d)}$ (Tsybakov, 2009). For multilayer composite Hölder classes, Schmidt-Hieber (2020) established matching minimax lower and upper bounds (up to polylogarithmic factors), showing that the optimal rate is governed by $\gamma_n = \max_{\ell=0, \dots, q} n^{-\tilde{\alpha}_\ell/(2\tilde{\alpha}_\ell + \tilde{d}_\ell)}$, where the effective smoothness indices are $\tilde{\alpha}_\ell = \alpha_\ell \prod_{k=\ell+1}^q (\alpha_k \wedge 1)$, $a \wedge b = \min\{a, b\}$. Thus, each layer contributes to the overall difficulty of the estimation problem through its intrinsic dimension $\tilde{d}_\ell$ and effective smoothness $\tilde{\alpha}_\ell$. Taken together, Theorems 1 and 2 show that γ_n is minimax-optimal up to polylogarithmic factors: Theorem 1 gives the upper bound for our estimator, and Theorem 2 gives the lower bound.

2.2. Neural Network Class. To approximate the nuisance function $g_o(t, \mathbf{X})$, we use the popular class of sparse deep ReLU neural networks (Schmidt-Hieber, 2020, and references therein). This choice is motivated by two considerations: (i) ReLU networks are universal approximators capable of representing complex, high-dimensional functions; and (ii) their structure lends itself to sharp theoretical control through entropy bounds and sparsity constraints. We define a general $(K + 1)$ -layer feedforward ReLU network with depth $K \in \mathbb{N}_+$ and width vector $\mathbf{p}^\top = (p_0, \dots, p_{K+1})$, where $p_0 = r = d + 1$ denotes the input dimension and $p_{K+1} = 1$ the scalar output. Specifically, the network is defined by

$$g(\mathbf{u}) = \mathbf{W}_K \sigma(\mathbf{W}_{K-1} \sigma(\dots \sigma(\mathbf{W}_0 \mathbf{u} + \mathbf{v}_0) \dots) + \mathbf{v}_{K-1}) + \mathbf{v}_K,$$

where $\mathbf{u} \in \mathbb{R}^r$, $\sigma(x) = \max\{x, 0\}$ is applied componentwise, $\mathbf{W}_\ell \in \mathbb{R}^{p_{\ell+1} \times p_\ell}$ are weight matrices, and $\mathbf{v}_\ell \in \mathbb{R}^{p_{\ell+1}}$ are bias vectors, $\ell = 0, \dots, K$.

For theoretical analysis we first restrict attention to networks with uniformly bounded parameters,

$$\mathcal{G}(K, \mathbf{p}) = \left\{ g : \max_{\ell=0, \dots, K} (\|\mathbf{W}_\ell\|_\infty \vee \|\mathbf{v}_\ell\|_\infty) \leq 1 \right\},$$

which ensures controlled Lipschitz behavior of the class, where for a matrix $\mathbf{A}$ and a vector $\mathbf{a}$ the entrywise sup-norms are adopted, $\|\mathbf{A}\|_\infty = \max_{i,j} |A_{ij}|$, $\|\mathbf{a}\|_\infty = \max_j |a_j|$. Following Schmidt-Hieber (2020) we further impose sparsity. For sparsity level $s \in \mathbb{N}_+$ and range bound $D > 0$, the sparse deep ReLU class is defined as

$$\mathcal{G}(K, s, \mathbf{p}, D) = \left\{ g \in \mathcal{G}(K, \mathbf{p}) : \sum_{\ell=0}^K (\|\mathbf{W}_\ell\|_0 + \|\mathbf{v}_\ell\|_0) \leq s, \quad \|g\|_\infty \leq D \right\},$$

where $\|\cdot\|_0$ counts nonzero entries. Sparsity serves to control network complexity and is essential for obtaining optimal convergence rates. In the subsequent analysis we set the depth, width, and sparsity levels to balance the bias-variance tradeoff. The bracketing entropy (van der Vaart and Wellner, 1996, Chs. 2–3) of this class is then bounded using results of Schmidt-Hieber (2020), and Dudley’s entropy integral (van der Vaart and Wellner, 1996, Thm. 2.14.1) is used to establish the required uniform empirical-process bounds.

3. FLEXI-Haz Procedure.

3.1. Estimation of (θ_o, g_o) . Estimation is based on maximizing the full likelihood induced by the hazard model (2.1). Under noninformative and conditionally independent censoring given $(\mathbf{X}, \mathbf{Z})$, the corresponding full log-likelihood is proportional to

$$(3.1) \quad \ell_n(\boldsymbol{\theta}, g) = \frac{1}{n} \sum_{i=1}^n \left[\Delta_i \{g(T_i, \mathbf{X}_i) + \boldsymbol{\theta}^\top \mathbf{Z}_i\} - \int_0^\tau Y_i(t) \exp\{g(t, \mathbf{X}_i) + \boldsymbol{\theta}^\top \mathbf{Z}_i\} dt \right].$$

where $Y_i(t) = \mathbf{1}(T_i \geq t)$, $t \in [0, \tau]$, $i = 1, \dots, n$, are at-risk processes, and τ is a pre-specified constant. Because g is modeled by a DNN, the predictable cumulative hazard in (3.1) contains a time-dependent integral that lacks a closed-form expression and must therefore be approximated numerically during optimization. We therefore introduce a grid $0 = t_0 < t_1 < \dots < t_{m_n} = \tau$ and approximate the integral by the Riemann-type sum

$$\int_0^\tau Y_i(t) \exp\{g(t, \mathbf{X}_i) + \boldsymbol{\theta}^\top \mathbf{Z}_i\} dt \approx \sum_{j: Y_i(t_{j-1})=1} (\min\{T_i, t_j\} - t_{j-1}) \exp\{g(t_j, \mathbf{X}_i) + \boldsymbol{\theta}^\top \mathbf{Z}_i\}.$$

Defining $\Delta_{ij} = \mathbf{1}\{T_i \in (t_{j-1}, t_j], \Delta_i = 1\}$, we obtain the discretized likelihood

$$(3.2) \quad \tilde{\ell}_n(\boldsymbol{\theta}, g) = \frac{1}{n} \sum_{i=1}^n \sum_{j: Y_i(t_{j-1})=1} \left[\Delta_{ij} \{g(t_j, \mathbf{X}_i) + \boldsymbol{\theta}^\top \mathbf{Z}_i\} - \{\min(T_i, t_j) - t_{j-1}\} \exp\{g(t_j, \mathbf{X}_i) + \boldsymbol{\theta}^\top \mathbf{Z}_i\} \right].$$

In practice, discretizing the time axis on a fine grid and augmenting it with all observed times (event and censoring times) provides an accurate and computationally efficient approximation, as detailed in Section 5.1. This construction automatically produces a dense grid in regions where failures occur, precisely where the integrand contributes most to the likelihood.

We consider two estimators. The oracle maximum-likelihood estimator is defined as

$$(3.3) \quad (\hat{\boldsymbol{\theta}}, \hat{g}) \in \arg \max_{g \in \mathcal{G}(K, s, \mathbf{p}, D), \boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}^p} \ell_n(\boldsymbol{\theta}, g).$$

The estimator that is actually implemented replaces the at-risk integral with its discretized version $\tilde{\ell}_n$. We therefore define the numerically integrated likelihood estimator as

$$(3.4) \quad (\tilde{\boldsymbol{\theta}}, \tilde{g}) \in \arg \max_{g \in \mathcal{G}(K, s, \mathbf{p}, D), \boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}^p} \tilde{\ell}_n(\boldsymbol{\theta}, g).$$

If the maximizer is not unique, owing to the non-uniqueness of neural-network representations, we take an arbitrary element of the argmax set. Theorems 1–4 establish the asymptotic properties of (3.3), while Theorem 5 shows that under the chosen grid these properties carry over to the estimator (3.4) implemented in practice.

3.2. Estimation and Marginal Inference for the Survival of a New Subject. We now turn to estimation and inference on the survival function of a new subject. Let $(\mathbf{X}_0, \mathbf{Z}_0)$ be an independent draw from the covariate distribution, independent of the training sample, and define

$$H_o(t | \mathbf{X}_0, \mathbf{Z}_0) = \int_0^t h_o(u | \mathbf{X}_0, \mathbf{Z}_0) du, \quad S_o(t | \mathbf{X}_0, \mathbf{Z}_0) = \exp\{-H_o(t | \mathbf{X}_0, \mathbf{Z}_0)\}.$$

Our goal is to construct valid pointwise inference for these quantities. Inference for such functionals can be formulated within the semiparametric framework of pathwise differentiability and efficient influence functions (Bickel et al., 1993; Van der Vaart, 2000; Tsiatis, 2006). A natural estimator of $H_o(t | \mathbf{X}_0, \mathbf{Z}_0)$ is obtained by plugging the estimators $(\tilde{\boldsymbol{\theta}}, \tilde{g})$ into the hazard and then numerically integrating, yielding

$$\hat{H}(t | \mathbf{X}_0, \mathbf{Z}_0) = \sum_{j: t_j \leq t} \Delta t_j \exp\{\tilde{g}(t_j, \mathbf{X}_0) + \tilde{\boldsymbol{\theta}}^\top \mathbf{Z}_0\},$$

where $\Delta t_j = t_j - t_{j-1}$, and $\hat{S}(t | \mathbf{X}_0, \mathbf{Z}_0) = \exp\{-\hat{H}(t | \mathbf{X}_0, \mathbf{Z}_0)\}$. Although the plug-in estimator is natural, it is generally not the most convenient starting point for first-order inference. The reason is that the target depends on the infinite-dimensional nuisance function g_o , and the first-order effect of nuisance estimation error need not vanish automatically. In particular, the nuisance-rate theory established in Section 4 controls $\tilde{g} - g_o$ in an integrated norm, which is sufficient for estimation of $\boldsymbol{\theta}_o$ but does not by itself guarantee that the plug-in survival estimator admits an asymptotically linear expansion with an $o_p(n^{-1/2})$ remainder.

The purpose of the one-step construction is to remove this first-order nuisance bias (Bickel et al., 1993; Van der Vaart, 2000). Concretely, we will augment the plug-in estimator by an empirical average of an estimated efficient influence function (EIF). This correction yields an estimator whose leading error term is linear and whose remainder is second-order under the rate conditions imposed below. As a consequence, Wald-type inference becomes available for the cumulative hazard and, via the delta method, for the survival function.

In practice, we implement this correction using cross-fitting (Zheng and van der Laan, 2011; Chernozhukov et al., 2018).. The motivation for cross-fitting is twofold. First, it separates nuisance estimation from influence-function evaluation, which substantially weakens empirical-process requirements. Second, it makes the one-step correction more robust to overfitting in flexible nuisance estimators such as DNN. For this reason, we formulate the estimator below in cross-fitted form from the outset.

Two additional points are worth emphasizing. First, the target considered here is *marginal* in the sense that it is evaluated at an independent draw $(\mathbf{X}_0, \mathbf{Z}_0)$. This differs from conditional pointwise inference at a fixed covariate value $(\mathbf{x}_0, \mathbf{z}_0)$, which may be nonregular without additional smoothness or localization assumptions. Second, the construction below also applies to the hazard model without a separate linear component, obtained by absorbing $\mathbf{Z}$ into the nuisance function g_o . In that case, the parametric score and information terms disappear, and the EIF reduces to its nuisance–Riesz component alone. Equivalently, one may replace $\mathbf{X}$ throughout by the enlarged covariate vector $(\mathbf{X}^\top, \mathbf{Z}^\top)^\top$.

We first present the estimator at a high level, postponing all technical derivations to Section 5 below.

To define the one-step estimator, let $\phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0)$ denote the efficient influence function for $H_o(t | \mathbf{X}_0, \mathbf{Z}_0)$, where $\mathbf{W} = (T, \Delta, \mathbf{X}, \mathbf{Z})$, and let $\hat{\phi}_t^{(-k)}(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0)$ be its estimator computed using nuisance estimators trained without fold k . Let $\{\mathcal{I}_k\}_{k=1}^K$ be a partition of $\{1, \dots, n\}$ into folds. For $i \in \mathcal{I}_k$, the observation $\mathbf{W}_i$ is evaluated using nuisance estimators fitted on the complement of $\mathcal{I}_k$.

The cross-fitted one-step estimator of the cumulative hazard is

$$(3.5) \quad \hat{H}^{1,\text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0) = \hat{H}^{\text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0) + \frac{1}{n} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \hat{\phi}_t^{(-k)}(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0),$$

where

$$\hat{H}^{\text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0) = \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \hat{H}^{(-k)}(t | \mathbf{X}_0, \mathbf{Z}_0)$$

and the corresponding one-step estimator of the survival function is

$$(3.6) \quad \hat{S}^{1,\text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0) = \exp\{-\hat{H}^{1,\text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0)\}.$$

At an intuitive level, (3.5) augments the plug-in estimator by a correction term that removes the first-order effect of nuisance estimation. Under the conditions stated below, the resulting estimator is asymptotically linear. The practical construction therefore reduces to two ingredients: estimation of the nuisance objects entering the plug-in hazard; and estimation of the efficient influence function $\phi_t^0(\cdot; \mathbf{X}_0, \mathbf{Z}_0)$.

The one-step correction is not shape-constrained. Hence, in finite samples, $\hat{H}^{1,\text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0)$ need not be monotone nondecreasing in t , and $\hat{S}^{1,\text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0) = \exp\{-\hat{H}^{1,\text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0)\}$ need not be monotone nonincreasing. This is not a problem for the pointwise asymptotic inference developed here, which is for each fixed t . If a monotone curve is desired for display, the estimated survival curve can be post-processed by monotone rearrangement or isotonic projection.

The next subsection describes the resulting estimator explicitly in the finite discrete-covariate case, where the influence function admits a closed-form representation. The explicit construction below focuses on the case where bX has finite support, since in that setting the nuisance Riesz representer admits a closed form. A more general high-level formulation for arbitrary covariate distributions is possible, based on an abstract $L^2(P_{\mathbf{X}, \mathbf{Z}})$ -valued pathwise differentiability argument and an associated representer condition, but this leads to a less explicit estimator and is therefore deferred to the Appendix.

3.2.1. *Explicit form under finite discrete covariates.* We now specialize to the case where $\mathbf{X}$ has finite support $\mathcal{X} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(J)}\}$ with $\Pr(\mathbf{X} = \mathbf{x}^{(j)}) = \pi_j > 0$, $j = 1, \dots, J$. In this setting, the nuisance Riesz representer underlying the efficient influence function admits a closed form, and therefore the one-step estimator becomes fully explicit.

For $u \in [0, \tau]$ and $j = 1, \dots, J$, define

$$A_j(u) = \mathbb{E}_0 \left\{ Y(u) h_o(u \mid \mathbf{X}, \mathbf{Z}) \mid \mathbf{X} = \mathbf{x}^{(j)} \right\}.$$

Then the nuisance part of the efficient influence function is represented by a weight function of the form

$$(3.7) \quad \psi_t^0(u, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) = \mathbf{1}(u \leq t) \mathbf{1}(\mathbf{X} = \mathbf{X}_0) \frac{h_o(u \mid \mathbf{X}_0, \mathbf{Z}_0)}{\Pr(\mathbf{X} = \mathbf{X}_0) A_{\mathbf{X}_0}(u)}.$$

Accordingly, the efficient influence function takes the form

$$(3.8) \quad \phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) = \int_0^t \mathbf{1}(\mathbf{X} = \mathbf{X}_0) \frac{h_o(u \mid \mathbf{X}_0, \mathbf{Z}_0)}{\Pr(\mathbf{X} = \mathbf{X}_0) A_{\mathbf{X}_0}(u)} dM(u) + c_{0,t}(\mathbf{X}_0, \mathbf{Z}_0)^\top \mathbf{I}(\boldsymbol{\theta}_0)^{-1} \ell_{\theta_0}^*(\mathbf{W}),$$

where $M(t) = N(t) - \int_0^t Y(u) h_o(u \mid \mathbf{X}, \mathbf{Z}) du$ is the counting-process martingale under $\mathbb{P}_0$ with $N(t) = \Delta \mathbf{1}(T \leq t)$. The quantities $c_{0,t}(\mathbf{X}_0, \mathbf{Z}_0)$, $I(\boldsymbol{\theta}_0)$, and $\ell_{\theta_0}^*$ are defined formally in Section 5; for now, it suffices to note that they represent the sensitivity term, efficient information, and efficient score, respectively. Replacing the unknown quantities in (3.8) by their estimators yields an explicit estimator $\hat{\phi}_t^{(-k)}(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0)$ and therefore an explicit cross-fitted one-step estimator through (3.5)–(3.6).

This discrete-support case is especially convenient in practice because the Riesz representer is explicit. Hence the one-step correction can be implemented directly, without introducing an additional approximation step for the representer.

Let $\hat{\sigma}_{H,t}^2(\mathbf{X}_0, \mathbf{Z}_0)$ denote a consistent estimator of the asymptotic variance of $\hat{H}^{1,\text{cf}}(t \mid \mathbf{X}_0, \mathbf{Z}_0)$, for example the empirical variance of the cross-fitted influence-function estimates is given by

$$(3.9) \quad \hat{\sigma}_{H,t}^2(\mathbf{X}_0, \mathbf{Z}_0) = \frac{1}{n} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \left\{ \hat{\phi}_t^{(-k)}(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0) \right\}^2.$$

An asymptotic $(1 - \alpha)$ confidence interval for $H_o(t \mid \mathbf{X}_0, \mathbf{Z}_0)$ is therefore

$$(3.10) \quad \hat{H}^{1,\text{cf}}(t \mid \mathbf{X}_0, \mathbf{Z}_0) \pm z_{1-\alpha/2} \frac{\hat{\sigma}_{H,t}(\mathbf{X}_0, \mathbf{Z}_0)}{\sqrt{n}},$$

and, by the delta method, an asymptotic $(1 - \alpha)$ confidence interval for $S_o(t \mid \mathbf{X}_0, \mathbf{Z}_0)$ is

$$(3.11) \quad \hat{S}^{1,\text{cf}}(t \mid \mathbf{X}_0, \mathbf{Z}_0) \pm z_{1-\alpha/2} \frac{\hat{S}^{1,\text{cf}}(t \mid \mathbf{X}_0, \mathbf{Z}_0) \hat{\sigma}_{H,t}(\mathbf{X}_0, \mathbf{Z}_0)}{\sqrt{n}}.$$

The asymptotic validity of these intervals is established in Section 5.

4. Asymptotic Theory for $(\tilde{\boldsymbol{\theta}}, \tilde{g})$.

4.1. *Assumptions.* Some additional notation will be used throughout. We write $a_n \lesssim b_n$ if there exists a constant $c > 0$ such that $a_n \leq c b_n$ for all n , and $a_n \asymp b_n$ if both $a_n \lesssim b_n$ and $b_n \lesssim a_n$ hold. For a column vector $\mathbf{a}$, $\mathbf{a}^{\otimes 2} = \mathbf{a} \mathbf{a}^\top$. Let $\mathbb{P}_0$ denote the true distribution

of a generic observation $(T, \Delta, \mathbf{X}, \mathbf{Z})$. For any measurable function $\mu = \mu(T, \mathbf{X}, \mathbf{Z})$, define the unweighted $L^2_{\mathbb{P}_0}$ norm $\|\mu\|_{L^2(\mathbb{P}_0)}^2 = \mathbb{E}_0\{\mu(T, \mathbf{X}, \mathbf{Z})^2\}$ and the event-weighted norm $\|\mu\|_{L^2(\mathbb{P}_0, \Delta)}^2 = \mathbb{E}_0\{\Delta \mu(T, \mathbf{X}, \mathbf{Z})^2\}$. Let $\eta = (\boldsymbol{\theta}, g)$, write

$$\chi_\eta(t, \mathbf{X}, \mathbf{Z}) = g(t, \mathbf{X}) + \boldsymbol{\theta}^\top \mathbf{Z},$$

and denote the true parameter by $\eta_o = (\boldsymbol{\theta}_o, g_o)$. For two parameters η_1, η_2 , we measure their respective event-weighted score distance by

$$d(\eta_1, \eta_2) = \left(\mathbb{E}_0[\Delta \{\chi_{\eta_1}(T, \mathbf{X}, \mathbf{Z}) - \chi_{\eta_2}(T, \mathbf{X}, \mathbf{Z})\}^2] \right)^{1/2} = \|\chi_{\eta_1} - \chi_{\eta_2}\|_{L^2(\mathbb{P}_0, \Delta)}.$$

Finally, the population loss is defined as

$$(4.1) \quad \ell_0(\eta) = \mathbb{E}_0 \left[\Delta \chi_\eta(T, \mathbf{X}, \mathbf{Z}) - \int_0^\tau Y(t) \exp\{\chi_\eta(t, \mathbf{X}, \mathbf{Z})\} dt \right].$$

A1. *The failure time U and censoring time C are conditionally independent given $(\mathbf{X}^\top, \mathbf{Z}^\top)$, and the censoring mechanism is noninformative in the sense that the censoring process is noninformative for $(\boldsymbol{\theta}, g)$. The true nuisance function satisfies $g_o \in \mathcal{H}(q, \boldsymbol{\alpha}, \mathbf{d}^*, \tilde{\mathbf{d}}, M)$.*

A2. *The covariate vector $(\mathbf{X}, \mathbf{Z})$ takes values in a bounded subset of $\mathbb{R}^{d+p}$, denoted $\mathcal{X} \times \mathcal{Z}$, where both sets are compact. Without loss of generality, assume $\mathcal{X} \subset [0, 1]^d$ and that $\|\mathbf{Z}\|_2 \leq B_Z$ almost surely. The joint density of $(\mathbf{X}, \mathbf{Z})$ is bounded away from zero on $\mathcal{X} \times \mathcal{Z}$. The parameter space for the finite-dimensional component is restricted to the compact set $\Theta = \{\boldsymbol{\vartheta} \in \mathbb{R}^p : \|\boldsymbol{\vartheta}\| \leq M_\Theta\}$. Finally, the study follows participants over a fixed, finite time horizon $\tau > 0$.*

A3. *The network architecture satisfies $K = O(\log n)$, $s = O(n\gamma_n^2 \log n)$, and $n\gamma_n^2 \lesssim \min_{0 \leq \ell \leq K} p_\ell \lesssim \max_{0 \leq \ell \leq K} p_\ell \lesssim n$. In addition, the sieve is uniformly bounded, $\|g\|_\infty \leq D_n$ for a known sieve radius $D_n = O(1)$.*

A4. *There exist $\delta_1, \delta_2, \delta_3 \in (0, 1]$ such that, almost surely, $\Pr(\Delta = 1 \mid T, \mathbf{X}, \mathbf{Z}) \geq \delta_1$, $\Pr(U \geq \tau \mid \mathbf{X}, \mathbf{Z}) \geq \delta_2$, and $\Pr(Y(t) = 1 \mid \mathbf{X}, \mathbf{Z}) \geq \delta_3$ for all $t \in [0, \tau]$.*

A5. *The covariate vector $\mathbf{Z}$ does not include a constant (intercept) term. Furthermore, the residual covariance matrix of $\mathbf{Z}$ given $\mathbf{T}, \mathbf{X}$ among failures, is positive definite, that is,*

$$\Sigma_{\mathbf{Z}} = \mathbb{E} \left[\left\{ \mathbf{Z} - \mathbb{E}(\mathbf{Z} \mid T, \mathbf{X}, \Delta = 1) \right\} \left\{ \mathbf{Z} - \mathbb{E}(\mathbf{Z} \mid T, \mathbf{X}, \Delta = 1) \right\}^\top \right] \succ 0.$$

Assumptions A1–A3 are also used in [Zhong, Mueller and Wang \(2022\)](#) under the partial linear PH regression model. A3 determines the structure of the neural network and provides a trade-off between the approximation error and estimation error ([Schmidt-Hieber, 2020](#)). Assumption A2 is a standard assumption for semi- or non-parametric regression. Moreover, A1 and A2 imply that there exist constants $0 < c < C < \infty$ (that depend on D_n, B_Z and M_Θ) such that for all $t \in (0, \tau]$ and $(\mathbf{X}, \mathbf{Z}) \in \mathcal{X} \times \mathcal{Z}$, $c_h \leq h_o(t|\mathbf{X}, \mathbf{Z}) \leq C_h$. Assumption A4 is required since in the hazard function (2.1) time is one of the components in the function g_o estimated by the DNN and therefore the effects of the covariates $\mathbf{X}$ and time are not disentangled, in contrast to the partial linear Cox model of [Zhong, Mueller and Wang \(2022\)](#). Hence we need a non-vanishing chance to observe events over the entire study horizon for all $(\mathbf{x}, \mathbf{Z})$ in the support. While this assumption is stronger than in standard Cox regression,

it is natural in baseline-free hazard models and is comparable to positivity conditions commonly imposed in semiparametric and causal models with continuous time. Assumption A5 is the non-PH analogue of the standard partial-linear identifiability condition. Because the nuisance component is allowed to depend on $(t, \mathbf{X})$, the relevant residual variation for the linear covariates must be measured after conditioning on $(T, \mathbf{X})$ among failures, rather than on $\mathbf{X}$ alone.

4.2. Asymptotic Results. The proofs of all lemmas and theorems in this section appear in Appendix A. We begin by establishing several basic properties of the model. A key step is to understand how the counting-process framework allows us to translate expressions involving the at-risk processes into simpler quantities expressed directly in terms of the event indicator Δ . The next lemma formalizes this relationship, showing that these two representations are equivalent up to a multiplicative constant. This equivalence will be used repeatedly throughout the analysis.

LEMMA 1. *Under A1–A2, for any nonnegative, predictable function $f(t, \mathbf{X}, \mathbf{Z})$ with $\mathbb{E}[\int_0^\tau Y(t) f(t, \mathbf{X}, \mathbf{Z}) dt] < \infty$,*

$$\mathbb{E}\left\{\int_0^\tau Y(t) f(t, \mathbf{X}, \mathbf{Z}) dt\right\} \asymp \mathbb{E}\{\Delta f(T, \mathbf{X}, \mathbf{Z})\}.$$

The proof uses only the positivity of the hazard function together with the compensator identity for counting processes, and therefore applies uniformly to all functions that arise in our later analysis.

LEMMA 2. *Let $\chi_{\eta_j}(t, \mathbf{X}, \mathbf{Z})$, $j = 1, 2$, be bounded predictable processes satisfying*

$$\mathbb{E}\left\{\int_0^\tau \chi_{\eta_j}(t, \mathbf{X}, \mathbf{Z})^2 dt\right\} < \infty \quad j = 1, 2.$$

Define $\phi(t, \mathbf{X}, \mathbf{Z}) = \chi_{\eta_1}(t, \mathbf{X}, \mathbf{Z}) - \chi_{\eta_2}(t, \mathbf{X}, \mathbf{Z})$. Under Assumptions A1, A2 and A4, there exist finite constants $0 < c_1 \leq c_2 < \infty$ and $0 < c_3 \leq c_4 < \infty$, depending only on $(\tau, \delta_1, \delta_2, c_h, C_h)$, such that

$$(4.2) \quad c_1 \mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2\} \leq \mathbb{E}\{\phi(T, \mathbf{X}, \mathbf{Z})^2\} \leq c_2 \mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2\},$$

$$(4.3) \quad c_3 \mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2\} \leq \mathbb{E}_{\mathbf{X}, \mathbf{Z}}\left\{\int_0^\tau \phi(t, \mathbf{X}, \mathbf{Z})^2 dt\right\} \leq c_4 \mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2\}.$$

In particular, if $\mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2\} = 0$, then $\phi(T, \mathbf{X}, \mathbf{Z}) = 0$ $\mathbb{P}_0$ -a.s. and $\phi(t, \mathbf{X}, \mathbf{Z}) = 0$ for Lebesgue-a.e. $t \in [0, \tau]$ and $\mathbb{P}_{\mathbf{X}, \mathbf{Z}}$ -a.e. $(\mathbf{X}, \mathbf{Z})$, i.e. $\chi_{\eta_1}(t, \mathbf{X}, \mathbf{Z}) = \chi_{\eta_2}(t, \mathbf{X}, \mathbf{Z})$ for $dt \otimes \mathbb{P}_{\mathbf{X}, \mathbf{Z}}$ -almost all $(t, \mathbf{X}, \mathbf{Z})$.

Lemma 2 shows that the distance $d(\cdot, \cdot)$ is a valid metric on the parameter space: $d(\eta_1, \eta_2) = 0$ if and only if the induced linear score coincides almost everywhere. In addition, the lemma shows that the event-weighted norm and the corresponding unweighted norms are equivalent up to multiplicative constants. As a result, any bound or convergence statement expressed in terms of the metric d can be translated into an equivalent statement under the unweighted $L^2(dt \otimes P_{\mathbf{X}, \mathbf{Z}})$ norm and vice versa. This equivalence is used repeatedly throughout the theoretical development.

Next we study the shape of the population loss around $\eta_o = (\theta_o, g_o)$.

LEMMA 3. Assume A1–A2. For any $\eta \in \Theta \times \mathcal{H}(q, \alpha, \mathbf{d}^*, \tilde{\mathbf{d}}, M)$,

$$\ell_0(\eta) - \ell_0(\eta_o) \asymp -d(\eta, \eta_o)^2.$$

Lemma 3 shows that the population loss is well behaved in a neighbourhood of the truth: any deviation from η_o incurs a loss of order $d(\eta, \eta_o)^2$. Combined with Lemma 2, this implies that η_o is the unique minimizer of the population loss over the square-integrable class under consideration. This population curvature is a key ingredient in the analysis that will later be paired with empirical process bounds to obtain convergence rates for $\hat{g}$ and to derive the asymptotic distribution of $\hat{\theta}$.

We now move to the main asymptotic results. Our ultimate goal is to derive the asymptotic distribution of $\hat{\theta}$ and to show that it attains the semiparametric information bound. To this end, we first establish the convergence rate of the nuisance estimator $\hat{g}$, show that this rate is essentially optimal, characterize the efficient score and information for θ_o , and then combine these ingredients to obtain an asymptotic distribution of $\hat{\theta}$. Finally, we verify that the numerical approximation used in practice does not alter the limit law.

We begin by quantifying how well the estimator $\hat{g}$ recovers the unknown function g_o . The next theorem gives a nonparametric convergence rate for $\hat{g}$ in the event-weighted $L^2_{\mathbb{P}_0}$ norm that naturally appears in the loss and in the distance function $d(\eta, \eta_o)$. The rate is expressed in terms of the complexity parameter γ_n and will later be used to control remainder terms in the expansion of the efficient score.

THEOREM 1. Let $(\hat{\theta}, \hat{g})$ be the estimator defined in (3.3). Under Assumptions A1–A3,

$$(4.4) \quad \left(\mathbb{E} \left[\Delta \{ \hat{g}(T, \mathbf{X}) - g_o(T, \mathbf{X}) \}^2 \right] \right)^{1/2} = O_p(\tau_n)$$

where the expectation is taken with respect to an independent draw $(T, \Delta, \mathbf{X}, \mathbf{Z}) \sim \mathbb{P}_0$ from the true distribution, and $\tau_n = \gamma_n \log^2 n$. The quantity in (4.4) is random through its dependence on the training sample used to construct $\hat{g}$.

The conclusion of Theorem 1 is strengthened by Lemma 2, which shows that $d(\eta, \eta_o)^2$ is equivalent to the unweighted norm $\|\chi_\eta - \chi_{\eta_o}\|_{L^2(dt \otimes \mathbb{P}_{\mathbf{X}, \mathbf{Z}})}^2$. Consequently, the estimator converges in L^2 over the entire time domain $(0, \tau)$ with respect to Lebesgue measure, and not merely at observed event times or in a way that depends on the censoring pattern.

In Theorem 1, both Zhong, Mueller and Wang (2022) and our work rely on Schmidt-Hieber (2020) exclusively for deterministic approximation results of sparse ReLU neural networks over composite Hölder function classes, including the network size and sparsity scaling required to achieve optimal approximation rates. These results are purely analytic and do not involve any distributional assumptions. All stochastic components of the proof, such as control of the estimation error, empirical process bounds, and derivation of convergence rates, are fundamentally different and are derived from the survival-analysis framework. In Zhong, Mueller and Wang (2022), these arguments rely on the Cox partial likelihood and counting-process martingale theory, while in our setting they are based on the full likelihood and properties of the hazard function. Consequently, unlike Schmidt-Hieber (2020), no assumptions on an additive error distribution, including normality, are required or used at any stage of the analysis.

The next result shows that the rate obtained in Theorem 1 is essentially optimal, up to logarithmic factors. It establishes a minimax lower bound for estimation of g_o under the same event-weighted $L^2_{\mathbb{P}_0}$ loss, taken uniformly over the parameter space Θ and the nuisance function class $\mathcal{H}(q, \alpha, \mathbf{d}^*, \tilde{\mathbf{d}}, M)$.

THEOREM 2. Under [A1–A2](#) and [A4](#) there exists $c > 0$ such that

$$\inf_{\hat{g}} \sup_{\theta_o \in \Theta, g_o \in \mathcal{H}(q, \alpha, \mathbf{d}^*, \tilde{\mathbf{d}}, M)} \mathbb{E}_{\mathbb{P}_{g_o, \theta_o}} [\Delta \{\hat{g}(T, \mathbf{X}) - g_o(T, \mathbf{X})\}^2] \geq c \gamma_n^2.$$

We now shift from nuisance-rate optimality to semiparametric efficiency for the finite-dimensional parameter θ_o and characterize the efficient score and information bound in our semiparametric model. To characterize the nuisance tangent space associated with g_o , we consider differentiable one-dimensional submodels that pass through g_o . Let $\mathcal{H}_{g_o}$ denote the collection of all functions $g \in L^2([0, \tau] \times [0, 1]^r)$ for which there exist a submodel $\{g_b : b \in (-1, 1)\} \subset \mathcal{H}(q, \alpha, \mathbf{d}^*, \tilde{\mathbf{d}}, M)$ satisfying

$$\lim_{b \rightarrow 0} \|b^{-1}(g_b - g_o) - g\|_{L^2([0, \tau] \times [0, 1]^r)} = 0.$$

Here, the scalar b is a perturbation parameter indexing the submodel. The set of all attainable first-order directions at g_o is then defined as

$$\mathcal{T}_{g_o} = \left\{ g \in L^2([0, \tau] \times [0, 1]^r) : \exists \{g_b\} \in \mathcal{H}_{g_o} \text{ s.t. } \lim_{b \rightarrow 0} \|b^{-1}(g_b - g_o) - g\|_{L^2([0, \tau] \times [0, 1]^r)} = 0 \right\}.$$

Finally, semiparametric efficiency theory requires a closed linear subspace, so we take the closure of the linear span of $\mathcal{T}_{g_o}$ under the norm $g \mapsto \{\mathbb{E}[\Delta g(T, \mathbf{X})^2]\}^{1/2}$, and denote it by $\overline{\mathcal{T}}_{g_o}$. This forms the nuisance tangent space. Let $(\overline{\mathcal{T}}_{g_o})^p = \{\mathbf{g} = (g_1, \dots, g_p)^\top : g_j \in \overline{\mathcal{T}}_{g_o} \text{ for each } j\}$. Define $\mathbf{g}^* = (g_1^*, \dots, g_p^*)^\top \in (\overline{\mathcal{T}}_{g_o})^p$ componentwise by

$$g_j^* \in \arg \min_{g \in \overline{\mathcal{T}}_{g_o}} \mathbb{E}[\Delta \{Z_j - g(T, \mathbf{X})\}^2], \quad j = 1, \dots, p.$$

Equivalently,

$$\mathbf{g}^* \in \arg \min_{\mathbf{g} \in (\overline{\mathcal{T}}_{g_o})^p} \mathbb{E}[\Delta \|\mathbf{Z} - \mathbf{g}(T, \mathbf{X})\|_2^2].$$

Since $(\overline{\mathcal{T}}_{g_o})^p$ is a Cartesian product and the criterion separates across coordinates, this minimization is componentwise. Thus, $\mathbf{g}^*(t, \mathbf{X})$ is the Δ -weighted L_2 -projection of $\mathbf{Z}$ onto $(\overline{\mathcal{T}}_{g_o})^p$. Finally, under the counting-process representation of the model ([Andersen and Gill, 1982](#)), the counting process $N(t) = \Delta I(T \leq t)$ has compensator $\int_0^t Y(s) \exp\{\boldsymbol{\theta}_o^\top \mathbf{Z} + g_o(s, \mathbf{X})\} ds$ so the associated martingale is

$$M(t) = N(t) - \int_0^t Y(s) \exp\{\boldsymbol{\theta}_o^\top \mathbf{Z} + g_o(s, \mathbf{X})\} ds.$$

THEOREM 3. Under Assumptions [A1–A5](#), the efficient score for θ_o in the semiparametric model (2.1) is

$$\ell_{\theta_o}^*(T, \mathbf{X}, \mathbf{Z}, \Delta) = \int_0^\tau \{\mathbf{Z} - \mathbf{g}^*(t, \mathbf{X})\} dM(t),$$

where $\mathbf{g}^*(t, \mathbf{X})$ is the Δ -weighted L_2 -projection of $\mathbf{Z}$ onto $(\overline{\mathcal{T}}_{g_o})^p$, and $M(t)$ is the associated counting-process martingale. The semiparametric information bound for θ_o is

$$I(\theta_o) = \mathbb{E}\{\ell_{\theta_o}^*(T, \mathbf{X}, \mathbf{Z}, \Delta)^{\otimes 2}\} = \mathbb{E}[\Delta \{\mathbf{Z} - \mathbf{g}^*(T, \mathbf{X})\}^{\otimes 2}].$$

Consequently, the asymptotic variance of any regular asymptotically linear (RAL) estimator of θ_o is bounded below by $I(\theta_o)^{-1}$ in the Loewner sense.

With the curvature and metric properties of the population loss, the convergence rate from Theorem 1, and the efficient score representation of Theorem 3, we can now establish an asymptotic linear expansion for $\hat{\boldsymbol{\theta}}$. This, in turn, yields the asymptotic distribution of the estimator.

THEOREM 4. *Under Assumptions A1–A5, suppose that the efficient information matrix $I(\boldsymbol{\theta}_o)$ is nonsingular and that $\sqrt{n}\tau_n^2 \rightarrow 0$ as $n \rightarrow \infty$, where $\tau_n = \gamma_n \log^2 n$. Then,*

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) = I(\boldsymbol{\theta}_o)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \ell_{\boldsymbol{\theta}_o}^*(T_i, \mathbf{X}_i, \mathbf{Z}_i, \Delta_i) + o_p(1),$$

and hence

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) \xrightarrow{D} N(\mathbf{0}, I(\boldsymbol{\theta}_o)^{-1}).$$

The key technical observation is that, in the Taylor expansion of the population score around g_o , the first-order derivative in $\hat{g} - g_o$ direction vanishes because the efficient score is orthogonal to the nuisance tangent space. The remaining second-order term is $o_p(n^{-1/2})$ provided that $n\gamma_n^4 \rightarrow 0$ as $n \rightarrow \infty$.

Finally, we justify the numerical approximation used to compute the estimator in practice. The next theorem shows that, under mild conditions the estimator $\tilde{\boldsymbol{\theta}}$ has the same asymptotic distribution as the oracle estimator $\hat{\boldsymbol{\theta}}$.

THEOREM 5. *Let $(\tilde{\boldsymbol{\theta}}, \tilde{g})$ be the numerically integrated likelihood estimator defined in (3.4), where the integral $\int_0^\tau Y_i(t) \exp\{g(t, \mathbf{X}_i) + \boldsymbol{\theta}^\top \mathbf{Z}_i\} dt$ is approximated on a time grid $0 = t_0 < t_1 < \dots < t_{m_n} = \tau$ with mesh size $\delta_n = \max_{1 \leq j \leq m_n} (t_j - t_{j-1})$ and suppose that the grid contains all observed times $T_1, \dots, T_n$. Let*

$$\omega_n(\delta) = \sup_{\eta \in \mathcal{N}_n} \sup_{\mathbf{X} \in \mathcal{X}} \sup_{\substack{s, t \in [0, \tau] \\ |t-s| \leq \delta}} |g(t, \mathbf{X}) - g(s, \mathbf{X})|, \quad \mathcal{N}_n = \{\eta : d(\eta, \eta_o) \leq M\tau_n\},$$

for some fixed $M > 0$. Assume that $\omega_n(\delta_n) = o(\tau_n^2)$, $\sqrt{n}\omega_n(\delta_n) \rightarrow 0$, and that the conditions of Theorems 1 and 4 hold. Then,

1. *The numerically integrated likelihood estimator attains the same convergence rate as the oracle estimator:*

$$d(\tilde{\eta}, \eta_o) = O_p(\tau_n), \quad \|\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_o\| = O_p(\tau_n).$$

2. *The parametric estimators are asymptotically equivalent:*

$$\sqrt{n}(\tilde{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}}) \xrightarrow{P} \mathbf{0}.$$

Consequently,

$$\sqrt{n}(\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) = \sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) + o_p(1) \xrightarrow{D} N(\mathbf{0}, I(\boldsymbol{\theta}_o)^{-1}),$$

where $I(\boldsymbol{\theta}_o)$ is the efficient information matrix from Theorem 3.

Theorem 5 shows that the difference between the exact and the numerically integrated likelihood estimators is $o_p(n^{-1/2})$, so that both share the same asymptotic normal distribution. The second conclusion verifies that the required error bounds hold for the grid with n time points, where the per-observation quadrature error is of order $1/n$. This ensures that the asymptotic efficiency result for $\hat{\boldsymbol{\theta}}$ carries over directly to the implementable estimator $\tilde{\boldsymbol{\theta}}$ based on numerical integration.

5. Semiparametric Representation and Asymptotic Theory for $\widehat{S}^{1,cf}(t \mid \mathbf{X}_0, \mathbf{Z}_0)$.

After Section 4 established estimation and efficiency results for the linear parameter θ_o , this section turns to the harder problem of inference for $H_o(t \mid \mathbf{X}_0, \mathbf{Z}_0)$ and $S_o(t \mid \mathbf{X}_0, \mathbf{Z}_0)$. Our development follows the semiparametric approach based on pathwise differentiability and efficient influence functions (Bickel et al., 1993; Van der Vaart, 2000; Tsiatis, 2006). We first introduce the additional assumptions needed for this target, then derive the efficient influence function and the corresponding cross-fitted one-step estimator. We next show how this construction simplifies in the finite-support case for $\mathbf{X}$, where the Riesz representer becomes explicit, and finally use these results to obtain asymptotic normality and pointwise confidence intervals for the cumulative hazard and survival function in the finite-support case for $\mathbf{X}$. The proofs of all lemmas and theorems in this section appear in Appendix B.

5.1. Additional assumptions for survival inference. Fix $t \in [0, \tau]$, and let $(\mathbf{X}_0, \mathbf{Z}_0)$ be an independent draw from the covariate distribution, independent of the training sample. The assumptions below are used to establish pathwise differentiability of the target, derive its efficient influence function, and justify the cross-fitted one-step estimator introduced in Section 3.2. To state these assumptions, we equip nuisance directions with the predictable-variation inner product

$$(5.1) \quad \langle a, b \rangle_0 = \mathbb{E}_0 \left\{ \int_0^\tau a(u, \mathbf{X}) b(u, \mathbf{X}) Y(u) h_o(u \mid \mathbf{X}, \mathbf{Z}) du \right\}, \quad \|b\|_0^2 = \langle b, b \rangle_0.$$

A6. For the fixed time t , $\mathbb{E}_0 \int_0^t h_o(u \mid \mathbf{X}_0, \mathbf{Z}_0) du < \infty$, and there exists $\kappa_t > 0$ such that $\mathbb{P}_0(Y(u) = 1 \mid \mathbf{X}_0, \mathbf{Z}_0) \geq \kappa_t$ a.s. for a.e. $u \in [0, t]$.

A7. Define the linear operator $Q : \overline{\mathcal{T}}_{g_o} \rightarrow L^2(P_{\mathbf{X}, \mathbf{Z}})$,

$$(Qb)(\mathbf{X}_0, \mathbf{Z}_0) = \int_0^t h_o(u \mid \mathbf{X}_0, \mathbf{Z}_0) b(u, \mathbf{X}_0) du, \quad b \in \overline{\mathcal{T}}_{g_o}.$$

Assume that Q is bounded, i.e., there exists a constant $C_t < \infty$ such that

$$\|Qb\|_{L^2(P_{\mathbf{X}, \mathbf{Z}})}^2 = E_{\mathbf{X}_0, \mathbf{Z}_0} \{ (Qb)(\mathbf{X}_0, \mathbf{Z}_0)^2 \} \leq C_t \|b\|_0^2, \quad \forall b \in \overline{\mathcal{T}}_{g_o}.$$

Assume further that there exists a random element $\psi_t^0(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0) \in L^2(P_{\mathbf{X}, \mathbf{Z}}; \overline{\mathcal{T}}_{g_o})$, such that for every $b \in \overline{\mathcal{T}}_{g_o}$,

$$(5.2) \quad (Qb)(\mathbf{X}_0, \mathbf{Z}_0) = \langle \psi_t^0(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0), b \rangle_0 \quad \text{in } L^2(P_{\mathbf{X}, \mathbf{Z}}).$$

A8. The nuisance covariate $\mathbf{X}$ takes values in a finite set $\mathcal{X} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(J)}\}$, with $\Pr(\mathbf{X} = \mathbf{x}^{(j)}) = \pi_j > 0$, $j = 1, \dots, J$. When pointwise inference is stated in explicit closed form below, we work under this finite-support assumption.

A9. Let $\{\mathcal{I}_k\}_{k=1}^K$ be a fixed K -fold partition of $\{1, \dots, n\}$, and for each k let $(\tilde{\theta}^{(-k)}, \tilde{g}^{(-k)})$ and $\widehat{\mathbf{g}}^{*,(-k)}$ be estimated using only the training sample $\mathcal{I}_k^c$. Assume these estimators are measurable with respect to the training sigma-field and, conditional on the training sample, are independent of the validation observations $\{\mathbf{W}_i : i \in \mathcal{I}_k\}$. Assume moreover that, uniformly in k , $\|\tilde{g}^{(-k)} - g_o\|_0 = O_p(\tau_n)$, $\|\widehat{\mathbf{g}}^{*,(-k)} - \mathbf{g}^*\|_0 = O_p(\tau_n)$, $\|\tilde{\theta}^{(-k)} - \theta_o\| = O_p(n^{-1/2})$, where $\tau_n = \gamma_n \log^2 n$, and $\sqrt{n} \tau_n^2 \rightarrow 0$.

A10. The efficient information matrix $I(\theta_o)$ is nonsingular and the smallest eigenvalue satisfies $\lambda_{\min} I(\theta_o) \geq c_0$ for some constant $c_0 > 0$. In addition, $E_0 \|\ell_{\theta_o}^*(\mathbf{W})\|^2 < \infty$, and $E_0 \left[\{\phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0)\}^2 \right] < \infty$.

A11. Let $0 = t_0 < t_1 < \dots < t_{m_n} = \tau$ be the numerical grid used in the estimation procedure, with mesh size $\delta_n = \max_{1 \leq j \leq m_n} (t_j - t_{j-1})$. Assume the grid contains all observed event and censoring times, that $\sqrt{n} \delta_n \rightarrow 0$, and that the corresponding Riemann-approximation error is $O(\delta_n)$ uniformly over a neighborhood of (θ_o, g_o) .

Under A6–A11, the efficient influence function and the cross-fitted one-step estimator admit a tractable representation. Assumptions A6–A7 guarantee pathwise differentiability of the target functional $H_o(t \mid \mathbf{X}_0, \mathbf{Z}_0)$, A8 yields an explicit closed-form Riesz representer, A9 provides the foldwise nuisance-rate conditions needed for replacing oracle quantities by estimated ones, A10 ensures regularity of the efficient score and information, and A11 controls the numerical integration error.

Under A6 and A8, the operator Q admits the explicit Riesz representer given in Lemma 6; hence A7 holds in the finite-support case. Assumption A8 is used only to obtain an explicit closed-form Riesz representer, and hence an explicit one-step correction, for pointwise survival-curve inference. It is not required for estimation of g_o , for semiparametric efficiency of θ_o , or for inference on θ_o . Moreover, A8 concerns only the nuisance covariates entering the flexible component $g(t, \mathbf{X})$, not the low-dimensional covariates $\mathbf{Z}$ of primary scientific interest. In applications with continuous nuisance covariates, one practical route is to replace them by a sufficiently fine grouped version when implementing the explicit one-step correction. Because this grouping acts only on the nuisance inputs of a flexible DNN, it is often not expected to materially affect the fitted hazard or resulting survival predictions, although this should be assessed empirically and may become less convenient when the induced joint support is very large.

5.2. Efficient influence function and one-step representation. We now provide the semi-parametric ingredients underlying the estimator introduced in Section 3.2. Consider nuisance perturbations of the form $g_\varepsilon = g_o + \varepsilon b$, with predictable $b = b(u, \mathbf{X})$. The corresponding nuisance score equals

$$(5.3) \quad S_{g,b}(\mathbf{W}) = \left. \frac{\partial}{\partial \varepsilon} \ell_0(\theta_o, g_\varepsilon) \right|_{\varepsilon=0} = \int_0^\tau b(u, \mathbf{X}) dM(u).$$

Since $\overline{\mathcal{T}}_{g_o}$ denote the closure of predictable directions under $\|\cdot\|_0$, elements of $\overline{\mathcal{T}}_{g_o}$ are effectively functions on $[0, \tau] \times \mathcal{X}$. As earlier, $\mathbf{g}^*(u, \mathbf{X}) = (g_1^*, \dots, g_p^*)^\top \in (\overline{\mathcal{T}}_{g_o})^p$ denote the componentwise L_0^2 -projection of $\mathbf{Z}$ onto $(\overline{\mathcal{T}}_{g_o})^p$, characterized by

$$(5.4) \quad \mathbb{E}_0 \left\{ \int_0^\tau \{ \mathbf{Z} - \mathbf{g}^*(u, \mathbf{X}) \} b(u, \mathbf{X}) Y(u) h_o(u \mid \mathbf{X}, \mathbf{Z}) du \right\} = 0, \quad \forall b \in \overline{\mathcal{T}}_{g_o}.$$

The efficient score for θ_o is therefore

$$\ell_{\theta_o}^*(\mathbf{W}) = \int_0^\tau \{ \mathbf{Z} - \mathbf{g}^*(u, \mathbf{X}) \} dM(u),$$

and $I(\theta_o) = \mathbb{E}_0 \{ \ell_{\theta_o}^*(\mathbf{W})^{\otimes 2} \}$. For $\mathbf{a} \in \mathbb{R}^p$, consider the parametric path $\theta_\varepsilon = \theta_o + \varepsilon \mathbf{a}$. To construct a least favorable submodel, we simultaneously perturb the nuisance function as

$$(5.5) \quad g_{\varepsilon, \mathbf{a}}(u, \mathbf{X}) = g_o(u, \mathbf{X}) - \varepsilon \mathbf{a}^\top \mathbf{g}^*(u, \mathbf{X}).$$

More generally, allowing an additional nuisance direction $b \in \overline{\mathcal{T}}_{g_o}$, we consider

$$g_\varepsilon(u, \mathbf{X}) = g_o(u, \mathbf{X}) - \varepsilon \mathbf{a}^\top \mathbf{g}^*(u, \mathbf{X}) + \varepsilon b(u, \mathbf{X}),$$

while keeping $\theta_\varepsilon = \theta_o + \varepsilon \mathbf{a}$. The score decomposes as

$$(5.6) \quad S(\mathbf{W}; \mathbf{a}, b) = \mathbf{a}^\top \ell_{\theta_o}^*(\mathbf{W}) + \int_0^\tau b(u, \mathbf{X}) dM(u).$$

Fix $t \in (0, \tau]$ and define the linear functional

$$(5.7) \quad L_t(b) = \int_0^t h_o(u \mid \mathbf{X}_0, \mathbf{Z}_0) b(u, \mathbf{X}_0) du, \quad b \in \overline{\mathcal{T}}_{g_o}.$$

When applied to a vector-valued function $\mu(u, \mathbf{X}) = (\mu_1, \dots, \mu_p)^\top$, we define $L_t(\mu) = (L_t(\mu_1), \dots, L_t(\mu_p))^\top$. Define

$$(5.8) \quad c_{0,t}(\mathbf{X}_0, \mathbf{Z}_0) = \int_0^t h_o(u \mid \mathbf{X}_0, \mathbf{Z}_0) \{\mathbf{Z}_0 - \mathbf{g}^*(u, \mathbf{X}_0)\} du = \mathbf{Z}_0 H_o(t \mid \mathbf{X}_0, \mathbf{Z}_0) - L_t(\mathbf{g}^*).$$

The following lemma computes the pathwise derivative of the target cumulative hazard $H_\varepsilon(t \mid \mathbf{X}_0, \mathbf{Z}_0)$ along a regular submodel. It shows the derivative splits cleanly into a nuisance part $L_t(b)$ (how perturbations of the nuisance component move the target) plus a parametric part $\mathbf{a}^\top c_{0,t}(\mathbf{X}_0, \mathbf{Z}_0)$ (how perturbations of $\boldsymbol{\theta}$ move the target). This is the basic Gateaux derivative input needed to identify the EIF.

LEMMA 4. *Let P_ε be a regular submodel with score (5.6). Then*

$$(5.9) \quad \left. \frac{d}{d\varepsilon} H_\varepsilon(t \mid \mathbf{X}_0, \mathbf{Z}_0) \right|_{\varepsilon=0} = L_t(b) + \mathbf{a}^\top c_{0,t}(\mathbf{X}_0, \mathbf{Z}_0).$$

The following lemma gives an explicit EIF decomposition: a nuisance correction term plus a parametric correction term.

LEMMA 5. *Fix $t \in [0, \tau]$ and let $(\mathbf{X}_0, \mathbf{Z}_0)$ be an independent draw from the covariate distribution, independent of the training sample. Assume A6–A7. Then ψ_t^0 is unique in $L^2(P_{\mathbf{X}, \mathbf{Z}}; \overline{\mathcal{T}}_{g_o})$ and the map $P \mapsto H_P(t \mid \mathbf{X}_0, \mathbf{Z}_0)$ is pathwise differentiable at $\mathbb{P}_0$ as an element of $L^2(P_{\mathbf{X}, \mathbf{Z}})$; that is, for every regular submodel $\mathbb{P}_\varepsilon$ through $\mathbb{P}_0$ with score $S(\mathbf{W}; \mathbf{a}, b)$,*

$$\left. \frac{d}{d\varepsilon} H_\varepsilon(t \mid \mathbf{X}_0, \mathbf{Z}_0) \right|_{\varepsilon=0} = \mathbb{E}_0\{\phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) S(\mathbf{W}; \mathbf{a}, b) \mid \mathbf{X}_0, \mathbf{Z}_0\} \quad \text{in } L^2(P_{\mathbf{X}, \mathbf{Z}}).$$

Equivalently, (5.2) can be stated as a unique (up to $P_{\mathbf{X}, \mathbf{Z}}$ -null sets) random element $\psi_t^0(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0) \in L^2(P_{\mathbf{X}, \mathbf{Z}}; \overline{\mathcal{T}}_{g_o})$ such that for all $b \in \overline{\mathcal{T}}_{g_o}$,

$$(5.10) \quad \mathbb{E}_{(\mathbf{X}_0, \mathbf{Z}_0)} \left[\left\{ \langle \psi_t^0(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0), b \rangle_0 - L_t(b) \right\}^2 \right] = 0.$$

Finally, an EIF for $H_o(t \mid \mathbf{X}_0, \mathbf{Z}_0)$ is

$$(5.11) \quad \phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) = \int_0^\tau \psi_t^0(u, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) dM(u) + c_{0,t}(\mathbf{X}_0, \mathbf{Z}_0)^\top I(\boldsymbol{\theta}_o)^{-1} \ell_{\boldsymbol{\theta}_o}^*(\mathbf{W}).$$

We emphasize that the finite-support assumption on $\mathbf{X}$ is introduced only to obtain an explicit and implementable one-step estimator. More generally, survival inference for a new subject can also be developed under arbitrary covariate distributions via a high-level semi-parametric characterization of the efficient influence function. Since that approach is more technical and does not yield an equally explicit estimator without additional strong approximation arguments, we present it in Appendix C.

5.3. *Explicit form under finite-support $\mathbf{X}$ and asymptotic theory for the cross-fitted estimator.* In the following lemma we show that under discrete finite-support $\mathbf{X}$, the abstract Riesz representer from Lemma 5 becomes an explicit weight function. Therefore, the EIF and one-step estimator become fully explicit.

LEMMA 6. *Under Assumptions A6–A8, the Riesz representer $\psi_t^0(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0)$ satisfying (5.10) admits the closed form*

$$(5.12) \quad \psi_t^0(u, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) = \mathbf{1}(u \leq t) \mathbf{1}(\mathbf{X} = \mathbf{X}_0) \frac{h_o(u | \mathbf{X}_0, \mathbf{Z}_0)}{\Pr(\mathbf{X} = \mathbf{X}_0) A_{\mathbf{X}_0}(u)}.$$

The following is a cross-fitted one-step estimator. Let $\{\mathcal{I}_1, \dots, \mathcal{I}_K\}$ be a fixed K -fold partition of $\{1, \dots, n\}$ and let $\mathcal{I}_k^c$ denote the complement of $\mathcal{I}_k$. For each fold k , let $(\tilde{\theta}^{(-k)}, \tilde{g}^{(-k)})$ be the estimators trained on $\mathcal{I}_k^c$. Define the fold-specific hazard estimator

$$\hat{h}^{(-k)}(u | \mathbf{X}, \mathbf{Z}) = \exp\left\{\tilde{g}^{(-k)}(u, \mathbf{X}) + (\tilde{\theta}^{(-k)})^\top \mathbf{Z}\right\}.$$

For $t \in [0, \tau]$, define the plug-in cumulative hazard at $(\mathbf{X}_0, \mathbf{Z}_0)$ by

$$\hat{H}^{(-k)}(t | \mathbf{X}_0, \mathbf{Z}_0) = \sum_{j: t_j \leq t} (t_j - t_{j-1}) \hat{h}^{(-k)}(t_j | \mathbf{X}_0, \mathbf{Z}_0).$$

For $i \in \mathcal{I}_k$, define $\Delta N_{ij} = \mathbf{1}\{T_i \in (t_{j-1}, t_j], \Delta_i = 1\}$, $Y_{ij} = Y_i(t_{j-1}) = \mathbf{1}\{T_i \geq t_{j-1}\}$, and $\Delta A_{ij}^{(-k)} = (t_j - t_{j-1}) Y_{ij} \hat{h}^{(-k)}(t_j | \mathbf{X}_i, \mathbf{Z}_i)$. The grid-based martingale increment is $\Delta \widehat{M}_{ij}^{(-k)} = \Delta N_{ij} - \Delta A_{ij}^{(-k)}$.

Let $\hat{\psi}_t(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0)^{(-k)}$ denote the fold-specific estimator of the Riesz representer. Under discrete finite support of $\mathbf{X}$,

$$\hat{\psi}_t^{(-k)}(t_j, \mathbf{X}_i; \mathbf{X}_0, \mathbf{Z}_0) = \mathbf{1}(t_j \leq t) \mathbf{1}(\mathbf{X}_i = \mathbf{X}_0) \frac{\hat{h}^{(-k)}(t_j | \mathbf{X}_0, \mathbf{Z}_0)}{\hat{\pi}_{j_0}^{(-k)} \hat{A}_{j_0}^{(-k)}(t_j)},$$

where j_0 satisfies $\mathbf{X}_0 = \mathbf{x}^{(j_0)}$, $\hat{\pi}_{j_0}^{(-k)} = |\mathcal{I}_k^c|^{-1} \sum_{i \in \mathcal{I}_k^c} \mathbf{1}(\mathbf{X}_i = \mathbf{x}^{(j_0)})$, and

$$\hat{A}_{j_0}^{(-k)}(t_j) = \frac{\sum_{i \in \mathcal{I}_k^c} Y_{ij} \hat{h}^{(-k)}(t_j | \mathbf{X}_i, \mathbf{Z}_i) \mathbf{1}(\mathbf{X}_i = \mathbf{x}^{(j_0)})}{\sum_{i \in \mathcal{I}_k^c} \mathbf{1}(\mathbf{X}_i = \mathbf{x}^{(j_0)})}.$$

The nuisance influence contribution is

$$\widehat{\phi}_{t,g}^{(-k)}(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0) = \sum_{j=1}^{m_n} \hat{\psi}_t^{(-k)}(t_j, \mathbf{X}_i; \mathbf{X}_0, \mathbf{Z}_0) \Delta \widehat{M}_{ij}^{(-k)}.$$

Let $\widehat{\mathbf{g}}^{*,(-k)}$ denote the estimator of the projection $\mathbf{g}^*$ and define

$$\hat{c}_t^{(-k)}(\mathbf{X}_0, \mathbf{Z}_0) = \sum_{j: t_j \leq t} (t_j - t_{j-1}) \hat{h}^{(-k)}(t_j | \mathbf{X}_0, \mathbf{Z}_0) \left\{ \mathbf{Z}_0 - \widehat{\mathbf{g}}^{*,(-k)}(t_j, \mathbf{X}_0) \right\}.$$

Let $\widehat{I}^{(-k)}$ and $\widehat{\ell}_\theta^{*,(-k)}$ be the out-of-fold efficient information and score estimators. Define

$$\widehat{\phi}_{t,\theta}^{(-k)}(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0) = \hat{c}_t^{(-k)}(\mathbf{X}_0, \mathbf{Z}_0)^\top \left\{ \widehat{I}^{(-k)} \right\}^{-1} \widehat{\ell}_\theta^{*,(-k)}(\mathbf{W}_i).$$

The fold-specific influence estimator is

$$\widehat{\phi}_t^{(-k)} = \widehat{\phi}_{t,g}^{(-k)} + \widehat{\phi}_{t,\theta}^{(-k)}.$$

The cross-fitted one-step estimator is

$$\widehat{H}^{1,\text{cf}}(t \mid \mathbf{X}_0, \mathbf{Z}_0) = \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \widehat{H}^{(-k)}(t \mid \mathbf{X}_0, \mathbf{Z}_0) + \frac{1}{n} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \widehat{\phi}_t^{(-k)}(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0).$$

Finally,

$$\widehat{S}^{1,\text{cf}}(t \mid \mathbf{X}_0, \mathbf{Z}_0) = \exp\{-\widehat{H}^{1,\text{cf}}(t \mid \mathbf{X}_0, \mathbf{Z}_0)\}.$$

The following lemma shows that in empirical sums, the oracle efficient score $\ell_{\theta_o}^*$ can be replaced by the cross-fitted plug-in score $\widehat{\ell}_{\theta_o}^{*,(-k)}$ without changing the $n^{1/2}$ -limit, provided the nuisance estimates converge at a suitable rate and cross-fitting ensures conditional independence. This is the key technical step that lets us use estimated nuisances inside the EIF expansion.

LEMMA 7. *For $i \in \mathcal{I}_k$ define the plug-in martingale*

$$\widehat{M}_i^{(-k)}(u) = N_i(u) - \int_0^u Y_i(s) \widehat{h}^{(-k)}(s \mid \mathbf{X}_i, \mathbf{Z}_i) ds,$$

and the plug-in efficient score

$$(5.13) \quad \widehat{\ell}_{\theta_o}^{*,(-k)}(\mathbf{W}_i) = \int_0^\tau \{\mathbf{Z}_i - \widehat{\mathbf{g}}^{*,(-k)}(u, \mathbf{X}_i)\} d\widehat{M}_i^{(-k)}(u).$$

Assume A1–A7 and A9. Then

$$(5.14) \quad \frac{1}{\sqrt{n}} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \left\{ \widehat{\ell}_{\theta_o}^{*,(-k)}(\mathbf{W}_i) - \ell_{\theta_o}^*(\mathbf{W}_i) \right\} = o_p(1).$$

Consequently, in any root- n expansion in which the oracle score $\ell_{\theta_o}^$ appears inside an empirical sum, it may be replaced by the cross-fitted plug-in score $\widehat{\ell}_{\theta_o}^{*,(-k)}$ without affecting the first-order limit.*

The following lemma proves the out-of-fold estimator $\widehat{I}^{(-k)}$ of the efficient information converges to $I(\theta_o)$, and that the inverse $\{\widehat{I}^{(-k)}\}^{-1}$ is well-behaved when $I(\theta_o)$ is nonsingular. This justifies replacing $I(\theta_o)^{-1}$ in the EIF by a cross-fitted plug-in inverse.

LEMMA 8. *Define the out-of-fold (efficient) information estimator*

$$\widehat{I}^{(-k)} = \frac{1}{|\mathcal{I}_k^c|} \sum_{j \in \mathcal{I}_k^c} \Delta_j \left\{ \mathbf{Z}_j - \widehat{\mathbf{g}}^{*,(-k)}(T_j, \mathbf{X}_j) \right\}^{\otimes 2}, \quad k = 1, \dots, K.$$

Assume A1–A7 and A9–A10. Then,

- (a) $\max_{1 \leq k \leq K} \|\widehat{I}^{(-k)} - I(\theta_o)\| = o_p(1).$
- (b) *With probability tending to one, each $\widehat{I}^{(-k)}$ is invertible and*

$$\max_{1 \leq k \leq K} \|\{\widehat{I}^{(-k)}\}^{-1} - I(\theta_o)^{-1}\| = o_p(1).$$

- (c) *If, in addition, $E_0\{\|\ell_{\theta_o}^*(\mathbf{W})\|^2\} < \infty$, then*

$$\frac{1}{\sqrt{n}} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \left(\{\widehat{I}^{(-k)}\}^{-1} - I(\theta_o)^{-1} \right) \ell_{\theta_o}^*(\mathbf{W}_i) = o_p(1).$$

Because the estimators use a time grid, the following lemma compares the true EIF ϕ_t^0 , its grid version $\phi_{t,n}^0$, and the fully plug-in cross-fitted estimate $\widehat{\phi}_t^{(-k)}$. It shows $\widehat{\phi}_t^{(-k)}$ is $L_2(\mathbb{P}_0)$ -close to the grid EIF uniformly over folds, and since the grid error vanishes when $\sqrt{n}\delta_n \rightarrow 0$, this yields L_2 -consistency for the true EIF as well.

LEMMA 9. Assume A1–A11. Let $\phi_{t,n}^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0)$ denote the grid version of (5.11) obtained by replacing $\int_0^\tau \psi_t^0(u, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) dM(u)$ with $\sum_{j=1}^{m_n} \psi_t^0(t_j, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) \Delta M_j$ and $c_{0,t}(\mathbf{X}_0, \mathbf{Z}_0)$ with its grid analogue. Then, uniformly in $k = 1, \dots, K$,

$$\mathbb{E}_0 \left[\left(\widehat{\phi}_t^{(-k)}(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) - \phi_{t,n}^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) \right)^2 \right] = o_p(1),$$

and therefore (since $\sqrt{n}\delta_n \rightarrow 0$ implies $\phi_{t,n}^0$ and ϕ_t^0 are asymptotically equivalent),

$$\mathbb{E}_0 \left[\left(\widehat{\phi}_t^{(-k)}(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) - \phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) \right)^2 \right] = o_p(1), \quad \text{uniformly in } k.$$

THEOREM 6. Fix $t \in [0, \tau]$ and let $(\mathbf{X}_0, \mathbf{Z}_0)$ be an independent draw from the covariate distribution. Under Assumptions A1–A11, and assuming moreover that for each fold k ,

$$\|\widetilde{g}^{(-k)} - g_o\|_0 = O_p(\gamma_n \log^2 n), \quad \gamma_n \log^2 n = o(n^{-1/4}),$$

uniformly in k , and that $\widetilde{\boldsymbol{\theta}}^{(-k)}$ satisfies the same first-order expansion as established in Theorem 4, namely,

$$\sqrt{|\mathcal{I}_k^c|} (\widetilde{\boldsymbol{\theta}}^{(-k)} - \boldsymbol{\theta}_o) = I(\boldsymbol{\theta}_o)^{-1} \frac{1}{\sqrt{|\mathcal{I}_k^c|}} \sum_{i \in \mathcal{I}_k^c} \ell_{\boldsymbol{\theta}_o}^*(\mathbf{W}_i) + o_p(1),$$

uniformly in k . Then, conditional on the independent draw $(\mathbf{X}_0, \mathbf{Z}_0)$,

$$\sqrt{n} \left\{ \widehat{H}^{1,\text{cf}}(t \mid \mathbf{X}_0, \mathbf{Z}_0) - H_o(t \mid \mathbf{X}_0, \mathbf{Z}_0) \right\} \mid (\mathbf{X}_0, \mathbf{Z}_0) \rightarrow^{\mathcal{D}} \mathcal{N}(0, \sigma_H^2(t; \mathbf{X}_0, \mathbf{Z}_0)),$$

where $\sigma_H^2(t; \mathbf{X}_0, \mathbf{Z}_0) = \mathbb{V}_0(\phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) \mid \mathbf{X}_0, \mathbf{Z}_0)$. Moreover, with

$$\widehat{\sigma}_H^2(t; \mathbf{X}_0, \mathbf{Z}_0) = \frac{1}{n} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \left\{ \widehat{\phi}_t^{(-k)}(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0) \right\}^2,$$

we have $\widehat{\sigma}_H^2(t; \mathbf{X}_0, \mathbf{Z}_0) \xrightarrow{P} \sigma_H^2(t; \mathbf{X}_0, \mathbf{Z}_0)$. Finally,

$$\sqrt{n} \left\{ \widehat{S}^{1,\text{cf}}(t \mid \mathbf{X}_0, \mathbf{Z}_0) - S_o(t \mid \mathbf{X}_0, \mathbf{Z}_0) \right\} \mid (\mathbf{X}_0, \mathbf{Z}_0) \rightarrow^{\mathcal{D}} \mathcal{N}(0, \sigma_S^2(t; \mathbf{X}_0, \mathbf{Z}_0))$$

where $\sigma_S^2(t; \mathbf{X}_0, \mathbf{Z}_0) = S_o(t \mid \mathbf{X}_0, \mathbf{Z}_0)^2 \sigma_H^2(t; \mathbf{X}_0, \mathbf{Z}_0)$ and a consistent variance estimator is

$$\widehat{\sigma}_S^2(t; \mathbf{X}_0, \mathbf{Z}_0) = \left\{ \widehat{S}^{1,\text{cf}}(t \mid \mathbf{X}_0, \mathbf{Z}_0) \right\}^2 \widehat{\sigma}_H^2(t; \mathbf{X}_0, \mathbf{Z}_0).$$

Theorem 6 yields a Wald-type pointwise confidence interval for the survival of a new subject at any fixed time $t \in [0, \tau]$, that is

$$\widehat{S}^{1,\text{cf}}(t \mid X_0, Z_0) \pm z_{1-\alpha/2} \widehat{\sigma}_S(t; X_0, Z_0) / \sqrt{n}.$$

Theorem 6 is proved conditionally on the random draw $(\mathbf{X}_0, \mathbf{Z}_0)$; it is not a uniform statement over all fixed covariate values $(\mathbf{x}_0, \mathbf{z}_0)$. Hence, the associated coverage statement is first understood conditionally,

$$\Pr\left(S_o(t \mid \mathbf{X}_0, \mathbf{Z}_0) \in \left[\hat{S}^{1,\text{cf}}(t \mid \mathbf{X}_0, \mathbf{Z}_0) \pm z_{1-\alpha/2} \frac{\hat{\sigma}_S(t; \mathbf{X}_0, \mathbf{Z}_0)}{\sqrt{n}}\right] \mid \mathbf{X}_0, \mathbf{Z}_0\right) \rightarrow 1 - \alpha.$$

Let

$$A_n(t) = \left\{ S_o(t \mid \mathbf{X}_0, \mathbf{Z}_0) \in \left[\hat{S}^{1,\text{cf}}(t \mid \mathbf{X}_0, \mathbf{Z}_0) \pm z_{1-\alpha/2} \frac{\hat{\sigma}_S(t; \mathbf{X}_0, \mathbf{Z}_0)}{\sqrt{n}}\right] \right\},$$

for which Theorem 6 yields $\Pr(A_n(t) \mid \mathbf{X}_0, \mathbf{Z}_0) \rightarrow 1 - \alpha$, in probability. Then, since $0 \leq \Pr(A_n(t) \mid \mathbf{X}_0, \mathbf{Z}_0) \leq 1$, bounded convergence in probability implies convergence in L^1 and therefore $\mathbb{E}\{\Pr(A_n(t) \mid \mathbf{X}_0, \mathbf{Z}_0)\} \rightarrow 1 - \alpha$. Applying the law of iterated expectations gives $\Pr(A_n(t)) = \mathbb{E}\{\Pr(A_n(t) \mid \mathbf{X}_0, \mathbf{Z}_0)\} \rightarrow 1 - \alpha$. Thus the pointwise (in t) Wald interval has asymptotic marginal coverage under the joint law of the training data and the independent draw $(\mathbf{X}_0, \mathbf{Z}_0)$, namely,

$$\Pr\left(S_o(t \mid \mathbf{X}_0, \mathbf{Z}_0) \in \left[\hat{S}^{1,\text{cf}}(t \mid \mathbf{X}_0, \mathbf{Z}_0) \pm z_{1-\alpha/2} \frac{\hat{\sigma}_S(t; \mathbf{X}_0, \mathbf{Z}_0)}{\sqrt{n}}\right]\right) \rightarrow 1 - \alpha.$$

In our partially linear hazard model, the nuisance function $g_o(t, \mathbf{X})$ is assumed to belong to a composite Hölder class, $g_o \in \mathcal{H}(q, \alpha, \mathbf{d}^*, \tilde{\mathbf{d}}, M)$. Under this assumption, sparse ReLU networks can achieve the (near-)minimax rate $\|\hat{g} - g_o\| = O_p(\tau_n)$, where the complexity exponent γ_n is governed by the hardest layer of the composition. The condition on γ_n imposed in the theorem above is therefore a standard second-order remainder requirement for one-step inference, and it ensures $\sqrt{n} \|\hat{g} - g_o\|_0^2 = o_p(1)$, so that nuisance-estimation error enters only through $o_p(n^{-1/2})$ terms. Interpreting this condition in terms of the composite Hölder parameters, it is sufficient that the slowest layer satisfies

$$\frac{\tilde{\alpha}_\ell}{2\tilde{\alpha}_\ell + \tilde{d}_\ell} > \frac{1}{4} \quad \text{for the maximizing } \ell, \quad \text{equivalently} \quad 2\tilde{\alpha}_\ell > \tilde{d}_\ell,$$

up to the polylogarithmic factor $\log^2 n$. Thus, the assumption is mild when the intrinsic dimensions $\tilde{d}_\ell$ are small (structural sparsity or low-dimensional composition) and the effective smoothness levels $\tilde{\alpha}_\ell$ are not too low. It can fail only in very rough or high-intrinsic-dimension regimes where the best achievable γ_n is slower than $n^{-1/4}$, in which case Wald-type inference would require additional structure (e.g., stronger smoothness, further dimension reduction, or alternative debiasing or regularization).

6. Practical Implementation.

6.1. Applying DNN. In practice, we approximate g_o by a fully connected ReLU network and model the linear term $\boldsymbol{\theta}^\top \mathbf{Z}$ using a parallel linear layer. The network computes $\tilde{g}(t, \mathbf{X})$ and $\tilde{\boldsymbol{\theta}}^\top \mathbf{Z}$ and their sum is inserted into the loss based on (3.2). This architecture jointly learns the nuisance function g and a finite-dimensional linear effect of $\mathbf{Z}$ in a way that mirrors the model structure.

To evaluate the approximated log-likelihood (3.2) in practice, we discretize the time axis on a grid (e.g., n) and augment it with all observed times, yielding $0 = t_0 < t_1 < \dots < t_{m_n} = \tau$. The dataset is then expanded in the usual counting-process manner. Each subject i with data $(T_i, \Delta_i, \mathbf{X}_i, \mathbf{Z}_i)$ is represented by one dummy-observation for every interval $[t_k, t_{k+1})$ such that $t_k \leq T_i$. All dummy-observations inherit $(\mathbf{X}_i, \mathbf{Z}_i)$; the event indicator

$\Delta_{ik} = \mathbf{1}\{T_i \in (t_{k-1}, t_k], \Delta = 1\}$ is set to 0 except for the final interval containing T_i , where it takes the original value Δ_i . This choice of grid and data expansion enables efficient computation within standard deep-learning frameworks. The construction parallels the standard expanded-data representation used for Cox models with time-dependent covariates (Therneau and Grambsch, 2000; Hu and Nan, 2023).

In theory we allow for sparsity, but in practice we do not explicitly enforce it. Network hyperparameters (number of layers, hidden units, activation functions, learning rate, batch size) are selected via tuning on the training set. Training maximizes the empirical likelihood using the ADAM optimizer in TensorFlow. The data are split into training and validation sets, with 33% reserved for early stopping. Training continues until the validation loss stops improving, and the weights achieving the best validation performance are retained. The linear $\mathbf{Z}$ -branch is initialized using the coefficients from a standard Cox model fit on $(\mathbf{X}, \mathbf{Z})$, providing a good starting point for the linear component.

After training, we extract the weights from the $\mathbf{Z}$ -branch as an initial estimate of $\boldsymbol{\theta}$. To mitigate the effect of early stopping on the optimization of $\boldsymbol{\theta}$, we then perform a separate numerical maximization of the likelihood with respect to $\boldsymbol{\theta}$, holding $\tilde{g}$ fixed at its network estimate. Thus, we treat $\tilde{g}$ as known and solve a low-dimensional optimization problem for $\boldsymbol{\theta}$ using SciPy’s optimization routines. This post-hoc refinement consistently improved the finite-sample behavior of $\hat{\boldsymbol{\theta}}$ in our experiments. For the theoretical results, the critical requirement is that $\hat{g}$ achieves the convergence rate needed for Theorem 4.

6.2. Estimation of the Covariance Matrix $I(\boldsymbol{\theta}_o)^{-1}$. Inference on $\boldsymbol{\theta}_0$ relies on estimating the efficient information matrix $I(\boldsymbol{\theta}_o)$, whose inverse provides the asymptotic covariance of the estimator $\hat{\boldsymbol{\theta}}$. Recall that the efficient information is

$$I(\boldsymbol{\theta}_o) = \mathbb{E} \left[\Delta \{ \mathbf{Z} - \mathbf{g}^*(T, \mathbf{X}) \} \{ \mathbf{Z} - \mathbf{g}^*(T, \mathbf{X}) \}^\top \right],$$

where $\mathbf{g}^*$ is the event-weighted projection of $\mathbf{Z}$ onto the nuisance tangent space. In practice we estimate $\mathbf{g}^*$ using an additional set of neural networks and then construct a plug-in estimate of $I(\boldsymbol{\theta}_o)$ from the corresponding residuals. Specifically, we approximate $\mathbf{g}^*$ coordinatewise. For each $j = 1, \dots, p$ we fit a separate fully connected network $\hat{g}_j^*(T, \mathbf{X})$ by least squares using only uncensored observations, that is by $\min \sum_{i: \Delta_i=1} \{Z_{ij} - \hat{g}_j^*(T_i, \mathbf{X}_i)\}^2$. Each network j , $j = 1, \dots, p$, takes $(T_i, \mathbf{X}_i)$ as input, Z_{ij} as target, and uses the same architecture and hyperparameters as the main model; empirically, these choices produce stable estimates and are not sensitive to tuning.

To avoid overfitting, we employ 5-fold cross-fitting. The data are split into five folds, for each fold and each coordinate j , the corresponding network is trained on the remaining four folds (restricting to $\Delta = 1$) and predictions $\hat{g}_j^*(T_i, \mathbf{X}_i)$ are obtained for the held-out fold. Aggregating across folds yields, for each observation i , the estimated projection vector

$$\hat{\mathbf{g}}^*(T_i, \mathbf{X}_i) = (\hat{g}_1^*(T_i, \mathbf{X}_i), \dots, \hat{g}_p^*(T_i, \mathbf{X}_i))^\top.$$

We then compute the residuals $\mathbf{R}_i = \sqrt{\Delta_i} \{ \mathbf{Z}_i - \hat{\mathbf{g}}^*(T_i, \mathbf{X}_i) \} \in \mathbb{R}^p$, $i = 1, \dots, n$, and stack them into a matrix $\mathcal{R}$. Our estimator of the efficient information is $\hat{I} = \mathcal{R}^\top \mathcal{R} / n$, and the asymptotic covariance is $\hat{I}^{-1}$. In practice this procedure is numerically stable, requires no tuning beyond the main network fit, and produces reliable covariance estimates for $\hat{\boldsymbol{\theta}}$.

7. Simulation Study and Real-data application.

7.1. Simulation Study. We compare the finite-sample performance of the proposed FLEXI-Haz method with the partially linear neural PH estimator of [Zhong, Mueller and Wang \(2022\)](#). Our focus is on estimation of the finite-dimensional parameter θ and on the coverage of the associated Wald confidence intervals. We consider nuisance and primary covariates $\mathbf{X} = (X_1, X_2, X_3)^\top$ and $\mathbf{Z} = (Z_1, Z_2)^\top$, where all covariates are independently distributed as $\text{Unif}[-1, 1]$. The true parameter is $\theta_o = (2, -1)^\top$. The nuisance component includes $f(\mathbf{X}) = 0.2(X_1 + X_2) + 0.5X_1X_2 + X_3^2$, which induces nonlinear and interaction effects. Event times are generated from

$$h_o(t | \mathbf{X}, \mathbf{Z}) = 0.1 \exp\left\{(0.1 + f(\mathbf{X})^2)t + \theta_o^\top \mathbf{Z}\right\}, \quad t \in [0, \tau],$$

with fixed study horizon $\tau = 30$. Right censoring is introduced through an independent exponential censoring time with mean $1/30$, together with administrative censoring at time 30. The resulting censoring rate is approximately 30%.

For both FLEXI-Haz and the model of [Zhong, Mueller and Wang \(2022\)](#), the nuisance function is estimated using a fully connected feedforward ReLU network. The two methods differ only in the underlying hazard model; the training pipeline and tuning strategy are otherwise identical. Hyperparameters are selected once using a single simulated dataset. We construct the expanded dataset on a grid of $m_n = n$ time points augmented with all observed times, and perform 5-fold cross-validation over the number of hidden layers $\{1, 2, 3, 4, 5, 6\}$, width $\{20, 40, 80, 160, 320\}$, and learning rate $\{10^{-3}, 5 \times 10^{-3}, 10^{-2}, 5 \times 10^{-2}\}$. Training uses early stopping with patience 35 and batch size 10^5 on the expanded dataset. For each hyperparameter combination, the network is trained once and the average validation loss across the five folds is recorded; the configuration with the smallest average loss is selected. Cross-validation selects, for FLEXI-Haz, five hidden layers, width 20, and learning rate 10^{-3} , and for [Zhong, Mueller and Wang \(2022\)](#), four hidden layers, width 80, and learning rate 10^{-3} .

Table 1 reports results for $n = 8000$ based on 200 replications. FLEXI-Haz is essentially unbiased, and its empirical standard deviation closely matches the average estimated standard error, yielding coverage probabilities close to the nominal levels. In contrast, the partially linear PH estimator of [Zhong, Mueller and Wang \(2022\)](#) exhibits substantial bias in both components of θ_o , resulting in near-zero empirical coverage.

To assess the finite-sample performance of the proposed pointwise survival-curve inference, we conducted an additional simulation experiment under the same setup as above, except that each nuisance covariate was generated independently from a $\text{Poisson}(2)$ distribution truncated at 6. In each of 200 simulation runs, we generated a training sample of size $n = 8000$ and an additional evaluation sample of size 200. For each target level $s \in \{0.1, 0.2, \dots, 0.9\}$, we selected a fixed time point t_s such that the mean estimated survival in the training sample was approximately s , and evaluated the empirical coverage of the pointwise confidence interval for the survival probability at time t_s based on the evaluation samples.

Table 2 demonstrates that the proposed pointwise inference procedure is well calibrated in finite samples. Across the selected time points, the empirical coverage remains close to the nominal level for both the 90% and 95% confidence intervals, supporting the practical reliability of the method in the discrete-covariate setting considered here. The slight overcoverage at $t = 6.7126$ may be due to boundary effects, since this is a late time point where the true survival probability is close to zero for some values of $(\mathbf{X}_0, \mathbf{Z}_0)$.

7.2. Real-data application: Rotterdam breast cancer study. To illustrate the practical performance of FLEXI-Haz, we analyzed the Rotterdam Breast Cancer data set, available in the `survival` package in R. These data were studied by Foekens et al. (2000) to assess clinical factors associated with time to breast cancer recurrence or death, where the outcome

TABLE 1

Simulation results for $n = 8000$ observations, based on 200 replications. The true parameter is $\theta_o = (2, -1)^\top$. “Est” denotes the mean of the estimated coefficients; “Emp SD” the empirical standard deviation; “Est SE” the estimated theoretical standard error; and “90%CI, 95%CI” the empirical coverage rates of the corresponding Wald-type confidence intervals.

	FLEXI-Haz					Zhong, Mueller and Wang (2022)				
	Est	Emp SD	Est SE	90%CI	95%CI	Est	Emp SD	Est SE	90%CI	95%CI
θ_{o1}	1.997	0.040	0.035	0.878	0.935	1.764	0.039	0.034	0.000	0.000
θ_{o2}	-0.994	0.027	0.029	0.937	0.959	-0.886	0.030	0.028	0.015	0.035

TABLE 2

Simulation results of empirical pointwise coverage of the 90% and 95% confidence intervals for the survival curve, based on 200 replications with training sample size $n = 8000$ and testing sample size of 200. For each target survival time t_s , we evaluated whether the confidence interval covered the true survival value at t_s .

Marginal survival s	Target t_s	90%CI	SD of coverage	95%CI	SD of coverage
0.1	6.7126	0.9377	0.0741	0.9676	0.0531
0.2	4.2934	0.9120	0.0861	0.9548	0.0638
0.3	3.0192	0.9167	0.0769	0.9565	0.0595
0.4	2.2003	0.9121	0.0755	0.9595	0.0523
0.5	1.6035	0.8903	0.0889	0.9488	0.0589
0.6	1.2330	0.8997	0.0912	0.9520	0.0611
0.7	0.9476	0.8995	0.0824	0.9480	0.0614
0.8	0.6812	0.9123	0.0815	0.9577	0.0559
0.9	0.3900	0.9044	0.0987	0.9567	0.0638

is defined as the number of days from primary surgery until the earlier of recurrence or death. The data set contains $n = 2982$ patients, with an overall censoring proportion of 57.35%, and includes nine baseline covariates: age, progesterone receptor level, estrogen receptor level, number of positive lymph nodes, menopausal status, tumor size, tumor grade, chemotherapy, and hormonal therapy. Among these variables, the first four are continuous and the remaining five are discrete.

Our primary scientific goal in this analysis is to assess the effects of chemotherapy and hormonal therapy. Accordingly, we included these two treatment indicators in the linear component $\mathbf{Z}$, while modeling the remaining covariates nonparametrically through the DNN component.

We used five-fold cross-validation. Within each fold, the training portion was further split so that 20% of the training data served as a validation set for tuning hyperparameters, after which model performance was evaluated on the held-out test set. Table 3 summarizes the estimated treatment coefficients and corresponding 95% confidence intervals for the proposed method, together with the comparator methods.

TABLE 3

Estimated treatment effects and 95% confidence intervals for the Rotterdam breast cancer data.

Treatment	Method	Estimate (SE)	95% CI
Chemotherapy	Cox PH	-0.127 (0.071)	(-0.267, 0.013)
	Zhong et al. (2022)	-0.257 (0.095)	(-0.443, -0.071)
	FLEXI-Haz	-0.296 (0.094)	(-0.480, -0.112)
Hormonal therapy	Cox PH	-0.103 (0.083)	(-0.266, 0.061)
	Zhong et al. (2022)	-0.372 (0.080)	(-0.529, -0.215)
	FLEXI-Haz	-0.365 (0.080)	(-0.522, -0.209)

Table 3 shows negative estimated effects of both chemotherapy and hormonal therapy, with 95% confidence intervals excluding zero under FLEXI-Haz. The corresponding esti-

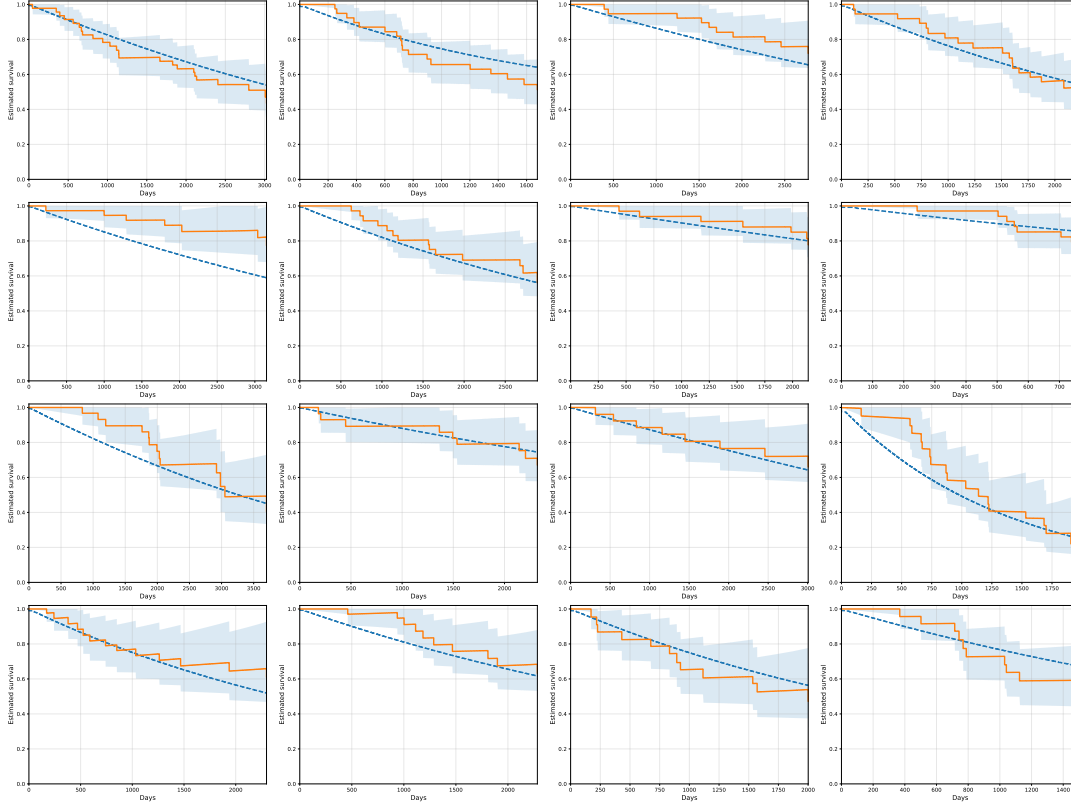


FIG 1. Estimated survival curves for 16 randomly selected subjects in the Rotterdam breast cancer data. The blue curve is the plug-in FLEXI-Haz estimator, evaluated on a dense grid. The red step function is the cross-fitted one-step estimator, and the shaded band gives the corresponding pointwise confidence interval. The visible steps in the one-step curves arise from the finite-support correction, which uses subjects with the same nuisance-covariate profile $\mathbf{X}_0$.

mates are qualitatively similar to those of [Zhong, Mueller and Wang \(2022\)](#), whereas the Cox PH model yields confidence intervals that include zero for both treatments. These findings indicate that FLEXI-Haz recovers interpretable treatment effects while accommodating flexible nonproportional nuisance effects.

Figure 1 displays estimated survival curves for 16 randomly selected subjects. The plug-in estimator is visually smooth because it is evaluated on a dense time grid, whereas the one-step estimator is piecewise constant, reflecting that its bias-correction term is constructed from subjects sharing the same discrete nuisance-covariate profile $\mathbf{X}_0$.

8. Discussion. We proposed FLEXI-Haz, a deep semiparametric survival model that combines a partially linear structure with a flexible non-proportional hazards formulation. By modeling the log-hazard as $\theta_o^\top \mathbf{Z} + g_o(t, \mathbf{X})$, the method preserves interpretability for the low-dimensional parameter θ_o while allowing the nuisance component to vary nonlinearly with both time and covariates. Our main theoretical results show that the DNN estimator of g_o attains the optimal composite minimax rate, that the estimator of θ_o is $\sqrt{n}$ -consistent, asymptotically normal, and semiparametrically efficient, and that pointwise frequentist asymptotic inference is available for the survival curve of a new subject through a cross-fitted one-step procedure. The simulation and real-data analyses support these conclusions and illustrate the practical advantages of FLEXI-Haz when proportional hazards may be too restrictive.

The main limitation of the current survival-inference theory is the additional assumption that the nuisance covariate vector $\mathbf{X}$ has discrete finite support, which is used to obtain an explicit Riesz representer and a tractable one-step correction. Although this restriction is often reasonable in applications with categorical covariates or grouped continuous features, extending the pointwise survival-inference theory to more general covariate distributions is an important direction for future work. Another natural extension is to develop analogous theory for richer network architectures beyond fully connected ReLU networks.

APPENDIX A: PROOFS FOR SECTION 4

LEMMA 1. *Under A1–A2, for any nonnegative, predictable function $f(t, \mathbf{X}, \mathbf{Z})$ with $\mathbb{E}[\int_0^\tau Y(t) f(t, \mathbf{X}, \mathbf{Z}) dt] < \infty$,*

$$\mathbb{E}\left\{\int_0^\tau Y(t) f(t, \mathbf{X}, \mathbf{Z}) dt\right\} \asymp \mathbb{E}\{\Delta f(T, \mathbf{X}, \mathbf{Z})\}.$$

PROOF OF LEMMA 1. Let $N(t) = \Delta \mathbf{1}(T \leq t)$ be the counting process associated with the event time, and let $M(t) = N(t) - \int_0^t Y(s) h_o(s | \mathbf{X}, \mathbf{Z}) ds$ denote its martingale decomposition under the true hazard (2.1). Because $f(t, \mathbf{X}, \mathbf{Z})$ is predictable, the stochastic integral $\int_0^\tau f(t, \mathbf{X}, \mathbf{Z}) dM(t)$ has expectation zero. Consequently,

$$\mathbb{E}\left\{\int_0^\tau f(t, \mathbf{X}, \mathbf{Z}) dN(t)\right\} = \mathbb{E}\left\{\int_0^\tau f(t, \mathbf{X}, \mathbf{Z}) Y(t) h_o(t | \mathbf{X}, \mathbf{Z}) dt\right\}.$$

Since $\int_0^\tau f(t, \mathbf{X}, \mathbf{Z}) dN(t) = \Delta f(T, \mathbf{X}, \mathbf{Z})$, we obtain the identity

$$(A.1) \quad \mathbb{E}\{\Delta f(T, \mathbf{X}, \mathbf{Z})\} = \mathbb{E}\left\{\int_0^\tau f(t, \mathbf{X}, \mathbf{Z}) Y(t) h_o(t | \mathbf{X}, \mathbf{Z}) dt\right\}.$$

Under Assumptions A1–A2, the hazard is uniformly bounded and bounded away from zero,

$$0 < c_h \leq h_o(t | \mathbf{X}, \mathbf{Z}) \leq C_h < \infty.$$

Applying these bounds to the right-hand side of (A.1) yields

$$c_h \mathbb{E}\left\{\int_0^\tau f(t, \mathbf{X}, \mathbf{Z}) Y(t) dt\right\} \leq \mathbb{E}\{\Delta f(T, \mathbf{X}, \mathbf{Z})\} \leq C_h \mathbb{E}\left\{\int_0^\tau f(t, \mathbf{X}, \mathbf{Z}) Y(t) dt\right\}.$$

This implies the desired equivalence:

$$\mathbb{E}\left\{\int_0^\tau f(t, \mathbf{X}, \mathbf{Z}) Y(t) dt\right\} \asymp \mathbb{E}\{\Delta f(T, \mathbf{X}, \mathbf{Z})\}.$$

□

LEMMA 2. *Let $\chi_{\eta_j}(t, \mathbf{X}, \mathbf{Z})$, $j = 1, 2$, be bounded predictable processes satisfying*

$$\mathbb{E}\left\{\int_0^\tau \chi_{\eta_j}(t, \mathbf{X}, \mathbf{Z})^2 dt\right\} < \infty \quad j = 1, 2.$$

Define $\phi(t, \mathbf{X}, \mathbf{Z}) = \chi_{\eta_1}(t, \mathbf{X}, \mathbf{Z}) - \chi_{\eta_2}(t, \mathbf{X}, \mathbf{Z})$. Under Assumptions A1, A2 and A4, there exist finite constants $0 < c_1 \leq c_2 < \infty$ and $0 < c_3 \leq c_4 < \infty$, depending only on $(\tau, \delta_1, \delta_2, c_h, C_h)$, such that

$$(4.2) \quad c_1 \mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2\} \leq \mathbb{E}\{\phi(T, \mathbf{X}, \mathbf{Z})^2\} \leq c_2 \mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2\},$$

$$(4.3) \quad c_3 \mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2\} \leq \mathbb{E}_{\mathbf{X}, \mathbf{Z}}\left\{\int_0^\tau \phi(t, \mathbf{X}, \mathbf{Z})^2 dt\right\} \leq c_4 \mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2\}.$$

In particular, if $\mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2\} = 0$, then $\phi(T, \mathbf{X}, \mathbf{Z}) = 0$ $\mathbb{P}_0$ -a.s. and $\phi(t, \mathbf{X}, \mathbf{Z}) = 0$ for Lebesgue-a.e. $t \in [0, \tau]$ and $\mathbb{P}_{\mathbf{X}, \mathbf{Z}}$ -a.e. $(\mathbf{X}, \mathbf{Z})$, i.e. $\chi_{\eta_1}(t, \mathbf{X}, \mathbf{Z}) = \chi_{\eta_2}(t, \mathbf{X}, \mathbf{Z})$ for $dt \otimes \mathbb{P}_{\mathbf{X}, \mathbf{Z}}$ -almost all $(t, \mathbf{X}, \mathbf{Z})$.

PROOF OF LEMMA 2. By the tower property of conditional expectation,

$$\mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2\} = \mathbb{E}\left\{\phi(T, \mathbf{X}, \mathbf{Z})^2 \Pr(\Delta = 1 \mid T, \mathbf{X}, \mathbf{Z})\right\}.$$

Assumption A4 states that

$$0 < \delta_1 \leq \Pr(\Delta = 1 \mid T, \mathbf{X}, \mathbf{Z}) \leq 1 \quad \text{a.s.}$$

Since $\phi(T, \mathbf{X}, \mathbf{Z})^2 \geq 0$, we obtain

$$\delta_1 \mathbb{E}\{\phi(T, \mathbf{X}, \mathbf{Z})^2\} \leq \mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2\} \leq \mathbb{E}\{\phi(T, \mathbf{X}, \mathbf{Z})^2\}.$$

Rearranging the inequalities gives

$$\mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2\} \leq \mathbb{E}\{\phi(T, \mathbf{X}, \mathbf{Z})^2\} \leq \frac{1}{\delta_1} \mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2\}.$$

Thus (4.2) holds with $c_1 \leq 1$ and $c_2 = 1/\delta_1$.

Next, we relate the event-weighted quantity $\mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2\}$ to the time-integrated L^2 norm of ϕ . Write,

$$(A.2) \quad \mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2\} = \mathbb{E}_{\mathbf{X}, \mathbf{Z}}\left[\mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2 \mid \mathbf{X}, \mathbf{Z}\}\right],$$

and let $f_{T, \Delta=1|\mathbf{X}, \mathbf{Z}}(t \mid \mathbf{X}, \mathbf{Z})$ denote the conditional density of T given $(\mathbf{X}, \mathbf{Z})$ on the event $\{\Delta = 1\}$. Then,

$$\mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2 \mid \mathbf{X}, \mathbf{Z}\} = \int_0^\tau \phi(t, \mathbf{X}, \mathbf{Z})^2 f_{T, \Delta=1|\mathbf{X}, \mathbf{Z}}(t \mid \mathbf{X}, \mathbf{Z}) dt.$$

Under Assumptions A1 and A2, the event-time hazard $h_o(t, \mathbf{X}, \mathbf{Z})$ is bounded away from zero and infinity. Similarly, Assumption A4 implies the censoring mechanism is well behaved. Therefore,

$$\Pr(T > t \mid \mathbf{X}, \mathbf{Z}) = \exp\left[-\int_0^t \{h_o(s \mid \mathbf{X}, \mathbf{Z}) + \mu(s \mid \mathbf{X}, \mathbf{Z})\} ds\right]$$

where $\mu(t \mid \mathbf{X}, \mathbf{Z})$ is the censoring hazard bounded away from zero and infinity, and

$$f_{T, \Delta=1|\mathbf{X}, \mathbf{Z}}(t \mid \mathbf{X}, \mathbf{Z}) = h_o(t \mid \mathbf{X}, \mathbf{Z}) \exp\left[-\int_0^t \{h_o(s \mid \mathbf{X}, \mathbf{Z}) + \mu(s \mid \mathbf{X}, \mathbf{Z})\} ds\right]$$

is bounded above and below by positive constants that depend only on $(\tau, \delta_1, \delta_2, c_h, C_h)$. Specifically, there exist $0 < c_f \leq C_f < \infty$ such that, almost surely,

$$c_f \leq f_{T, \Delta=1|\mathbf{X}, \mathbf{Z}}(t \mid \mathbf{X}, \mathbf{Z}) \leq C_f, \quad t \in [0, \tau].$$

Using these bounds inside the conditional expectation (A.2), we obtain, that given $(\mathbf{X}, \mathbf{Z})$,

$$c_f \int_0^\tau \phi(t, \mathbf{X}, \mathbf{Z})^2 dt \leq \mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2 \mid \mathbf{X}, \mathbf{Z}\} \leq C_f \int_0^\tau \phi(t, \mathbf{X}, \mathbf{Z})^2 dt.$$

Taking expectation over $(\mathbf{X}, \mathbf{Z})$ yields

$$c_f \mathbb{E}_{\mathbf{X}, \mathbf{Z}}\left\{\int_0^\tau \phi(t, \mathbf{X}, \mathbf{Z})^2 dt\right\} \leq \mathbb{E}\{\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2\} \leq C_f \mathbb{E}_{\mathbf{X}, \mathbf{Z}}\left\{\int_0^\tau \phi(t, \mathbf{X}, \mathbf{Z})^2 dt\right\}.$$

Thus (4.3) holds with $c_3 = 1/C_f$ and $c_4 = 1/c_f$ (or equivalently with c_3, c_4 chosen directly as above).

Finally, suppose $\mathbb{E}\{\Delta\phi(T, \mathbf{X}, \mathbf{Z})^2\} = 0$. Since $\Delta\phi(T, \mathbf{X}, \mathbf{Z})^2 \geq 0$, this implies $\Delta\phi(T, \mathbf{X}, \mathbf{Z})^2 = 0$ almost surely, hence $\phi(T, \mathbf{X}, \mathbf{Z}) = 0$ $\mathbb{P}_0$ -a.s. on $\{\Delta = 1\}$. From (4.2) we also have

$$0 \leq \mathbb{E}\{\phi(T, \mathbf{X}, \mathbf{Z})^2\} \leq c_2 \mathbb{E}\{\Delta\phi(T, \mathbf{X}, \mathbf{Z})^2\} = 0,$$

so $\mathbb{E}\{\phi(T, \mathbf{X}, \mathbf{Z})^2\} = 0$, which forces $\phi(T, \mathbf{X}, \mathbf{Z}) = 0$ $\mathbb{P}_0$ -almost surely. Likewise, from (4.3),

$$0 \leq \mathbb{E}_{\mathbf{X}, \mathbf{Z}}\left\{\int_0^\tau \phi(t, \mathbf{X}, \mathbf{Z})^2 dt\right\} \leq c_4 \mathbb{E}\{\Delta\phi(T, \mathbf{X}, \mathbf{Z})^2\} = 0,$$

so

$$\mathbb{E}_{\mathbf{X}, \mathbf{Z}}\left\{\int_0^\tau \phi(t, \mathbf{X}, \mathbf{Z})^2 dt\right\} = 0.$$

Since the integrand $\phi(t, \mathbf{X}, \mathbf{Z})^2$ is nonnegative, this implies

$$\int_0^\tau \phi(t, \mathbf{X}, \mathbf{Z})^2 dt = 0 \quad \text{for } \mathbb{P}_{\mathbf{X}, \mathbf{Z}}\text{-a.e. } (\mathbf{X}, \mathbf{Z}),$$

and hence $\phi(t, \mathbf{X}, \mathbf{Z}) = 0$ for Lebesgue-a.e. $t \in [0, \tau]$ and $P_{\mathbf{X}, \mathbf{Z}}$ -a.e. $(\mathbf{X}, \mathbf{Z})$. Equivalently,

$$\chi_{\eta_1}(t, \mathbf{X}, \mathbf{Z}) = \chi_{\eta_2}(t, \mathbf{X}, \mathbf{Z}) \quad \text{for } dt \otimes \mathbb{P}_{\mathbf{X}, \mathbf{Z}}\text{-almost all } (t, \mathbf{X}, \mathbf{Z}).$$

□

LEMMA 3. Assume A1–A2. For any $\eta \in \Theta \times \mathcal{H}(q, \alpha, \mathbf{d}^*, \tilde{\mathbf{d}}, M)$,

$$\ell_0(\eta) - \ell_0(\eta_o) \asymp -d(\eta, \eta_o)^2.$$

PROOF OF LEMMA 3. Recall that $\ell_0(\eta)$ denotes the population (expected) log-likelihood,

$$\ell_0(\eta) = \mathbb{E}\left\{\Delta\chi_\eta(T, \mathbf{X}, \mathbf{Z}) - \int_0^\tau Y(t)h_\eta(t, \mathbf{X}, \mathbf{Z})dt\right\},$$

where χ_η is the linear predictor and $h_\eta(t, \mathbf{X}, \mathbf{Z})$ is the corresponding instantaneous hazard process. Let $\eta_o = (g_o, \theta_o)$ denote the true parameter, with true hazard $h_o(t, \mathbf{X}, \mathbf{Z}) = h_{\eta_o}(t, \mathbf{X}, \mathbf{Z})$.

We study the behavior of ℓ_0 along a one-dimensional path through η_o . Fix an arbitrary direction $w = (w_g, w_\theta)$ and define

$$\eta_\varepsilon = \eta_o + \varepsilon w, \quad \varepsilon \in \mathbb{R}.$$

Assume that ε is restricted to a small neighbourhood of 0 so that $\eta_\varepsilon \in \Theta \times \mathcal{H}(q, \alpha, \mathbf{d}^*, \tilde{\mathbf{d}}, M)$. Because χ_η is linear in (g, θ) , the Gateaux derivative of χ_η in direction w at η_o is simply

$$\dot{\chi}_h(t, \mathbf{X}, \mathbf{Z}) = \frac{d}{d\varepsilon} \chi_{\eta_\varepsilon}(t, \mathbf{X}, \mathbf{Z}) \Big|_{\varepsilon=0},$$

and we can write

$$\chi_{\eta_\varepsilon}(t, \mathbf{X}, \mathbf{Z}) = \chi_{\eta_o}(t, \mathbf{X}, \mathbf{Z}) + \varepsilon \dot{\chi}_w(t, \mathbf{X}, \mathbf{Z}).$$

For notational simplicity, when we fix a particular η , we also write

$$q(t, \mathbf{X}, \mathbf{Z}) = \chi_\eta(t, \mathbf{X}, \mathbf{Z}) - \chi_{\eta_o}(t, \mathbf{X}, \mathbf{Z}),$$

so q is the difference in linear predictors between η and η_o . Along the path η_ε , q is proportional to the direction $\dot{\chi}_w$.

Next, note that

$$h_{\eta_\varepsilon}(t \mid \mathbf{X}, \mathbf{Z}) = h_o(t \mid \mathbf{X}, \mathbf{Z}) \exp\{\chi_{\eta_\varepsilon}(t, \mathbf{X}, \mathbf{Z}) - \chi_{\eta_o}(t, \mathbf{X}, \mathbf{Z})\} = h_o(t \mid \mathbf{X}, \mathbf{Z}) \exp\{\varepsilon \dot{\chi}_w(t, \mathbf{X}, \mathbf{Z})\},$$

so at $\varepsilon = 0$ we have $h_{\eta_o} = h_o$ and

$$\begin{aligned} \frac{d}{d\varepsilon} h_{\eta_\varepsilon}(t, \mathbf{X}, \mathbf{Z}) \Big|_{\varepsilon=0} &= h_o(t \mid \mathbf{X}, \mathbf{Z}) \dot{\chi}_w(t, \mathbf{X}, \mathbf{Z}), \\ \frac{d^2}{d\varepsilon^2} h_{\eta_\varepsilon}(t, \mathbf{X}, \mathbf{Z}) \Big|_{\varepsilon=0} &= h_o(t \mid \mathbf{X}, \mathbf{Z}) \dot{\chi}_w(t, \mathbf{X}, \mathbf{Z})^2. \end{aligned}$$

By the definition of the Gateaux derivative,

$$\ell'_0(\eta_o)[w] = \frac{d}{d\varepsilon} \ell_0(\eta_\varepsilon) \Big|_{\varepsilon=0}.$$

Using the explicit form of ℓ_0 and differentiating under the expectation,

$$\begin{aligned} \ell'_0(\eta_o)[w] &= \frac{d}{d\varepsilon} \mathbb{E} \left\{ \Delta \chi_{\eta_\varepsilon}(T, \mathbf{X}, \mathbf{Z}) - \int_0^\tau Y(t) h_{\eta_\varepsilon}(t \mid \mathbf{X}, \mathbf{Z}) dt \right\} \Big|_{\varepsilon=0} \\ &= \mathbb{E} \left\{ \Delta \dot{\chi}_w(T, \mathbf{X}, \mathbf{Z}) - \int_0^\tau Y(t) \frac{d}{d\varepsilon} h_{\eta_\varepsilon}(t \mid \mathbf{X}, \mathbf{Z}) \Big|_{\varepsilon=0} dt \right\} \\ &= \mathbb{E} \left\{ \Delta \dot{\chi}_w(T, \mathbf{X}, \mathbf{Z}) - \int_0^\tau Y(t) h_o(t \mid \mathbf{X}, \mathbf{Z}) \dot{\chi}_w(t, \mathbf{X}, \mathbf{Z}) dt \right\}. \end{aligned}$$

Since

$$\Delta \dot{\chi}_w(T, \mathbf{X}, \mathbf{Z}) = \int_0^\tau \dot{\chi}_w(t, \mathbf{X}, \mathbf{Z}) dN(t),$$

the Doob–Meyer decomposition for the true model (2.1) gives

$$dN(t) = Y(t) h_o(t \mid \mathbf{X}, \mathbf{Z}) dt + dM(t),$$

where $M(t)$ is a martingale. Therefore,

$$\Delta \dot{\chi}_w(T, \mathbf{X}, \mathbf{Z}) = \int_0^\tau \dot{\chi}_w(t, \mathbf{X}, \mathbf{Z}) Y(t) h_o(t \mid \mathbf{X}, \mathbf{Z}) dt + \int_0^\tau \dot{\chi}_w(t, \mathbf{X}, \mathbf{Z}) dM(t).$$

Substituting this into the expression for $\ell'_0(\eta_o)[w]$ and cancelling the compensator term,

$$\ell'_0(\eta_o)[w] = \mathbb{E} \left\{ \int_0^\tau \dot{\chi}_w(t, \mathbf{X}, \mathbf{Z}) dM(t) \right\}.$$

The stochastic integral $\int_0^\tau \dot{\chi}_w(t, \mathbf{X}, \mathbf{Z}) dM(t)$ is a martingale with mean zero. Hence,

$$\ell'_0(\eta_o)[w] = 0 \quad \text{for every direction } w.$$

In particular, when we fix η and set $q(t, \mathbf{X}, \mathbf{Z}) = \chi_\eta(t, \mathbf{X}, \mathbf{Z}) - \chi_{\eta_o}(t, \mathbf{X}, \mathbf{Z})$, we can summarize this step as

$$\ell'_0(\eta_o)[w] = \mathbb{E} \left\{ \Delta q(T, \mathbf{X}, \mathbf{Z}) - \int_0^\tau Y(t) h_o(t \mid \mathbf{X}, \mathbf{Z}) q(t, \mathbf{X}, \mathbf{Z}) dt \right\} = 0.$$

To control $\ell_0(\eta) - \ell_0(\eta_o)$, we apply a one-dimensional Taylor expansion of $\ell_0(\eta_\varepsilon)$ around $\varepsilon = 0$. Write $\eta = \eta_1$ and define the line segment $\eta_v = \eta_o + v(\eta - \eta_o)$ for $v \in [0, 1]$. The mean value theorem yields

$$\ell_0(\eta) = \ell_0(\eta_o) + \ell'_0(\eta_o)[w] + \frac{1}{2} \ell''_0(\eta_{\bar{v}})[w, w]$$

for some $\bar{v} \in (0, 1)$, where $w = \eta - \eta_o$. Since we have already shown that $\ell'_0(\eta_o)[w] = 0$, it remains to analyse the second derivative term.

For a fixed $v \in [0, 1]$, consider the path $\varepsilon \mapsto \eta_v + \varepsilon w$ and write

$$q_v(t, \mathbf{X}, \mathbf{Z}) = \left. \frac{d}{d\varepsilon} \chi_{\eta_v + \varepsilon w}(t, \mathbf{X}, \mathbf{Z}) \right|_{\varepsilon=0},$$

which is again linear in w . Repeating the differentiation steps above but now starting from η_v instead of η_o gives

$$\begin{aligned} \ell''_0(\eta_v)[w, w] &= \left. \frac{d^2}{d\varepsilon^2} \ell_0(\eta_v + \varepsilon w) \right|_{\varepsilon=0} \\ &= \mathbb{E} \left\{ \Delta \left. \frac{d^2}{d\varepsilon^2} \chi_{\eta_v + \varepsilon w}(T, \mathbf{X}, \mathbf{Z}) \right|_{\varepsilon=0} - \int_0^\tau Y(t) \left. \frac{d^2}{d\varepsilon^2} h_{\eta_v + \varepsilon w}(t, \mathbf{X}, \mathbf{Z}) \right|_{\varepsilon=0} dt \right\}. \end{aligned}$$

Since χ_η is linear in η , its second derivative in ε vanishes, while the second derivative of the hazard is

$$\left. \frac{d^2}{d\varepsilon^2} h_{\eta_v + \varepsilon w}(t | \mathbf{X}, \mathbf{Z}) \right|_{\varepsilon=0} = h_{\eta_v}(t | \mathbf{X}, \mathbf{Z}) q_v(t, \mathbf{X}, \mathbf{Z})^2.$$

Hence

$$\ell''_0(\eta_v)[w, w] = -\mathbb{E} \left\{ \int_0^\tau Y(t) q_v(t, \mathbf{X}, \mathbf{Z})^2 h_{\eta_v}(t | \mathbf{X}, \mathbf{Z}) dt \right\}.$$

For notational simplicity (since we will ultimately evaluate this at $\varepsilon = 1$), we drop the subscript v and simply write $q(t, \mathbf{X}, \mathbf{Z}) = \chi_\eta(t, \mathbf{X}, \mathbf{Z}) - \chi_{\eta_o}(t, \mathbf{X}, \mathbf{Z})$ along the line, so that

$$\ell''_0(\eta_v)[w, w] = -\mathbb{E} \left\{ \int_0^\tau Y(t) q(t, \mathbf{X}, \mathbf{Z})^2 h_{\eta_v}(t | \mathbf{X}, \mathbf{Z}) dt \right\}.$$

Next, for relating the second derivative to $d(\eta, \eta_o)^2$, we apply Lemma 1 for

$$f_v(t, \mathbf{X}, \mathbf{Z}) = q(t, \mathbf{X}, \mathbf{Z})^2 \frac{h_{\eta_v}(t | \mathbf{X}, \mathbf{Z})}{h_o(t | \mathbf{X}, \mathbf{Z})}.$$

Indeed, Lemma 1 gives

$$\mathbb{E} \left\{ \int_0^\tau Y(t) q(t, \mathbf{X}, \mathbf{Z})^2 h_{\eta_v}(t | \mathbf{X}, \mathbf{Z}) dt \right\} \asymp \mathbb{E} \left\{ \Delta q(T, \mathbf{X}, \mathbf{Z})^2 \frac{h_{\eta_v}(T | \mathbf{X}, \mathbf{Z})}{h_o(T | \mathbf{X}, \mathbf{Z})} \right\}.$$

Under Assumptions A1–A2, the processes χ_η are uniformly bounded, and hence

$$0 < c_\star \leq \frac{h_{\eta_v}(t | \mathbf{X}, \mathbf{Z})}{h_o(t | \mathbf{X}, \mathbf{Z})} \leq C_\star < \infty, \quad \text{for all } (t, \mathbf{X}, \mathbf{Z}), v \in [0, 1].$$

Therefore,

$$c_\star \mathbb{E}[\Delta q(T, \mathbf{X}, \mathbf{Z})^2] \leq \mathbb{E} \left[\Delta q(T, \mathbf{X}, \mathbf{Z})^2 \frac{h_{\eta_v}(T | \mathbf{X}, \mathbf{Z})}{h_o(T | \mathbf{X}, \mathbf{Z})} \right] \leq C_\star \mathbb{E}[\Delta q(T, \mathbf{X}, \mathbf{Z})^2].$$

Combining this with the expression for $\ell''_0(\eta_v)[w, w]$ yields the two-sided bound

$$-C_\star \mathbb{E}[\Delta q(T, \mathbf{X}, \mathbf{Z})^2] \leq \ell''_0(\eta_v)[w, w] \leq -c_\star \mathbb{E}[\Delta q(T, \mathbf{X}, \mathbf{Z})^2].$$

Recall that the distance $d(\eta, \eta_o)$ is defined by

$$d(\eta, \eta_o)^2 = \mathbb{E} \left\{ \Delta \{ \chi_\eta(T, \mathbf{X}, \mathbf{Z}) - \chi_{\eta_o}(T, \mathbf{X}, \mathbf{Z}) \}^2 \right\} = \mathbb{E} \{ \Delta q(T, \mathbf{X}, \mathbf{Z})^2 \}.$$

Hence

$$-C_\star d(\eta, \eta_o)^2 \leq \ell_0''(\eta_v)[w, w] \leq -c_\star d(\eta, \eta_o)^2.$$

Therefore, by the Taylor expansion with the mean value theorem,

$$\ell_0(\eta) - \ell_0(\eta_o) = \frac{1}{2} \ell_0''(\eta_{\bar{v}})[w, w]$$

for some $\bar{v} \in (0, 1)$, and we have just shown that $\ell_0''(\eta_{\bar{v}})[w, w]$ is comparable to $-d(\eta, \eta_o)^2$. Thus,

$$\ell_0(\eta) - \ell_0(\eta_o) \asymp -d(\eta, \eta_o)^2,$$

which completes the proof. $\square$

In the following, $\mathbb{P}_n$ and $\mathbb{P}$ denote the empirical and probability measures of $(T, \Delta, \mathbf{X}, \mathbf{Z})$, and $\mathbb{G}_n = \sqrt{n}(\mathbb{P}_n - \mathbb{P})$ is the empirical process. The symbol $\mathbb{E}^*$ stands for outer expectation, used to avoid measurability issues in the supremum over an uncountable class of functions.

LEMMA 10. *Consider the following classes of functions indexed by the parameter η :*

- *The log-likelihood increment*

$$f_\eta(T, \mathbf{X}, \mathbf{Z}, \Delta) = \Delta \chi_\eta(T, \mathbf{X}, \mathbf{Z}) - \int_0^T Y(t) \exp\{\chi_\eta(t, \mathbf{X}, \mathbf{Z})\} dt,$$

- *The efficient score-type term*

$$\psi_\eta(T, \mathbf{X}, \mathbf{Z}, \Delta) = \int_0^T \{\mathbf{Z} - \mathbf{g}^*(t, \mathbf{X})\} dM_\eta(t),$$

where $\mathbf{g}^*$ denotes the efficient projection defined in Theorem 3.

For $0 < \delta \leq 1$, define the localized classes

$$\mathcal{F}_\delta = \{f_\eta - f_{\eta_o} : d(\eta, \eta_o) \leq \delta\}, \quad \mathcal{G}_\delta = \{\psi_\eta - \psi_{\eta_o} : d(\eta, \eta_o) \leq \delta\}.$$

Under Assumptions A1–A3 and A5, there exists a constant $C > 0$ such that for every $\delta \in (0, 1]$,

$$\mathbb{E}^* \left\{ \sup_{\mu \in \mathcal{F}_\delta \cup \mathcal{G}_\delta} |\sqrt{n}(\mathbb{P}_n - \mathbb{P})\mu| \right\} \leq C \left\{ \delta \sqrt{s \log(\mathcal{U}/\delta)} + n^{-1/2} s \log(\mathcal{U}/\delta) \right\},$$

where s is the sparsity of the neural network class, and

$$\mathcal{U} = K \prod_{k=0}^K (p_k + 1) \sum_{k=0}^K p_k p_{k+1}.$$

PROOF OF LEMMA 10. Fix η_1, η_2 and denote $q(t, \mathbf{X}, \mathbf{Z}) = \chi_{\eta_1}(t, \mathbf{X}, \mathbf{Z}) - \chi_{\eta_2}(t, \mathbf{X}, \mathbf{Z})$. For notational convenience, we write $\chi_\eta(t) = \chi_\eta(t, \mathbf{X}, \mathbf{Z})$ and $q(t) = \chi_{\eta_1}(t) - \chi_{\eta_2}(t)$. Because χ_η is uniformly bounded under A1–A2 and depends on the network parameters in a Lipschitz way, we have

$$|\exp\{\chi_{\eta_1}(t, \mathbf{X}, \mathbf{Z})\} - \exp\{\chi_{\eta_2}(t, \mathbf{X}, \mathbf{Z})\}| \lesssim |q(t, \mathbf{X}, \mathbf{Z})|.$$

Using the definition of f_η and the triangle and Cauchy–Schwarz inequalities,

$$|f_{\eta_1} - f_{\eta_2}| \leq \Delta |q(T)| + \int_0^T Y(t) |\exp\{\chi_{\eta_1}(t)\} - \exp\{\chi_{\eta_2}(t)\}| dt.$$

Plugging the bound, squaring and applying the inequality $(a + b)^2 \leq 2a^2 + 2b^2$ gives

$$(f_{\eta_1} - f_{\eta_2})^2 \lesssim 2\Delta q(T)^2 + 2 \int_0^\tau Y(t) q(t)^2 dt,$$

where all quantities are uniform in $(\mathbf{X}, \mathbf{Z})$ by A1–A2. By Lemma 1 we get

$$E \left[\int_0^\tau Y(t) q(t)^2 dt \right] \asymp E[\Delta q(T)^2].$$

The right-hand side is exactly the squared metric $d^2(\eta_1, \eta_2)$. Thus,

$$\|f_{\eta_1} - f_{\eta_2}\|_{L^2(\mathbb{P})} \lesssim d(\eta_1, \eta_2).$$

Repeating the argument for ψ_η requires only replacing the exponential term with the stochastic integral. Assumption A5 ensures boundedness of the predictable integrand $\mathbf{Z} - g^*(t, \mathbf{X})$ and square-integrability of the martingale increments; A1–A2 control the compensator. The same calculation shows

$$\|\psi_{\eta_1} - \psi_{\eta_2}\|_{L^2(\mathbb{P})} \lesssim d(\eta_1, \eta_2).$$

Hence both localized classes $\mathcal{F}_\delta$ and $\mathcal{G}_\delta$ lie in a common $L^2(\mathbb{P})$ -ball of radius $O(\delta)$ and admit a common envelope of order $O(\delta)$. The parametrization $\eta = (\boldsymbol{\theta}, g)$ is s -sparse in the neural-network coordinates, and Assumption A4 ensures that the weights are uniformly bounded and that the architecture has fixed depth and layer widths. In particular, the map $\eta \mapsto \chi_\eta$ belongs to the class of sparse feedforward neural networks considered in Zhong, Mueller and Wang (2022). Moreover, the maps $\eta \mapsto f_\eta$ and $\eta \mapsto \psi_\eta$ are Lipschitz transformations of χ_η in the $L^2(\mathbb{P})$ -metric, with Lipschitz constant bounded uniformly under Assumptions A1–A3 and A5. Since Lipschitz images of a function class cannot increase its covering number by more than a multiplicative constant, the metric entropy of the localized classes $\mathcal{F}_\delta$ and $\mathcal{G}_\delta$ is therefore controlled by the entropy of the underlying sparse neural-network ball

$$\mathcal{H}_\delta = \{\chi_\eta - \chi_{\eta_o} : d(\eta, \eta_o) \leq \delta\}.$$

Lemma 6 of Zhong, Mueller and Wang (2022) establishes that for such a sparse network class,

$$\log N_{[]}(\varepsilon, \mathcal{H}_\delta, L^2(P)) \lesssim s \log \left(\frac{\mathcal{U}\delta}{\varepsilon} \right),$$

where $\mathcal{U}$ is the architectural complexity factor. Because $\mathcal{F}_\delta$ and $\mathcal{G}_\delta$ lie in a common $L^2(\mathbb{P})$ -ball of radius $O(\delta)$ and share a common envelope of order $O(\delta)$, the same entropy bound applies to their union. Consequently,

$$\log N_{[]}(\varepsilon, \mathcal{F}_\delta \cup \mathcal{G}_\delta, L^2(\mathbb{P})) \lesssim s \log \left(\frac{\mathcal{U}\delta}{\varepsilon} \right).$$

Next, we bound the bracketing entropy integral. Using the entropy estimate derived above, the associated Dudley integral satisfies

$$J_{[]}(\delta) = \int_0^\delta \sqrt{\log N_{[]}(\varepsilon)} d\varepsilon \leq C_2 \delta \sqrt{s \log \left(\frac{\mathcal{U}}{\delta} \right)}.$$

Finally, applying Lemma 3.4.2 of van der Vaart and Wellner (1996) to the empirical process $\mathbb{G}_n = \sqrt{n}(\mathbb{P}_n - \mathbb{P})$, with envelope $F_\delta = O(\delta)$, gives

$$\mathbb{E}^* \left\{ \sup_{\mu \in \mathcal{F}_\delta \cup \mathcal{G}_\delta} |\mathbb{G}_n \mu| \right\} \leq C_3 \left\{ J_{[]}(\delta) + n^{-1/2} \|F_\delta\|_{L^2(\mathbb{P})} \log N_{[]}(\delta) \right\}.$$

Substituting the bound on $J_{\square}(\delta)$ and noting that $\|F_{\delta}\|_{L^2(\mathbb{P})} = O(\delta)$ and $\log N_{\square}(\delta) \lesssim s \log(\mathcal{U}/\delta)$ yields the asserted maximal inequality. $\square$

The following lemma establishes equivalence of expectations under two models whose hazard functions differ only through the nuisance function g . Because the hazards are uniformly bounded above and below, the distributions under g_0 and g_1 are mutually absolutely continuous with Radon–Nikodym derivatives uniformly bounded. This equivalence allows us to control expectations under one model using expectations under another.

LEMMA 11. *Suppose Assumptions A1–A2 and A4 hold. Let $g_0, g_1 \in \mathcal{H}(q, \alpha, \mathbf{d}^*, \tilde{\mathbf{d}}, M)$ and fix $\theta \in \Theta$. For $j = 0, 1$, let $\mathbb{P}_{g_j}$ denote the distribution of $(T, \mathbf{X}, \mathbf{Z}, \Delta)$ under event hazard $h_{g_j}(t | \mathbf{X}, \mathbf{Z}) = \exp\{g_j(t, \mathbf{X}) + \theta^\top \mathbf{Z}\}$ and a common censoring hazard $\mu(t | \mathbf{X}, \mathbf{Z})$. As a consequence of Assumptions A1–A2, there exists a constant $0 < D < \infty$ such that, almost surely,*

$$e^{-D} \leq h_{g_j}(t | \mathbf{X}, \mathbf{Z}) \leq e^D, \quad \text{for all } t \in [0, \tau], \quad j = 0, 1.$$

Define $c = e^{-D}$ and $C = e^D$. Then, for any nonnegative measurable function $q : [0, \tau] \times \mathcal{X} \rightarrow [0, \infty)$ with $\mathbb{E}_{\mathbb{P}_{g_0}}[q(T, \mathbf{X})] < \infty$,

$$K_- \mathbb{E}_{\mathbb{P}_{g_0}}\{q(T, \mathbf{X})\} \leq \mathbb{E}_{\mathbb{P}_{g_1}}\{q(T, \mathbf{X})\} \leq K_+ \mathbb{E}_{\mathbb{P}_{g_0}}\{q(T, \mathbf{X})\},$$

where

$$K_- = \frac{c}{C} e^{-\tau(C-c)}, \quad K_+ = \frac{C}{c} e^{\tau(C-c)}.$$

In particular, $\mathbb{E}_{\mathbb{P}_{g_1}}[q(T, \mathbf{X})] \asymp \mathbb{E}_{\mathbb{P}_{g_0}}[q(T, \mathbf{X})]$ with constants depending only on (c, C, τ) .

PROOF OF LEMMA 11. We work conditional on the covariates $(\mathbf{x}, \mathbf{z})$ and then integrate over their distribution. Under model g_j , the event time U_j has density and survival

$$f_j^U(t | \mathbf{x}, \mathbf{z}) = h_j(t | \mathbf{x}, \mathbf{z}) S_j^U(t | \mathbf{x}, \mathbf{z}), \quad S_j^U(t | \mathbf{x}, \mathbf{z}) = \exp\left\{-\int_0^t h_j(s | \mathbf{x}, \mathbf{z}) ds\right\}.$$

The censoring time C has density $f^C(t | \mathbf{x}, \mathbf{z}) = \mu(t | \mathbf{x}, \mathbf{z}) S^C(t | \mathbf{x}, \mathbf{z})$ and survival $S^C(t | \mathbf{x}, \mathbf{z})$, common to both models. The observed time and event indicator are $T_j = \min(U_j, C)$ and $\Delta_j = \mathbf{1}\{U_j \leq C\}$. Given $(\mathbf{x}, \mathbf{z})$, their joint density under g_j is

$$f_j(t, \delta | \mathbf{x}, \mathbf{z}) = \begin{cases} h_j(t | \mathbf{x}, \mathbf{z}) S_j^U(t | \mathbf{x}, \mathbf{z}) S^C(t | \mathbf{x}, \mathbf{z}), & \delta = 1 \\ \mu(t | \mathbf{x}, \mathbf{z}) S^C(t | \mathbf{x}, \mathbf{z}) S_j^U(t | \mathbf{x}, \mathbf{z}), & \delta = 0. \end{cases}$$

By Assumptions A1–A2,

$$ct \leq \int_0^t h_j(s | \mathbf{x}, \mathbf{z}) ds \leq Ct, \quad e^{-Ct} \leq S_j^U(t | \mathbf{x}, \mathbf{z}) \leq e^{-ct}.$$

The likelihood ratio of (T_j, Δ_j) under g_1 versus g_0 , conditional on $(\mathbf{x}, \mathbf{z})$, is

$$L(t, \delta | \mathbf{x}, \mathbf{z}) = \frac{f_1(t, \delta | \mathbf{x}, \mathbf{z})}{f_0(t, \delta | \mathbf{x}, \mathbf{z})} = \left\{ \frac{h_1(t | \mathbf{x}, \mathbf{z})}{h_0(t | \mathbf{x}, \mathbf{z})} \right\}^\delta \frac{S_1^U(t | \mathbf{x}, \mathbf{z})}{S_0^U(t | \mathbf{x}, \mathbf{z})}.$$

Since $c \leq h_j \leq C$,

$$\frac{c}{C} \leq \frac{h_1(t | \mathbf{x}, \mathbf{z})}{h_0(t | \mathbf{x}, \mathbf{z})} \leq \frac{C}{c}.$$

Moreover,

$$\frac{S_1^U(t | \mathbf{x}, \mathbf{z})}{S_0^U(t | \mathbf{x}, \mathbf{z})} = \exp \left[- \int_0^t \{h_1(s | \mathbf{x}, \mathbf{z}) - h_0(s | \mathbf{x}, \mathbf{z})\} ds \right],$$

and because $|h_1 - h_0| \leq C - c$,

$$e^{-(C-c)t} \leq \frac{S_1^U(t | \mathbf{x}, \mathbf{z})}{S_0^U(t | \mathbf{x}, \mathbf{z})} \leq e^{(C-c)t}.$$

For $t \leq \tau$ this yields

$$\frac{c}{C} e^{-(C-c)\tau} \leq L(t, \delta | \mathbf{x}, \mathbf{z}) \leq \frac{C}{c} e^{(C-c)\tau}.$$

Thus the Radon–Nikodým derivative satisfies

$$K_- \leq \frac{d\mathbb{P}_{g_1}}{d\mathbb{P}_{g_0}}(T, \Delta, \mathbf{X}, \mathbf{Z}) \leq K_+, \quad \mathbb{P}_{g_0}\text{-a.s.},$$

where $K_- = \frac{c}{C} e^{-\tau(C-c)}$ and $K_+ = \frac{C}{c} e^{\tau(C-c)}$. For any nonnegative measurable function q with finite expectation under $\mathbb{P}_{g_0}$,

$$\mathbb{E}_{\mathbb{P}_{g_1}} \{q(T, \mathbf{X})\} = \mathbb{E}_{\mathbb{P}_{g_0}} \left\{ q(T, \mathbf{X}) \frac{d\mathbb{P}_{g_1}}{d\mathbb{P}_{g_0}}(T, \Delta, \mathbf{X}, \mathbf{Z}) \right\}$$

and the bounds on the likelihood ratio give

$$K_- \mathbb{E}_{\mathbb{P}_{g_0}} \{q(T, \mathbf{X})\} \leq \mathbb{E}_{\mathbb{P}_{g_1}} \{q(T, \mathbf{X})\} \leq K_+ \mathbb{E}_{\mathbb{P}_{g_0}} \{q(T, \mathbf{X})\}.$$

□

LEMMA 12. Assume A2. Let $\eta_j = (\boldsymbol{\theta}_j, g_j)$, $j = 1, 2$, be two parameter values in $(\Theta \times \mathcal{H}(q, \boldsymbol{\alpha}, \mathbf{d}^*, \tilde{\mathbf{d}}, M)) \cup (\Theta \times \mathcal{G}(K, s, \mathbf{p}, D))$. Define

$$\chi_{\eta_j}(t, \mathbf{X}, \mathbf{Z}) = g_j(t, \mathbf{X}) + \boldsymbol{\theta}_j^\top \mathbf{Z}, \quad j = 1, 2.$$

Assume that $\|\mathbf{Z}\| \leq B_{\mathbf{Z}}$ a.s. for some finite constant $B_{\mathbf{Z}}$. Then

$$\mathbb{E} \left[\Delta \{g_1(T, \mathbf{X}) - g_2(T, \mathbf{X})\}^2 \right] \leq 2d(\eta_1, \eta_2)^2 + 2B_{\mathbf{Z}}^2 \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|^2.$$

PROOF OF LEMMA 12. Fix $(T, \Delta, \mathbf{X}, \mathbf{Z})$ and set $a = g_1(T, \mathbf{X}) - g_2(T, \mathbf{X})$, $b = (\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)^\top \mathbf{Z}$. Then $\chi_{\eta_1}(T, \mathbf{X}, \mathbf{Z}) - \chi_{\eta_2}(T, \mathbf{X}, \mathbf{Z}) = a + b$. Use the elementary inequality

$$(u - v)^2 \leq 2u^2 + 2v^2 \quad \text{for all } u, v \in \mathbb{R},$$

with $u = a + b$ and $v = b$. This gives

$$a^2 = (a + b - b)^2 \leq 2(a + b)^2 + 2b^2.$$

Multiplying by $\Delta \in \{0, 1\}$ preserves the inequality

$$\Delta a^2 \leq 2\Delta (a + b)^2 + 2\Delta b^2.$$

Taking expectations and using the definition of $d(\eta_1, \eta_2)$,

$$\begin{aligned} \mathbb{E} \left[\Delta \{g_1(T, \mathbf{X}) - g_2(T, \mathbf{X})\}^2 \right] &= \mathbb{E}(\Delta a^2) \\ &\leq 2\mathbb{E}\{\Delta (a + b)^2\} + 2\mathbb{E}(\Delta b^2) \\ &= 2d(\eta_1, \eta_2)^2 + 2\mathbb{E}(\Delta b^2). \end{aligned}$$

For the second term, by boundedness of $\mathbf{Z}$,

$$|b| = |(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)^\top \mathbf{Z}| \leq \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\| \|\mathbf{Z}\| \leq B_{\mathbf{Z}} \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|,$$

and hence

$$b^2 \leq B_{\mathbf{Z}}^2 \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|^2.$$

Since $\Delta \leq 1$,

$$\mathbb{E}(\Delta b^2) \leq \mathbb{E}(b^2) \leq B_{\mathbf{Z}}^2 \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|^2.$$

Putting everything together,

$$\mathbb{E} \left[\Delta \{g_1(T, \mathbf{X}) - g_2(T, \mathbf{X})\}^2 \right] \leq 2d(\eta_1, \eta_2)^2 + 2B_{\mathbf{Z}}^2 \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|^2.$$

□

LEMMA 13. *If Assumptions A1–A5 hold, then there exists a constant $c_\theta > 0$ such that, for all parameters $\eta_1 = (\boldsymbol{\theta}_1, g_1)$ and $\eta_2 = (\boldsymbol{\theta}_2, g_2)$,*

$$d(\eta_1, \eta_2)^2 \geq c_\theta \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|^2.$$

In particular, one may take

$$c_\theta = \lambda_{\min}(\Sigma_{\mathbf{Z}}) > 0.$$

PROOF OF LEMMA 13. Let $\boldsymbol{\beta} = \boldsymbol{\theta}_1 - \boldsymbol{\theta}_2$, $a(T, \mathbf{X}) = g_1(T, \mathbf{X}) - g_2(T, \mathbf{X})$, $b = \boldsymbol{\beta}^\top \mathbf{Z}$. Then

$$\chi_{\eta_1}(T, \mathbf{X}, \mathbf{Z}) - \chi_{\eta_2}(T, \mathbf{X}, \mathbf{Z}) = a(T, \mathbf{X}) + b,$$

and hence

$$d(\eta_1, \eta_2)^2 = \mathbb{E}_0 [\Delta \{a(T, \mathbf{X}) + b\}^2].$$

Because $a(T, \mathbf{X})$ is measurable with respect to $(T, \mathbf{X})$, we have

$$\mathbb{E}_0 [\Delta \{a + b\}^2 | T, \mathbf{X}] = \Pr(\Delta = 1 | T, \mathbf{X}) \mathbb{E}_0 [\{a + b\}^2 | T, \mathbf{X}, \Delta = 1].$$

By the conditional variance decomposition,

$$\mathbb{E}_0 [\{a + b\}^2 | T, \mathbf{X}, \Delta = 1] = \left(a + \mathbb{E}_0[b | T, \mathbf{X}, \Delta = 1] \right)^2 + \text{Var}_0(b | T, \mathbf{X}, \Delta = 1).$$

Therefore

$$\mathbb{E}_0 [\Delta \{a + b\}^2] = \mathbb{E}_0 \left[\Pr(\Delta = 1 | T, \mathbf{X}) \left(a + \mathbb{E}_0[b | T, \mathbf{X}, \Delta = 1] \right)^2 \right] + \mathbb{E}_0 [\Delta \text{Var}_0(b | T, \mathbf{X}, \Delta = 1)].$$

Since the first term is nonnegative,

$$d(\eta_1, \eta_2)^2 \geq \mathbb{E}_0 [\Delta \text{Var}_0(b | T, \mathbf{X}, \Delta = 1)].$$

Now, because $b = \boldsymbol{\beta}^\top \mathbf{Z}$,

$$\text{Var}_0(b | T, \mathbf{X}, \Delta = 1) = \boldsymbol{\beta}^\top \text{Var}_0(\mathbf{Z} | T, \mathbf{X}, \Delta = 1) \boldsymbol{\beta},$$

and thus

$$\mathbb{E}_0 [\Delta \text{Var}_0(b | T, \mathbf{X}, \Delta = 1)] = \boldsymbol{\beta}^\top \mathbb{E}_0 \left[\Delta \left\{ \mathbf{Z} - \mathbb{E}_0(\mathbf{Z} | T, \mathbf{X}, \Delta = 1) \right\}^{\otimes 2} \right] \boldsymbol{\beta} = \boldsymbol{\beta}^\top \Sigma_{\mathbf{Z}} \boldsymbol{\beta}.$$

By Assumption A5,

$$\boldsymbol{\beta}^\top \Sigma_{\mathbf{Z}} \boldsymbol{\beta} \geq \lambda_{\min}(\Sigma_{\mathbf{Z}}) \|\boldsymbol{\beta}\|^2.$$

Hence

$$d(\eta_1, \eta_2)^2 \geq \lambda_{\min}(\Sigma_{\mathbf{Z}}) \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|^2.$$

This proves the lemma with $c_\theta = \lambda_{\min}(\Sigma_{\mathbf{Z}})$. $\square$

THEOREM 1. *Let $(\hat{\boldsymbol{\theta}}, \hat{g})$ be the estimator defined in (3.3). Under Assumptions A1–A3,*

$$(4.4) \quad \left(\mathbb{E} \left[\Delta \{ \hat{g}(T, \mathbf{X}) - g_o(T, \mathbf{X}) \}^2 \right] \right)^{1/2} = O_p(\tau_n)$$

where the expectation is taken with respect to an independent draw $(T, \Delta, \mathbf{X}, \mathbf{Z}) \sim \mathbb{P}_0$ from the true distribution, and $\tau_n = \gamma_n \log^2 n$. The quantity in (4.4) is random through its dependence on the training sample used to construct $\hat{g}$.

PROOF OF THEOREM 1. Let $\ell(\eta; \mathbf{W})$, $\mathbf{W} = (T, \Delta, \mathbf{X}^\top, \mathbf{Z}^\top)^\top$, denote the per-observation log-likelihood contribution, so that $\ell_n(\eta) = \mathbb{P}_n \ell(\eta; \cdot)$ is the empirical loss and $\ell_0(\eta) = \mathbb{P}_0 \ell(\eta; \cdot)$ is the population loss. Let $\mathbb{G}_n = \sqrt{n}(\mathbb{P}_n - \mathbb{P}_0)$ be the empirical process. Recall the distance

$$d(\eta_1, \eta_2) = \left(\mathbb{E}_0 [\Delta \{ \chi_{\eta_1}(T, \mathbf{X}, \mathbf{Z}) - \chi_{\eta_2}(T, \mathbf{X}, \mathbf{Z}) \}^2] \right)^{1/2}, \quad \eta = (\boldsymbol{\theta}, g),$$

and let $\eta_o = (\boldsymbol{\theta}_o, g_o)$ denote the true parameter. By Lemma 10, there exists a constant $C > 0$ such that for every $0 < \delta \leq 1$,

$$(A.3) \quad \mathbb{E}^* \sup_{d(\eta, \eta_o) \leq \delta} |\mathbb{G}_n \{ \ell(\eta; \cdot) - \ell(\eta_o; \cdot) \}| \leq \Psi_n(\delta) \equiv \delta \sqrt{s \log \frac{\mathcal{U}}{\delta}} + \frac{s}{\sqrt{n}} \log \frac{\mathcal{U}}{\delta},$$

and by Markov's inequality, this implies

$$\sup_{d(\eta, \eta_o) \leq \delta} |\mathbb{G}_n \{ \ell(\eta; \cdot) - \ell(\eta_o; \cdot) \}| = O_p(\Psi_n(\delta)).$$

Set $\tau_n = \gamma_n \log^2 n$ as in the theorem. By Assumption A3 and the specific choice of $(K, s, \mathbf{p})$ (cf. the construction in Schmidt-Hieber (2020) and Zhong, Mueller and Wang (2022)), we have

$$s \log \left(\frac{\mathcal{U}}{\tau_n} \right) = O(n \gamma_n^2 \log n),$$

so that

$$\frac{1}{\sqrt{n}} \tau_n^{-2} \Psi_n(\tau_n) = \tau_n^{-1} \sqrt{\frac{s}{n} \log \frac{\mathcal{U}}{\tau_n}} + \frac{s}{n \tau_n^2} \log \frac{\mathcal{U}}{\tau_n} \lesssim 1$$

for n large enough. Hence, there exists a constant $c > 0$ such that

$$(A.4) \quad \Psi_n(\tau_n) \leq \frac{c}{2} \sqrt{n} \tau_n^2 \quad \text{and} \quad \frac{1}{\sqrt{n}} \Psi_n(\tau_n) \leq \frac{c}{2} \tau_n^2.$$

Consequently,

$$(A.5) \quad \sup_{d(\eta, \eta_o) \leq \tau_n} |(\ell_n - \ell_0)(\eta) - (\ell_n - \ell_0)(\eta_o)| = \sup_{d(\eta, \eta_o) \leq \tau_n} \frac{1}{\sqrt{n}} |\mathbb{G}_n \{ \ell(\eta; \cdot) - \ell(\eta_o; \cdot) \}| = O_p\left(\frac{1}{\sqrt{n}} \Psi_n(\tau_n)\right) = O_p(\tau_n^2),$$

where we used (A.3) and (A.4).

By Theorem 1 of [Schmidt-Hieber \(2020\)](#), applied to the composite Hölder class $\mathcal{H}(q, \alpha, \mathbf{d}^*, \tilde{\mathbf{d}}, M)$ with the choice of $(K, s, \mathbf{p}, D)$ given in Assumption A3, there exists $\tilde{g}_1 \in \mathcal{G}(K, s, \mathbf{p}, D)$ such that

$$\mathbb{E}\{\tilde{g}_1(T, \mathbf{X}) - g_o(T, \mathbf{X})\}^2 \leq C_1 \tau_n^2, \quad \mathbb{E}\{\Delta(\tilde{g}_1(T, \mathbf{X}) - g_o(T, \mathbf{X}))\}^2 \leq C_1 \tau_n^2$$

for some constant $C_1 > 0$ independent of n . Define $\tilde{\eta}_1 = (\boldsymbol{\theta}_o, \tilde{g}_1)$. Then,

$$d^2(\tilde{\eta}_1, \eta_o) = \mathbb{E}[\Delta\{\chi_{\tilde{\eta}_1}(T, \mathbf{X}, \mathbf{Z}) - \chi_{\eta_o}(T, \mathbf{X}, \mathbf{Z})\}^2] = \mathbb{E}[\Delta\{\tilde{g}_1(T, \mathbf{X}) - g_o(T, \mathbf{X})\}^2] \leq C_1 \tau_n^2,$$

since the $\boldsymbol{\theta}$ -component is the same in both $\tilde{\eta}_1$ and η_o . Therefore,

$$(A.6) \quad d(\tilde{\eta}_1, \eta_o) \leq \sqrt{C_1} \tau_n,$$

and in particular $\tilde{\eta}_1$ lies inside a ball of radius proportional to τ_n around η_o .

Lemma 3 states that, for any $\eta \in \mathcal{F}_D$,

$$\ell_0(\eta) - \ell_0(\eta_o) \asymp -d(\eta, \eta_o)^2,$$

that is, there exist constants $0 < c_-, c_+ < \infty$ such that

$$-c_+ d(\eta, \eta_o)^2 \leq \ell_0(\eta) - \ell_0(\eta_o) \leq -c_- d(\eta, \eta_o)^2.$$

Applying this to $\eta = \tilde{\eta}_1$ and using (A.6), we obtain

$$(A.7) \quad \ell_0(\tilde{\eta}_1) - \ell_0(\eta_o) \geq -c_+ d(\tilde{\eta}_1, \eta_o)^2 \geq -c_+ C_1 \tau_n^2 = -C_2 \tau_n^2,$$

for some constant $C_2 > 0$.

By combining (A.5) and (A.7) we get

$$\begin{aligned} \ell_n(\tilde{\eta}_1) - \ell_n(\eta_o) &= \{\ell_n(\tilde{\eta}_1) - \ell_0(\tilde{\eta}_1)\} - \{\ell_n(\eta_o) - \ell_0(\eta_o)\} + \{\ell_0(\tilde{\eta}_1) - \ell_0(\eta_o)\} \\ &\geq - \sup_{d(\eta, \eta_o) \leq \tau_n} |(\ell_n - \ell_0)(\eta) - (\ell_n - \ell_0)(\eta_o)| - C_2 \tau_n^2 \\ (A.8) \quad &= -O_p(\tau_n^2). \end{aligned}$$

Since $\hat{\eta} = (\hat{\boldsymbol{\theta}}, \hat{g})$ is the maximizer of the empirical likelihood over the sieve, then

$$(A.9) \quad \ell_n(\hat{\eta}) = \max_{\eta} \ell_n(\eta) \geq \ell_n(\tilde{\eta}_1) \geq \ell_n(\eta_o) - O_p(\tau_n^2).$$

On the other hand, by Lemma 3, the population loss satisfies

$$\ell_0(\eta) - \ell_0(\eta_o) \leq -c_- d(\eta, \eta_o)^2 \quad \text{for all } \eta \in \mathcal{F}_D.$$

The empirical-process bound (A.5) and the likelihood chain (A.9) verify the conditions (i)–(ii) of Theorem 3.4.1 in [van der Vaart and Wellner \(1996\)](#) with radius of order τ_n . Therefore,

$$d(\hat{\eta}, \eta_o) = O_p(\tau_n).$$

Finally, by writing

$$\chi_{\hat{\eta}}(t, \mathbf{X}, \mathbf{Z}) - \chi_{\eta_o}(t, \mathbf{X}, \mathbf{Z}) = \{\hat{g}(t, \mathbf{X}) - g_o(t, \mathbf{X})\} + (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o)^\top \mathbf{Z},$$

and applying Lemma 12 with $\eta_1 = \hat{\eta} = (\hat{\boldsymbol{\theta}}, \hat{g})$ and $\eta_2 = \eta_o = (\boldsymbol{\theta}_o, g_o)$, we obtain

$$(A.10) \quad \mathbb{E}[\Delta\{\hat{g}(T, \mathbf{X}) - g_o(T, \mathbf{X})\}^2] \leq C \left(d(\hat{\eta}, \eta_o)^2 + \|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o\|^2 \right).$$

Applying Lemma 13 with $\eta_1 = \hat{\eta}$ and $\eta_2 = \eta_o$ yields

$$\|\hat{\theta} - \theta_o\|^2 \leq c_{\hat{\theta}}^{-1} d(\hat{\eta}, \eta_o)^2 = O_p(\tau_n^2).$$

Hence, $\|\hat{\theta} - \theta_o\| = O_p(\tau_n)$, and the right-hand side of (A.10) is $O_p(\tau_n^2)$, which is exactly the claim of Theorem 1. The expectation is taken with respect to a new observation $(T, \Delta, \mathbf{X}, \mathbf{Z}) \sim \mathbb{P}_0$, while the $O_p(\cdot)$ is with respect to the randomness in the training sample used to construct $\hat{g}$. $\square$

THEOREM 2. *Under A1–A2 and A4 there exists $c > 0$ such that*

$$\inf_{\hat{g}} \sup_{\theta_o \in \Theta, g_o \in \mathcal{H}(q, \alpha, \mathbf{d}^*, \tilde{\mathbf{d}}, M)} \mathbb{E}_{\mathbb{P}_{g_o, \theta_o}} [\Delta \{\hat{g}(T, \mathbf{X}) - g_o(T, \mathbf{X})\}^2] \geq c \gamma_n^2.$$

PROOF OF THEOREM 2. For $g \in \mathcal{H}(q, \alpha, \mathbf{d}^*, \tilde{\mathbf{d}}, M)$ and $\theta_o \in \Theta$, let $\mathbb{P}_{g, \theta_o}$ denote the joint distribution of $(T, \Delta, \mathbf{X}, \mathbf{Z})$ induced by the hazard $h_{g, \theta_o}(t | \mathbf{X}, \mathbf{Z}) = \exp\{g(t, \mathbf{X}) + \theta_o^\top \mathbf{Z}\}$ together with a censoring hazard that is held fixed across all g . By the standard Kullback-Leibler (KL) distance for bounded hazards (Assumptions A1–A2), there exists $C > 0$ such that for any $g_0, g_1 \in \mathcal{H}(q, \alpha, \mathbf{d}^*, \tilde{\mathbf{d}}, M)$,

$$\text{KL}(\mathbb{P}_{g_1, \theta_o} \| \mathbb{P}_{g_0, \theta_o}) \leq C n \mathbb{E}_{\mathbb{P}_{g_1, \theta_o}} [\Delta \{g_1(T, \mathbf{X}) - g_0(T, \mathbf{X})\}^2].$$

Since $\Delta \leq 1$,

$$\text{KL}(\mathbb{P}_{g_1, \theta_o} \| \mathbb{P}_{g_0, \theta_o}) \leq C n \mathbb{E}_{\mathbb{P}_{g_1, \theta_o}} [\{g_1(T, \mathbf{X}) - g_0(T, \mathbf{X})\}^2].$$

Lemma 11 yields the equivalence

$$\text{KL}(\mathbb{P}_{g_1, \theta_o} \| \mathbb{P}_{g_0, \theta_o}) \leq C' n \mathbb{E}_{\mathbb{P}_{g_0, \theta_o}} [\{g_1(T, \mathbf{X}) - g_0(T, \mathbf{X})\}^2]$$

for a constant C' depending only on A1–A2.

By the packing construction for composite Hölder networks (Schmidt-Hieber, 2020), there exist functions $g^{(0)}, \dots, g^{(N)} \in \mathcal{H}(q, \alpha, \mathbf{d}^*, \tilde{\mathbf{d}}, M)$ and constants $c_1, c_2 > 0$ such that

$$(A.11) \quad \|g^{(j)} - g^{(k)}\|_{L^2(P_{g^{(0)}, \theta_o})} \geq 2c_1 \gamma_n, \quad j \neq k,$$

and

$$(A.12) \quad \frac{1}{N} \sum_{j=1}^N \|g^{(j)} - g^{(0)}\|_{L^2(P_{g^{(0)}, \theta_o})}^2 \leq \frac{c_2 \log N}{n}.$$

Applying the KL bound with $g_0 = g^{(0)}$ and $g_1 = g^{(j)}$ and using (A.12),

$$\frac{1}{N} \sum_{j=1}^N \text{KL}(\mathbb{P}_{g^{(j)}, \theta_o} \| \mathbb{P}_{g^{(0)}, \theta_o}) \leq C' c_2 \log N.$$

Choose $\alpha \in (0, 1/8)$ so that $C' c_2 \leq \alpha$. Then,

$$\frac{1}{N} \sum_{j=1}^N \text{KL}(\mathbb{P}_{g^{(j)}, \theta_o} \| \mathbb{P}_{g^{(0)}, \theta_o}) \leq \alpha \log N.$$

Applying Tsybakov's Fano inequality to $\{g^{(0)}, \dots, g^{(N)}\}$ with separation (A.11), we obtain that there exists $c > 0$ (depending only on c_1 and α) such that for any estimator $\hat{g}$,

$$\sup_{j=0, \dots, N} \mathbb{E}_{\mathbb{P}_{g^{(j)}, \theta_o}} \left(\|\hat{g} - g^{(j)}\|_{L^2(\mathbb{P}_{g^{(0)}, \theta_o})}^2 \right) \geq c \gamma_n^2.$$

Since the supremum over $g_o \in \mathcal{H}$ dominates this finite set,

$$\inf_{\hat{g}} \sup_{g_o \in \mathcal{H}} \mathbb{E}_{\mathbb{P}_{g_o, \theta_o}} \left(\|\hat{g} - g_o\|_{L^2(\mathbb{P}_{g(0), \theta_o})}^2 \right) \geq c \gamma_n^2.$$

By Lemma 11, there exists a constant $C_{\text{eq}} > 0$ (depending only on A1–A2) such that, uniformly over $g_o \in \mathcal{H}$ and all sieve estimators $\hat{g}$,

$$\mathbb{E}_{\mathbb{P}_{g(0), \theta_o}} [\{\hat{g}(T, \mathbf{X}) - g_o(T, \mathbf{X})\}^2] \leq C_{\text{eq}} \mathbb{E}_{\mathbb{P}_{g_o, \theta_o}} [\{\hat{g}(T, \mathbf{X}) - g_o(T, \mathbf{X})\}^2].$$

Combining this with the previous display yields

$$\inf_{\hat{g}} \sup_{g_o \in \mathcal{H}} \mathbb{E}_{\mathbb{P}_{g_o, \theta_o}} [\{\hat{g}(T, \mathbf{X}) - g_o(T, \mathbf{X})\}^2] \geq \frac{c}{C_{\text{eq}}} \gamma_n^2.$$

Finally, since the supremum in the theorem is also taken over $\theta_o \in \Theta$, and the above bound holds for each fixed θ_o , we obtain

$$(A.13) \quad \inf_{\hat{g}} \sup_{\theta_o \in \Theta, g_o \in \mathcal{H}} \mathbb{E}_{\mathbb{P}_{g_o, \theta_o}} [\{\hat{g}(T, \mathbf{X}) - g_o(T, \mathbf{X})\}^2] \geq c_1 \gamma_n^2$$

for some $c_1 > 0$. By Lemma 2 and Assumptions A1, A2, A4, there exists a constant $C_\Delta > 0$ such that, for all $g_o \in \mathcal{H}$ and all $\hat{g}$,

$$\mathbb{E}_{\mathbb{P}_{g_o, \theta_o}} [\{\hat{g}(T, \mathbf{X}) - g_o(T, \mathbf{X})\}^2] \leq C_\Delta \mathbb{E}_{\mathbb{P}_{g_o, \theta_o}} [\Delta \{\hat{g}(T, \mathbf{X}) - g_o(T, \mathbf{X})\}^2].$$

Combining this with (A.13) gives

$$\inf_{\hat{g}} \sup_{\theta_o \in \Theta, g_o \in \mathcal{H}} \mathbb{E}_{\mathbb{P}_{g_o, \theta_o}} [\Delta \{\hat{g}(T, \mathbf{X}) - g_o(T, \mathbf{X})\}^2] \geq \frac{c_1}{C_\Delta} \gamma_n^2.$$

Renaming the constant $c = c_1/C_\Delta$ completes the proof. $\square$

THEOREM 3. *Under Assumptions A1–A5, the efficient score for θ_o in the semiparametric model (2.1) is*

$$\ell_{\theta_o}^*(T, \mathbf{X}, \mathbf{Z}, \Delta) = \int_0^\tau \{\mathbf{Z} - \mathbf{g}^*(t, \mathbf{X})\} dM(t),$$

where $\mathbf{g}^*(t, \mathbf{X})$ is the Δ -weighted L_2 -projection of $\mathbf{Z}$ onto $(\overline{\mathcal{T}}_{g_o})^p$, and $M(t)$ is the associated counting-process martingale. The semiparametric information bound for θ_o is

$$I(\theta_o) = \mathbb{E} \{ \ell_{\theta_o}^*(T, \mathbf{X}, \mathbf{Z}, \Delta)^{\otimes 2} \} = \mathbb{E} [\Delta \{ \mathbf{Z} - \mathbf{g}^*(T, \mathbf{X}) \}^{\otimes 2}].$$

Consequently, the asymptotic variance of any regular asymptotically linear (RAL) estimator of θ_o is bounded below by $I(\theta_o)^{-1}$ in the Loewner sense.

PROOF OF THEOREM 3. Let $\mathbf{u} \in \mathbb{R}^p$ and let $\{g_s : s \in (-1, 1)\} \in \mathcal{H}_{g_o}$ be a one-dimensional regular submodel through g_o , with

$$\dot{g}_0 = \left. \frac{\partial g_s}{\partial s} \right|_{s=0} \in \mathcal{T}_{g_o}.$$

Consider the semiparametric submodel

$$\{\mathbb{P}_{\theta_o + s\mathbf{u}, g_s} : s \in (-1, 1)\}.$$

For one observation $(T, \Delta, \mathbf{X}, \mathbf{Z})$,

$$\ell(\theta, g) = \Delta \{g(T, \mathbf{X}) + \theta^\top \mathbf{Z}\} - \int_0^\tau Y(t) \exp\{g(t, \mathbf{X}) + \theta^\top \mathbf{Z}\} dt.$$

Differentiating at $s = 0$ gives

$$\left. \frac{d}{ds} \ell(\boldsymbol{\theta}_o + s\mathbf{u}, g_s) \right|_{s=0} = \mathbf{u}^\top \dot{\ell}_{\theta_o} + \dot{\ell}_{g_o, \dot{g}_o},$$

where

$$\dot{\ell}_{\theta_o} = \int_0^\tau \mathbf{Z} dM(t), \quad \dot{\ell}_{g_o, g} = \int_0^\tau g(t, \mathbf{X}) dM(t).$$

Hence the nuisance score space is

$$\mathcal{T}_{g_o}^{\text{nuis}} = \{\dot{\ell}_{g_o, g} : g \in \mathcal{T}_{g_o}\},$$

and the efficient score is

$$\ell_{\theta_o}^* = \dot{\ell}_{\theta_o} - \Pi_{g_o}(\dot{\ell}_{\theta_o}),$$

where Π_{g_o} denotes projection onto $\overline{\mathcal{T}_{g_o}^{\text{nuis}}}$ in $L_2(\mathbb{P}_0)$.

By Lemma 1 of [Sasieni \(1992\)](#), for any square-integrable predictable functions μ_1, μ_2 ,

$$E_0 \left[\left(\int_0^\tau \mu_1 dM \right) \left(\int_0^\tau \mu_2 dM \right) \right] = E_0[\Delta \mu_1(T, \mathbf{X}) \mu_2(T, \mathbf{X})].$$

Therefore the map

$$J : h \mapsto \int_0^\tau h(t, \mathbf{X}) dM(t)$$

is an isometry from $L_2(\mathbb{P}_0, \Delta)$ onto its image. Since $\overline{\mathcal{T}_{g_o}}$ is a closed linear subspace of $L_2(\mathbb{P}_0, \Delta)$ and $Z_j \in L_2(\mathbb{P}_0, \Delta)$ by Assumption A2, the Hilbert projection theorem implies that for each $j = 1, \dots, p$ there exists a unique $g_j^* \in \overline{\mathcal{T}_{g_o}}$ such that

$$\Pi_{g_o}(\dot{\ell}_{\theta_o, j}) = \dot{\ell}_{g_o, g_j^*}.$$

Let $\mathbf{g}^* = (g_1^*, \dots, g_p^*)^\top$. Then

$$\ell_{\theta_o}^*(T, \mathbf{X}, \mathbf{Z}, \Delta) = \int_0^\tau \{\mathbf{Z} - \mathbf{g}^*(t, \mathbf{X})\} dM(t).$$

The orthogonality condition for the projection is

$$E_0 \left[\{\dot{\ell}_{\theta_o, j} - \dot{\ell}_{g_o, g_j^*}\} \dot{\ell}_{g_o, g} \right] = 0 \quad \forall g \in \mathcal{T}_{g_o},$$

which, by the isometry above, is equivalent to

$$E_0 \left[\Delta \{Z_j - g_j^*(T, \mathbf{X})\} g(T, \mathbf{X}) \right] = 0 \quad \forall g \in \mathcal{T}_{g_o}.$$

By density, the same holds for all $g \in \overline{\mathcal{T}_{g_o}}$. Thus g_j^* is the Δ -weighted $L_2(\mathbb{P}_0)$ projection of Z_j onto $\overline{\mathcal{T}_{g_o}}$, and equivalently

$$\mathbf{g}^* \in \arg \min_{\mathbf{g} \in (\overline{\mathcal{T}_{g_o}})^p} E_0 \left[\Delta \|\mathbf{Z} - \mathbf{g}(T, \mathbf{X})\|_2^2 \right].$$

Finally, by the martingale isometry,

$$I(\boldsymbol{\theta}_o) = E_0[\ell_{\theta_o}^{*\otimes 2}] = E_0[\Delta \{\mathbf{Z} - \mathbf{g}^*(T, \mathbf{X})\}^{\otimes 2}].$$

Finally, under the regularity conditions in Assumptions (A1)–(A4), existence and uniqueness of the weighted projection $\mathbf{g}^*$ in $\overline{\mathcal{T}_{g_o}}^p$ follow from the approximation theorems of [Stone \(1985\)](#) and the general semiparametric projection theory in Appendix A.4 of [Bickel et al. \(1993\)](#). Therefore, both the efficient score and the efficient information matrix are well-defined and given by the formulas above. $\square$

THEOREM 4. *Under Assumptions A1–A5, suppose that the efficient information matrix $I(\boldsymbol{\theta}_o)$ is nonsingular and that $\sqrt{n}\tau_n^2 \rightarrow 0$ as $n \rightarrow \infty$, where $\tau_n = \gamma_n \log^2 n$. Then,*

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) = I(\boldsymbol{\theta}_o)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \ell_{\boldsymbol{\theta}_o}^*(T_i, \mathbf{X}_i, \mathbf{Z}_i, \Delta_i) + o_p(1),$$

and hence

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) \xrightarrow{D} N(\mathbf{0}, I(\boldsymbol{\theta}_o)^{-1}).$$

PROOF OF THEOREM 4. For any $(\boldsymbol{\theta}, g)$, define the per-observation score

$$r_{\boldsymbol{\theta},g}(T, \mathbf{X}, \mathbf{Z}, \Delta) = \Delta \{\mathbf{Z} - \mathbf{g}^*(T, \mathbf{X})\} - \int_0^\tau \{\mathbf{Z} - \mathbf{g}^*(t, \mathbf{X})\} Y(t) \exp\{g(t, \mathbf{X}) + \boldsymbol{\theta}^\top \mathbf{Z}\} dt,$$

where $\mathbf{g}^* = (g_1^*, \dots, g_p^*)^\top$ is the p -dimensional minimizer from Theorem 3, which depends only on $(\boldsymbol{\theta}_o, g_o)$. By Theorem 3, at the true parameter $(\boldsymbol{\theta}_o, g_o)$ this coincides with the efficient score,

$$r_{\boldsymbol{\theta}_o, g_o}(T, \mathbf{X}, \mathbf{Z}, \Delta) = \ell_{\boldsymbol{\theta}_o}^*(T, \mathbf{X}, \mathbf{Z}, \Delta), \quad \mathbb{P} r_{\boldsymbol{\theta}_o, g_o} = \mathbf{0},$$

and the efficient information matrix is

$$I(\boldsymbol{\theta}_o) = \mathbb{P} \{r_{\boldsymbol{\theta}_o, g_o}^{\otimes 2}\} = \mathbb{E}[\Delta \{\mathbf{Z} - \mathbf{g}^*(T, \mathbf{X})\}^{\otimes 2}].$$

The empirical efficient score at $(\boldsymbol{\theta}, g)$ is $\mathbb{P}_n r_{\boldsymbol{\theta},g}$.

In the following it will be shown that $\mathbb{P}_n r_{\hat{\boldsymbol{\theta}}, \hat{g}} = \mathbf{0}$. For the j -th coordinate of $r_{\boldsymbol{\theta},g}$, we have

$$\begin{aligned} r_{\boldsymbol{\theta},g}(T, \mathbf{X}, \mathbf{Z}, \Delta; j) &= \Delta \{Z_j - g_j^*(T, \mathbf{X})\} - \int_0^\tau \{Z_j - g_j^*(t, \mathbf{X})\} Y(t) \exp\{g(t, \mathbf{X}) + \boldsymbol{\theta}^\top \mathbf{Z}\} dt \\ &= \left[\Delta Z_j - \int_0^\tau Z_j Y(t) \exp\{g(t, \mathbf{X}) + \boldsymbol{\theta}^\top \mathbf{Z}\} dt \right] \\ &\quad - \left[\Delta g_j^*(T, \mathbf{X}) - \int_0^\tau g_j^*(t, \mathbf{X}) Y(t) \exp\{g(t, \mathbf{X}) + \boldsymbol{\theta}^\top \mathbf{Z}\} dt \right]. \end{aligned}$$

Therefore,

$$\begin{aligned} \mathbb{P}_n[r_{\boldsymbol{\theta},g}(\cdot; j)] &= \mathbb{P}_n \left[\Delta Z_j - \int_0^\tau Z_j Y(t) \exp\{g(t, \mathbf{X}) + \boldsymbol{\theta}^\top \mathbf{Z}\} dt \right] \\ &\quad - \mathbb{P}_n \left[\Delta g_j^*(T, \mathbf{X}) - \int_0^\tau g_j^*(t, \mathbf{X}) Y(t) \exp\{g(t, \mathbf{X}) + \boldsymbol{\theta}^\top \mathbf{Z}\} dt \right] \\ &= \dot{\ell}_{n,\boldsymbol{\theta},j}(\boldsymbol{\theta}, g) - \dot{\ell}_{n,g}(\boldsymbol{\theta}, g)[g_j^*], \end{aligned}$$

where

$$\dot{\ell}_{n,\boldsymbol{\theta}}(\boldsymbol{\theta}, g) = \mathbb{P}_n \left[\Delta \mathbf{Z} - \int_0^\tau \mathbf{Z} Y(t) \exp\{g(t, \mathbf{X}) + \boldsymbol{\theta}^\top \mathbf{Z}\} dt \right],$$

and, for any admissible nuisance direction $\mu \in \overline{\mathcal{T}}_{g_o}$, the directional Gateaux derivative score

$$\dot{\ell}_{n,g}(\boldsymbol{\theta}, g)[\mu] = \mathbb{P}_n \left[\Delta \mu(T, \mathbf{X}) - \int_0^\tau \mu(t, \mathbf{X}) Y(t) \exp\{g(t, \mathbf{X}) + \boldsymbol{\theta}^\top \mathbf{Z}\} dt \right].$$

Thus the projected score is the difference between the ordinary parametric score and the nuisance directional score taken in the direction $\mathbf{g}^*$. Since $(\hat{\boldsymbol{\theta}}, \hat{g})$ maximizes $\ell_n(\boldsymbol{\theta}, g)$, the

empirical first-order conditions give $\dot{\ell}_{n,\theta}(\hat{\boldsymbol{\theta}}, \hat{g}) = \mathbf{0}$ and $\dot{\ell}_{n,g}(\hat{\boldsymbol{\theta}}, \hat{g})[\mu] = 0$ for every admissible scalar nuisance direction $\mu \in \overline{\mathcal{T}}_{g_o}$. Since each g_j^* belongs to $\overline{\mathcal{T}}_{g_o}$ by Theorem 3, applying this with $\mu = g_j^*$ yields

$$\mathbb{P}_n r_{\hat{\boldsymbol{\theta}}, \hat{g}}(\cdot; j) = \dot{\ell}_{n,\theta,j}(\hat{\boldsymbol{\theta}}, \hat{g}) - \dot{\ell}_{n,g}(\hat{\boldsymbol{\theta}}, \hat{g})[g_j^*] = 0,$$

for every $j = 1, \dots, p$. Hence $\mathbb{P}_n r_{\hat{\boldsymbol{\theta}}, \hat{g}} = \mathbf{0}$.

Now, write

$$\mathbb{P}_n \{r_{\hat{\boldsymbol{\theta}}, \hat{g}} - r_{\theta_o, g_o}\} = \mathbb{P} \{r_{\hat{\boldsymbol{\theta}}, \hat{g}} - r_{\theta_o, g_o}\} + (\mathbb{P}_n - \mathbb{P}) \{r_{\hat{\boldsymbol{\theta}}, \hat{g}} - r_{\theta_o, g_o}\}.$$

By Theorem 1, we have $d(\hat{\eta}, \eta_o) = O_p(\tau_n)$, where $\eta = (\boldsymbol{\theta}, g)$ and $\eta_o = (\boldsymbol{\theta}_o, g_o)$. Using the same entropy bound and maximal inequality as in Lemma 10, one shows that the class

$$\mathcal{R}_n = \{r_{\theta, g} : d(\eta, \eta_o) \leq C\tau_n\}$$

is stochastically equicontinuous, and in particular

$$\sup_{r \in \mathcal{R}_n} |(\mathbb{P}_n - \mathbb{P})r| = o_p(n^{-1/2}).$$

Since $(\hat{\boldsymbol{\theta}}, \hat{g})$ lies in this neighbourhood with probability tending to one, we obtain

$$(\mathbb{P}_n - \mathbb{P}) \{r_{\hat{\boldsymbol{\theta}}, \hat{g}} - r_{\theta_o, g_o}\} = o_p(n^{-1/2}).$$

Using $\mathbb{P}_n r_{\hat{\boldsymbol{\theta}}, \hat{g}} = \mathbf{0}$, this identity can be written as

$$(A.14) \quad \mathbb{P} \{r_{\hat{\boldsymbol{\theta}}, \hat{g}} - r_{\theta_o, g_o}\} = -\mathbb{P}_n r_{\theta_o, g_o} + o_p(n^{-1/2}).$$

Consider the map

$$m(\boldsymbol{\theta}, g) = \mathbb{P} r_{\boldsymbol{\theta}, g}.$$

Since $m(\boldsymbol{\theta}_o, g_o) = \mathbf{0}$ and $r_{\theta, g}$ is smooth in $(\boldsymbol{\theta}, g)$ in a neighbourhood of $(\boldsymbol{\theta}_o, g_o)$, a first-order expansion gives

$$m(\hat{\boldsymbol{\theta}}, \hat{g}) - m(\boldsymbol{\theta}_o, g_o) = \dot{m}_{\boldsymbol{\theta}}(\boldsymbol{\theta}_o, g_o)(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) + \dot{m}_g(\boldsymbol{\theta}_o, g_o)[\hat{g} - g_o] + R_n,$$

where $\dot{m}_{\boldsymbol{\theta}}(\boldsymbol{\theta}_o, g_o)$ is the derivative of m with respect to $\boldsymbol{\theta}$ at $(\boldsymbol{\theta}_o, g_o)$, $\dot{m}_g(\boldsymbol{\theta}_o, g_o)[\cdot]$ is the Gateaux derivative in the g -direction, and R_n is a second-order remainder.

Differentiating r_{θ, g_o} with respect to $\boldsymbol{\theta}$ and using the representation of the efficient score in Theorem 3 together with the martingale isometry, one obtains

$$\dot{m}_{\boldsymbol{\theta}}(\boldsymbol{\theta}_o, g_o) = \left. \frac{\partial}{\partial \boldsymbol{\theta}} \right|_{\boldsymbol{\theta}=\boldsymbol{\theta}_o} \mathbb{P} r_{\boldsymbol{\theta}, g_o} = -I(\boldsymbol{\theta}_o).$$

Hence the contribution of $\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o$ in the expansion is $-I(\boldsymbol{\theta}_o)(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o)$.

Fix a deterministic scalar direction $\mu \in \overline{\mathcal{T}}_{g_o}$ and consider $g_s = g_o + s\mu$. Differentiating under the integral sign and applying the martingale compensation identity yields

$$\left. \frac{d}{ds} \right|_{s=0} m(\boldsymbol{\theta}_o, g_s) = -\mathbb{E}[\Delta\{\mathbf{Z} - \mathbf{g}^*(T, \mathbf{X})\}\mu(T, \mathbf{X})].$$

By Theorem 3, $\mathbf{g}^*$ is the coordinatewise Δ -weighted $L^2(\mathbb{P})$ -projection of $\mathbf{Z}$ onto the nuisance tangent space $(\overline{\mathcal{T}}_{g_o})^p$. Therefore,

$$\mathbb{E}[\Delta\{\mathbf{Z} - \mathbf{g}^*(T, \mathbf{X})\}\mu(T, \mathbf{X})] = \mathbf{0} \quad \text{for all } \mu \in \overline{\mathcal{T}}_{g_o},$$

and hence $\dot{m}_g(\boldsymbol{\theta}_o, g_o)[\mu] = \mathbf{0}$ for every $\mu \in \bar{\mathcal{T}}_{g_o}$. Arguing as in [Zhong, Mueller and Wang \(2022\)](#) (Lemma 5), the sieve score equation together with the tangent-space approximation implies that

$$\dot{m}_g(\boldsymbol{\theta}_o, g_o)[\hat{g} - g_o] = o_p(n^{-1/2}).$$

The map $g \mapsto m(\boldsymbol{\theta}_o, g)$ is twice Fréchet differentiable, and its second derivative is bounded by a constant multiple of $\mathbb{E}[\Delta\{\hat{g}(T, \mathbf{X}) - g_o(T, \mathbf{X})\}^2]$. Consequently,

$$|R_n| \lesssim \mathbb{E}[\Delta\{\hat{g}(T, \mathbf{X}) - g_o(T, \mathbf{X})\}^2] = O_p(\tau_n^2),$$

where Theorem 1 gives $\|\hat{g} - g_o\|_{L^2(\Delta)} = O_p(\tau_n)$. Our assumption $\sqrt{n}\tau_n^2 \rightarrow 0$ implies $\tau_n^2 = o(n^{-1/2})$, so $R_n = o_p(n^{-1/2})$.

Combining the pieces above, the expansion of $m(\hat{\boldsymbol{\theta}}, \hat{g})$ becomes

$$\mathbb{P}\{r_{\hat{\boldsymbol{\theta}}, \hat{g}} - r_{\boldsymbol{\theta}_o, g_o}\} = -I(\boldsymbol{\theta}_o)(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) + o_p(n^{-1/2}).$$

Substituting this into (A.14) yields

$$-I(\boldsymbol{\theta}_o)(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) = -\mathbb{P}_n r_{\boldsymbol{\theta}_o, g_o} + o_p(n^{-1/2}),$$

or, equivalently,

$$\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o = I(\boldsymbol{\theta}_o)^{-1} \mathbb{P}_n r_{\boldsymbol{\theta}_o, g_o} + o_p(n^{-1/2}).$$

Recalling that $r_{\boldsymbol{\theta}_o, g_o} = \ell_{\boldsymbol{\theta}_o}^*$ from Theorem 3, we obtain the stochastic expansion

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) = I(\boldsymbol{\theta}_o)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \ell_{\boldsymbol{\theta}_o}^*(T_i, \mathbf{X}_i, \mathbf{Z}_i, \Delta_i) + o_p(1).$$

The efficient score $\ell_{\boldsymbol{\theta}_o}^*$ has mean zero and covariance $I(\boldsymbol{\theta}_o)$, again by Theorem 3. By the classical central limit theorem,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \ell_{\boldsymbol{\theta}_o}^*(T_i, \mathbf{X}_i, \mathbf{Z}_i, \Delta_i) \xrightarrow{D} N(\mathbf{0}, I(\boldsymbol{\theta}_o)).$$

Slutsky's theorem then gives

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) \xrightarrow{D} N(\mathbf{0}, I(\boldsymbol{\theta}_o)^{-1}),$$

which completes the proof. $\square$

LEMMA 14. *Let $0 = t_0 < t_1 < \dots < t_{m_n} = \tau$ be a grid with mesh size $\delta_n = \max_{1 \leq j \leq m_n} (t_j - t_{j-1})$, and suppose that the grid contains all observed times $T_1, \dots, T_n$. For $\eta = (\boldsymbol{\theta}, g)$, let $\mathcal{N}_n = \{\eta : d(\eta, \eta_o) \leq M\tau_n\}$ for some fixed $M > 0$. Assume A2–A3. Also define the uniform time-modulus*

$$\omega_n(\delta) = \sup_{\eta \in \mathcal{N}_n} \sup_{\mathbf{X} \in \mathcal{X}} \sup_{\substack{s, t \in [0, \tau] \\ |t-s| \leq \delta}} |g(t, \mathbf{X}) - g(s, \mathbf{X})|.$$

Then there exist constants $C_1, C_2 > 0$, depending only on the bounds in A2–A3, such that

$$\sup_{\eta \in \mathcal{N}_n} |\tilde{\ell}_n(\eta) - \ell_n(\eta)| \leq C_1 \omega_n(\delta_n),$$

and

$$\sup_{\eta \in \mathcal{N}_n} \|\nabla_{\boldsymbol{\theta}} \tilde{\ell}_n(\eta) - \nabla_{\boldsymbol{\theta}} \ell_n(\eta)\| \leq C_2 \omega_n(\delta_n).$$

In particular, if $\omega_n(\delta_n) = O(\delta_n)$, then both differences are $O(\delta_n)$ uniformly on $\mathcal{N}_n$.

PROOF OF LEMMA 14. Fix $\eta = (\boldsymbol{\theta}, g) \in \mathcal{N}_n$. Recall

$$\ell_n(\eta) = \frac{1}{n} \sum_{i=1}^n \left[\Delta_i \chi_\eta(T_i, \mathbf{X}_i, \mathbf{Z}_i) - \int_0^\tau Y_i(t) e^{\chi_\eta(t, \mathbf{X}_i, \mathbf{Z}_i)} dt \right],$$

and

$$\tilde{\ell}_n(\eta) = \frac{1}{n} \sum_{i=1}^n \sum_{j: Y_i(t_{j-1})=1} \left[\Delta_{ij} \chi_\eta(t_j, \mathbf{X}_i, \mathbf{Z}_i) - \{\min(T_i, t_j) - t_{j-1}\} e^{\chi_\eta(t_j, \mathbf{X}_i, \mathbf{Z}_i)} \right],$$

where $\chi_\eta(t, \mathbf{X}, \mathbf{Z}) = g(t, \mathbf{X}) + \boldsymbol{\theta}^\top \mathbf{Z}$. Because the grid contains all observed times, for each i there exists an index $j(i)$ such that $T_i = t_{j(i)}$. Hence

$$\sum_{j: Y_i(t_{j-1})=1} \Delta_{ij} \chi_\eta(t_j, \mathbf{X}_i, \mathbf{Z}_i) = \Delta_i \chi_\eta(T_i, \mathbf{X}_i, \mathbf{Z}_i),$$

so the event contribution is represented exactly. Therefore

$$\tilde{\ell}_n(\eta) - \ell_n(\eta) = -\frac{1}{n} \sum_{i=1}^n R_i(\eta),$$

where

$$R_i(\eta) = \sum_{j: Y_i(t_{j-1})=1} \{\min(T_i, t_j) - t_{j-1}\} e^{\chi_\eta(t_j, \mathbf{X}_i, \mathbf{Z}_i)} - \int_0^\tau Y_i(t) e^{\chi_\eta(t, \mathbf{X}_i, \mathbf{Z}_i)} dt.$$

For each i , define $\psi_{i,\eta}(t) = Y_i(t) e^{\chi_\eta(t, \mathbf{X}_i, \mathbf{Z}_i)}$, $t \in [0, \tau]$. On any interval $(t_{j-1}, t_j]$ such that $Y_i(t_{j-1}) = 1$, we have $Y_i(t) = 1$ for $t \in [t_{j-1}, \min(T_i, t_j)]$, and thus

$$\int_{t_{j-1}}^{\min(T_i, t_j)} \psi_{i,\eta}(t) dt = \int_{t_{j-1}}^{\min(T_i, t_j)} e^{\chi_\eta(t, \mathbf{X}_i, \mathbf{Z}_i)} dt.$$

Since Θ is compact, $\|\mathbf{Z}\| \leq B_{\mathbf{Z}}$ a.s. by A2, and $\|g\|_\infty \leq D_n = O(1)$ by A3, there exists a constant $C < \infty$ such that uniformly over $i, \eta \in \mathcal{N}_n$, and $t \in [0, \tau]$,

$$e^{\chi_\eta(t, \mathbf{X}_i, \mathbf{Z}_i)} \leq C.$$

Moreover, by the mean-value theorem,

$$|e^{\chi_\eta(t, \mathbf{X}_i, \mathbf{Z}_i)} - e^{\chi_\eta(s, \mathbf{X}_i, \mathbf{Z}_i)}| \leq C |g(t, \mathbf{X}_i) - g(s, \mathbf{X}_i)| \leq C \omega_n(\delta_n)$$

whenever $|t - s| \leq \delta_n$. Therefore, for each interval,

$$\left| \int_{t_{j-1}}^{\min(T_i, t_j)} e^{\chi_\eta(t, \mathbf{X}_i, \mathbf{Z}_i)} dt - \{\min(T_i, t_j) - t_{j-1}\} e^{\chi_\eta(t_j, \mathbf{X}_i, \mathbf{Z}_i)} \right| \leq (t_j - t_{j-1}) C \omega_n(\delta_n).$$

Summing over j and using $\sum_j (t_j - t_{j-1}) = \tau$, we obtain

$$|R_i(\eta)| \leq C \tau \omega_n(\delta_n)$$

uniformly in i and $\eta \in \mathcal{N}_n$. Hence

$$\sup_{\eta \in \mathcal{N}_n} |\tilde{\ell}_n(\eta) - \ell_n(\eta)| \leq \frac{1}{n} \sum_{i=1}^n \sup_{\eta \in \mathcal{N}_n} |R_i(\eta)| \leq C_1 \omega_n(\delta_n)$$

for some constant $C_1 > 0$.

Next consider the $\boldsymbol{\theta}$ -scores. Since g does not depend on θ ,

$$\nabla_{\boldsymbol{\theta}} \ell_n(\eta) = \frac{1}{n} \sum_{i=1}^n \left[\Delta_i \mathbf{Z}_i - \int_0^{\tau} \mathbf{Z}_i Y_i(t) e^{\chi_{\eta}(t, \mathbf{X}_i, \mathbf{Z}_i)} dt \right],$$

and

$$\nabla_{\boldsymbol{\theta}} \tilde{\ell}_n(\eta) = \frac{1}{n} \sum_{i=1}^n \left[\Delta_i \mathbf{Z}_i - \sum_{j: Y_i(t_{j-1})=1} \{ \min(T_i, t_j) - t_{j-1} \} \mathbf{Z}_i e^{\chi_{\eta}(t_j, \mathbf{X}_i, \mathbf{Z}_i)} \right].$$

Therefore,

$$\nabla_{\boldsymbol{\theta}} \tilde{\ell}_n(\eta) - \nabla_{\boldsymbol{\theta}} \ell_n(\eta) = -\frac{1}{n} \sum_{i=1}^n \mathbf{Z}_i R_i(\eta).$$

Using $\|\mathbf{Z}_i\| \leq B_Z$ a.s. from A2 together with the bound on $R_i(\eta)$,

$$\sup_{\eta \in \mathcal{N}_n} \|\nabla_{\boldsymbol{\theta}} \tilde{\ell}_n(\eta) - \nabla_{\boldsymbol{\theta}} \ell_n(\eta)\| \leq \frac{1}{n} \sum_{i=1}^n \|\mathbf{Z}_i\| \sup_{\eta \in \mathcal{N}_n} |R_i(\eta)| \leq B_Z C_1 \omega_n(\delta_n).$$

This proves the second bound with $C_2 = B_Z C_1$. $\square$

LEMMA 15. Let $S_n(\eta) = \nabla_{\boldsymbol{\theta}} \ell_n(\eta)$ and $\dot{S}_n(\eta) = \nabla_{\boldsymbol{\theta}} S_n(\eta)$ be the Hessian of the empirical log-likelihood with respect to $\boldsymbol{\theta}$. For a fixed $M > 0$, define the shrinking neighbourhood $\mathcal{N}_n = \{\eta : d(\eta, \eta_o) \leq M\tau_n\}$, where $\tau_n = \gamma_n \log^2 n$. Assume A1–A4. Then,

$$\sup_{\eta \in \mathcal{N}_n} \|\dot{S}_n(\eta) + I_0(\boldsymbol{\theta}_o)\| \xrightarrow{P} 0,$$

where

$$I_0(\boldsymbol{\theta}_o) = \mathbb{E}_0 \left[\int_0^{\tau} \mathbf{Z} \mathbf{Z}^{\top} Y(t) \exp\{\boldsymbol{\theta}_o^{\top} \mathbf{Z} + g_o(t, \mathbf{X})\} dt \right] = \mathbb{E}_0 [\Delta \mathbf{Z} \mathbf{Z}^{\top}].$$

If, in addition, $I_0(\boldsymbol{\theta}_o)$ is nonsingular, then

$$\sup_{\eta \in \mathcal{N}_n} \|\dot{S}_n(\eta)^{-1}\| = O_p(1).$$

PROOF OF LEMMA 15. For a single observation $(T, \Delta, \mathbf{X}, \mathbf{Z})$, the per-observation log-likelihood is

$$\ell(\boldsymbol{\theta}, g) = \Delta \{\boldsymbol{\theta}^{\top} \mathbf{Z} + g(T, \mathbf{X})\} - \int_0^{\tau} Y(t) \exp\{\boldsymbol{\theta}^{\top} \mathbf{Z} + g(t, \mathbf{X})\} dt.$$

The corresponding $\boldsymbol{\theta}$ -score is

$$s(\boldsymbol{\theta}, g) = \nabla_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta}, g) = \Delta \mathbf{Z} - \int_0^{\tau} \mathbf{Z} Y(t) \exp\{\boldsymbol{\theta}^{\top} \mathbf{Z} + g(t, \mathbf{X})\} dt,$$

and the $\boldsymbol{\theta}$ -Hessian is

$$\dot{s}(\boldsymbol{\theta}, g) = \nabla_{\boldsymbol{\theta}} s(\boldsymbol{\theta}, g) = - \int_0^{\tau} \mathbf{Z} \mathbf{Z}^{\top} Y(t) \exp\{\boldsymbol{\theta}^{\top} \mathbf{Z} + g(t, \mathbf{X})\} dt.$$

Therefore,

$$\dot{S}_n(\eta) = \mathbb{P}_n \dot{s}(\boldsymbol{\theta}, g).$$

Define the matrix-valued per-observation integrand

$$J(\eta; \mathbf{W}) = \int_0^\tau \mathbf{Z}\mathbf{Z}^\top Y(t) \exp\{\boldsymbol{\theta}^\top \mathbf{Z} + g(t, \mathbf{X})\} dt, \quad \mathbf{W} = (T, \Delta, \mathbf{X}, \mathbf{Z}),$$

so that

$$\dot{S}_n(\eta) = -\mathbb{P}_n J(\eta; \cdot), \quad \mathbb{P} \dot{s}(\boldsymbol{\theta}, g) = -\mathbb{P} J(\eta; \cdot).$$

By Assumptions A2–A3, the covariates are bounded and the sieve is uniformly bounded, so there exists a constant $C < \infty$ such that

$$\sup_{\eta \in \mathcal{N}_n} \|J(\eta; \mathbf{W})\| \leq C \quad \text{a.s.}$$

Since p is fixed, it suffices to control each matrix entry separately. The class

$$\mathcal{J}_n = \{J(\eta; \cdot) : \eta \in \mathcal{N}_n\}$$

inherits the same bracketing entropy bound as the localized likelihood/score classes used in Lemma 10. Hence it is $\mathbb{P}$ –Glivenko–Cantelli, and

$$(A.15) \quad \sup_{\eta \in \mathcal{N}_n} \|\mathbb{P}_n J(\eta; \cdot) - \mathbb{P} J(\eta; \cdot)\| \xrightarrow{P} 0.$$

Next we show continuity of $\eta \mapsto \mathbb{P} J(\eta; \cdot)$ in the metric d . Let $\eta_1 = (\boldsymbol{\theta}_1, g_1)$ and $\eta_2 = (\boldsymbol{\theta}_2, g_2)$, and write $\phi(t, \mathbf{X}, \mathbf{Z}) = \chi_{\eta_1}(t, \mathbf{X}, \mathbf{Z}) - \chi_{\eta_2}(t, \mathbf{X}, \mathbf{Z})$. Then, by boundedness of $\mathbf{Z}$ and the mean-value theorem for the exponential, there exists a constant $C_1 > 0$ such that

$$\|J(\eta_1; \mathbf{W}) - J(\eta_2; \mathbf{W})\| \leq C_1 \int_0^\tau Y(t) |\phi(t, \mathbf{X}, \mathbf{Z})| dt.$$

Taking expectations and applying Cauchy–Schwarz,

$$\begin{aligned} \|\mathbb{P} J(\eta_1; \cdot) - \mathbb{P} J(\eta_2; \cdot)\| &\leq C_1 \mathbb{E}_0 \left[\int_0^\tau Y(t) |\phi(t, \mathbf{X}, \mathbf{Z})| dt \right] \\ &\leq C_1 \tau^{1/2} \left\{ \mathbb{E}_0 \left[\int_0^\tau Y(t) \phi(t, \mathbf{X}, \mathbf{Z})^2 dt \right] \right\}^{1/2}. \end{aligned}$$

Under Assumptions A1–A4, the true hazard $h_o(t | \mathbf{X}, \mathbf{Z}) = \exp\{\boldsymbol{\theta}_o^\top \mathbf{Z} + g_o(t, \mathbf{X})\}$ is bounded away from zero, say $h_o \geq c_h > 0$. Therefore,

$$\begin{aligned} \mathbb{E}_0 \left[\int_0^\tau Y(t) \phi(t, \mathbf{X}, \mathbf{Z})^2 dt \right] &\leq c_h^{-1} \mathbb{E}_0 \left[\int_0^\tau Y(t) h_o(t | \mathbf{X}, \mathbf{Z}) \phi(t, \mathbf{X}, \mathbf{Z})^2 dt \right] \\ &= c_h^{-1} \mathbb{E}_0 [\Delta \phi(T, \mathbf{X}, \mathbf{Z})^2] = c_h^{-1} d(\eta_1, \eta_2)^2, \end{aligned}$$

where the equality follows from the compensator identity. Hence

$$\|\mathbb{P} J(\eta_1; \cdot) - \mathbb{P} J(\eta_2; \cdot)\| \leq C_2 d(\eta_1, \eta_2)$$

for some constant $C_2 > 0$. In particular,

$$(A.16) \quad \sup_{\eta \in \mathcal{N}_n} \|\mathbb{P} J(\eta; \cdot) - \mathbb{P} J(\eta_o; \cdot)\| \leq C_2 M \tau_n \longrightarrow 0.$$

At the true parameter,

$$\begin{aligned} \mathbb{P} J(\eta_o; \cdot) &= \mathbb{E}_0 \left[\int_0^\tau \mathbf{Z}\mathbf{Z}^\top Y(t) \exp\{\boldsymbol{\theta}_o^\top \mathbf{Z} + g_o(t, \mathbf{X})\} dt \right] \\ &= \mathbb{E}_0 [\Delta \mathbf{Z}\mathbf{Z}^\top] = I_0(\boldsymbol{\theta}_o), \end{aligned}$$

again by the compensator identity. Therefore, using (A.15) and (A.16),

$$\begin{aligned} \sup_{\eta \in \mathcal{N}_n} \|\dot{S}_n(\eta) + I_0(\boldsymbol{\theta}_o)\| &= \sup_{\eta \in \mathcal{N}_n} \|-\mathbb{P}_n J(\eta; \cdot) + I_0(\boldsymbol{\theta}_o)\| \\ &\leq \sup_{\eta \in \mathcal{N}_n} \|\mathbb{P}_n J(\eta; \cdot) - \mathbb{P} J(\eta; \cdot)\| + \sup_{\eta \in \mathcal{N}_n} \|\mathbb{P} J(\eta; \cdot) - \mathbb{P} J(\eta_o; \cdot)\| \\ &\xrightarrow{P} 0. \end{aligned}$$

If $I_0(\boldsymbol{\theta}_o)$ is nonsingular, let $\lambda_0 = \lambda_{\min}(I_0(\boldsymbol{\theta}_o)) > 0$. On the event

$$\sup_{\eta \in \mathcal{N}_n} \|\dot{S}_n(\eta) + I_0(\boldsymbol{\theta}_o)\| < \lambda_0/2,$$

all matrices $-\dot{S}_n(\eta)$ are nonsingular and

$$\sup_{\eta \in \mathcal{N}_n} \|\dot{S}_n(\eta)^{-1}\| \leq 2/\lambda_0.$$

Since this event has probability tending to one, we conclude

$$\sup_{\eta \in \mathcal{N}_n} \|\dot{S}_n(\eta)^{-1}\| = O_p(1).$$

This completes the proof. $\square$

THEOREM 5. *Let $(\tilde{\boldsymbol{\theta}}, \tilde{g})$ be the numerically integrated likelihood estimator defined in (3.4), where the integral $\int_0^\tau Y_i(t) \exp\{g(t, \mathbf{X}_i) + \boldsymbol{\theta}^\top \mathbf{Z}_i\} dt$ is approximated on a time grid $0 = t_0 < t_1 < \dots < t_{m_n} = \tau$ with mesh size $\delta_n = \max_{1 \leq j \leq m_n} (t_j - t_{j-1})$ and suppose that the grid contains all observed times $T_1, \dots, T_n$. Let*

$$\omega_n(\delta) = \sup_{\eta \in \mathcal{N}_n} \sup_{\mathbf{X} \in \mathcal{X}} \sup_{\substack{s, t \in [0, \tau] \\ |t-s| \leq \delta}} |g(t, \mathbf{X}) - g(s, \mathbf{X})|, \quad \mathcal{N}_n = \{\eta : d(\eta, \eta_o) \leq M\tau_n\},$$

for some fixed $M > 0$. Assume that $\omega_n(\delta_n) = o(\tau_n^2)$, $\sqrt{n} \omega_n(\delta_n) \rightarrow 0$, and that the conditions of Theorems 1 and 4 hold. Then,

1. *The numerically integrated likelihood estimator attains the same convergence rate as the oracle estimator:*

$$d(\tilde{\eta}, \eta_o) = O_p(\tau_n), \quad \|\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_o\| = O_p(\tau_n).$$

2. *The parametric estimators are asymptotically equivalent:*

$$\sqrt{n}(\tilde{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}}) \xrightarrow{P} \mathbf{0}.$$

Consequently,

$$\sqrt{n}(\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) = \sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) + o_p(1) \xrightarrow{\mathcal{D}} N(\mathbf{0}, I(\boldsymbol{\theta}_o)^{-1}),$$

where $I(\boldsymbol{\theta}_o)$ is the efficient information matrix from Theorem 3.

PROOF OF THEOREM 5. Recall that $\ell_n(\eta)$ and $\tilde{\ell}_n(\eta)$ denote the exact and numerically integrated log-likelihoods, respectively, and $S_n(\eta) = \nabla_{\boldsymbol{\theta}} \ell_n(\eta)$ and $\tilde{S}_n(\eta) = \nabla_{\boldsymbol{\theta}} \tilde{\ell}_n(\eta)$ their $\boldsymbol{\theta}$ -scores. By Lemma 14, for all η in the shrinking neighborhood $\mathcal{N}_n = \{\eta : d(\eta, \eta_o) \leq M\tau_n\}$ for some fixed $M \geq 1$, we have the uniform bounds

$$\sup_{\eta \in \mathcal{N}_n} |\tilde{\ell}_n(\eta) - \ell_n(\eta)| \leq C_1 \omega_n(\delta_n),$$

and

$$\sup_{\eta \in \mathcal{N}_n} \|\tilde{S}_n(\eta) - S_n(\eta)\| \leq C_2 \omega_n(\delta_n).$$

Since $\eta_o \in \mathcal{N}_n$ for all n large enough, subtracting the values at η_o gives

$$(A.17) \quad \sup_{d(\eta, \eta_o) \leq M\tau_n} |\ell_n(\eta) - \tilde{\ell}_n(\eta) - \{\ell_n(\eta_o) - \tilde{\ell}_n(\eta_o)\}| = O_p(\omega_n(\delta_n)),$$

and

$$(A.18) \quad \sup_{d(\eta, \eta_o) \leq M\tau_n} \|\tilde{S}_n(\eta) - S_n(\eta)\| = O_p(\omega_n(\delta_n)).$$

These bounds reflect that the per-subject approximation error is controlled by the local time-modulus $\omega_n(\delta_n)$ from Lemma 14, while the network outputs are uniformly bounded because the sieve $\mathcal{G}(K, s, \mathbf{p}, D)$ is constructed with bounded weights.

By (A.17) and Theorem 1, on the ball $\{d(\eta, \eta_o) \leq \tau_n\}$ we have

$$\begin{aligned} \sup_{d(\eta, \eta_o) \leq \tau_n} |(\tilde{\ell}_n - \ell_o)(\eta) - (\tilde{\ell}_n - \ell_o)(\eta_o)| &\leq \sup_{d(\eta, \eta_o) \leq \tau_n} |(\ell_n - \ell_o)(\eta) - (\ell_n - \ell_o)(\eta_o)| \\ &\quad + \sup_{d(\eta, \eta_o) \leq \tau_n} |\tilde{\ell}_n(\eta) - \ell_n(\eta) - \{\tilde{\ell}_n(\eta_o) - \ell_n(\eta_o)\}| \\ &= O_p(\tau_n^2) + O_p(\omega_n(\delta_n)) = O_p(\tau_n^2), \end{aligned}$$

because $\omega_n(\delta_n) = o(\tau_n^2)$ by assumption. The population loss satisfies $\ell_o(\eta) - \ell_o(\eta_o) \asymp -d(\eta, \eta_o)^2$ by Lemma 3. Therefore, the same M-estimation argument used in the proof of Theorem 1 for the oracle estimator (via Theorem 3.4.1 in [van der Vaart and Wellner, 1996](#)) applies to $\tilde{\ell}_n$, yielding $d(\tilde{\eta}, \eta_o) = O_p(\tau_n)$ and $\|\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_o\| = O_p(\tau_n)$, which proves part 1.

By the score equations, $S_n(\tilde{\eta}) = \mathbf{0}$ and $\tilde{S}_n(\tilde{\eta}) = \mathbf{0}$. Therefore,

$$\mathbf{0} = \tilde{S}_n(\tilde{\eta}) - S_n(\tilde{\eta}) = \{\tilde{S}_n(\tilde{\eta}) - S_n(\tilde{\eta})\} + \{S_n(\tilde{\eta}) - S_n(\hat{\eta})\}.$$

Since $d(\hat{\eta}, \eta_o) = O_p(\tau_n)$ and $d(\tilde{\eta}, \eta_o) = O_p(\tau_n)$, there exists a constant $C > 0$ such that, with probability tending to one, $\{d(\hat{\eta}, \eta_o) \leq C\tau_n, d(\tilde{\eta}, \eta_o) \leq C\tau_n\} \subseteq \mathcal{N}_n$, where $M \geq C$. On this event, since the two estimators satisfy the score equations

$$S_n(\hat{\eta}) = \mathbf{0}, \quad \tilde{S}_n(\tilde{\eta}) = \mathbf{0},$$

we also get

$$\mathbf{0} = S_n(\tilde{\eta}) + R_n(\tilde{\eta}).$$

where $R_n(\eta) = \tilde{S}_n(\eta) - S_n(\eta)$. Subtracting $S_n(\hat{\eta}) = \mathbf{0}$ yields

$$(A.19) \quad \mathbf{0} = \{S_n(\tilde{\eta}) - S_n(\hat{\eta})\} + R_n(\tilde{\eta}).$$

In addition,

$$S_n(\tilde{\eta}) - S_n(\hat{\eta}) = \underbrace{S_n(\tilde{\boldsymbol{\theta}}, \tilde{g}) - S_n(\hat{\boldsymbol{\theta}}, \tilde{g})}_A + \underbrace{S_n(\hat{\boldsymbol{\theta}}, \tilde{g}) - S_n(\hat{\boldsymbol{\theta}}, \hat{g})}_B.$$

Thus (A.19) becomes $\mathbf{0} = A + B + R_n(\tilde{\eta})$. By the mean value theorem in $\boldsymbol{\theta}$, there exists $\boldsymbol{\theta}^\dagger$ on the segment between $\hat{\boldsymbol{\theta}}$ and $\tilde{\boldsymbol{\theta}}$ such that

$$A = \dot{S}_n(\boldsymbol{\theta}^\dagger, \tilde{g})(\tilde{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}}).$$

By Lemma 15, $\dot{S}_n(\boldsymbol{\theta}^\dagger, \tilde{g}) \xrightarrow{P} -I_0(\boldsymbol{\theta}_o)$ and $\|\dot{S}_n(\boldsymbol{\theta}^\dagger, \tilde{g})^{-1}\| = O_p(1)$. Substituting this into (A.19), we obtain

$$\dot{S}_n(\boldsymbol{\theta}^\dagger, \tilde{g})(\tilde{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}}) = -B - R_n(\tilde{\eta}).$$

and

$$(A.20) \quad \sqrt{n}(\tilde{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}}) = -\dot{S}_n(\boldsymbol{\theta}^\dagger, \tilde{g})^{-1} \left[\sqrt{n}B + \sqrt{n}R_n(\tilde{\eta}) \right].$$

Lemma 14 gives uniformly over $\mathcal{N}_n$,

$$\|R_n(\eta)\| \leq C\omega_n(\delta_n),$$

so

$$\sqrt{n}R_n(\tilde{\eta}) = O_p(\sqrt{n}\omega_n(\delta_n)) = o_p(1),$$

because $\sqrt{n}\omega_n(\delta_n) \rightarrow 0$ by assumption.

We decompose

$$B = S_n(\hat{\boldsymbol{\theta}}, \tilde{g}) - S_n(\hat{\boldsymbol{\theta}}, \hat{g}) = \{\mathbb{P}S(\hat{\boldsymbol{\theta}}, \tilde{g}) - \mathbb{P}S(\hat{\boldsymbol{\theta}}, \hat{g})\} + \{(\mathbb{P}_n - \mathbb{P})S(\hat{\boldsymbol{\theta}}, \tilde{g}) - (\mathbb{P}_n - \mathbb{P})S(\hat{\boldsymbol{\theta}}, \hat{g})\},$$

where $S(\eta; \mathbf{W})$ is the per-observation score in $\boldsymbol{\theta}$ and $S_n(\eta) = \mathbb{P}_n S(\eta; \cdot)$. The second-order expansion of the population score (see the proof of Theorem 4) yields

$$\|\mathbb{P}S(\hat{\boldsymbol{\theta}}, \tilde{g}) - \mathbb{P}S(\hat{\boldsymbol{\theta}}, \hat{g})\| \lesssim \|\tilde{g} - g_o\|_{L^2(\Delta)}^2 + \|\hat{g} - g_o\|_{L^2(\Delta)}^2 = O_p(\tau_n^2).$$

Moreover, Lemma 10 applied to the localized score class $\{S(\hat{\boldsymbol{\theta}}, g; \cdot) : d((\hat{\boldsymbol{\theta}}, g), \eta_o) \leq C\tau_n\}$ implies

$$\sup_{d((\hat{\boldsymbol{\theta}}, g), \eta_o) \leq C\tau_n} |(\mathbb{P}_n - \mathbb{P})S(\hat{\boldsymbol{\theta}}, g; \cdot)| = O_p(\tau_n^2),$$

and since both $(\hat{\boldsymbol{\theta}}, \tilde{g})$ and $(\hat{\boldsymbol{\theta}}, \hat{g})$ lie in this ball with probability $1 - o(1)$, we obtain

$$\|B\| = O_p(\tau_n^2), \quad \sqrt{n}B = o_p(1),$$

because $\sqrt{n}\tau_n^2 \rightarrow 0$. Using (A.20) together with $\dot{S}_n(\boldsymbol{\theta}^\dagger, \tilde{g})^{-1} = O_p(1)$, $\sqrt{n}B = o_p(1)$, and $\sqrt{n}R_n(\tilde{\eta}) = o_p(1)$, we obtain

$$\sqrt{n}(\tilde{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}}) = o_p(1).$$

Finally, combining this with Theorem 4, which states that $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o)$ is asymptotically normal $N(\mathbf{0}, I(\boldsymbol{\theta}_o)^{-1})$, and using Slutsky's theorem yields

$$\sqrt{n}(\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) = \sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_o) + \sqrt{n}(\tilde{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}}) \xrightarrow{d} N(\mathbf{0}, I(\boldsymbol{\theta}_o)^{-1}),$$

as claimed. $\square$

APPENDIX B: PROOFS OF SECTION 5

LEMMA 4. *Let P_ε be a regular submodel with score (5.6). Then*

$$(5.9) \quad \left. \frac{d}{d\varepsilon} H_\varepsilon(t \mid \mathbf{X}_0, \mathbf{Z}_0) \right|_{\varepsilon=0} = L_t(b) + \mathbf{a}^\top c_{0,t}(\mathbf{X}_0, \mathbf{Z}_0).$$

PROOF OF LEMMA 4. Along the joint submodel,

$$h_\varepsilon(u \mid \mathbf{X}_0, \mathbf{Z}_0) = h_o(u \mid \mathbf{X}_0, \mathbf{Z}_0) \exp \left\{ \varepsilon \mathbf{a}^\top (\mathbf{Z}_0 - \mathbf{g}^*(u, \mathbf{X}_0)) + \varepsilon b(u, \mathbf{X}_0) \right\}.$$

Differentiating at $\varepsilon = 0$ gives

$$\left. \frac{d}{d\varepsilon} h_\varepsilon(u \mid \mathbf{X}_0, \mathbf{Z}_0) \right|_{\varepsilon=0} = h_o(u \mid \mathbf{X}_0, \mathbf{Z}_0) \left\{ \mathbf{a}^\top (\mathbf{Z}_0 - \mathbf{g}^*(u, \mathbf{X}_0)) + b(u, \mathbf{X}_0) \right\}.$$

Integration over $[0, t]$ yields the stated result. $\square$

LEMMA 5. Fix $t \in [0, \tau]$ and let $(\mathbf{X}_0, \mathbf{Z}_0)$ be an independent draw from the covariate distribution, independent of the training sample. Assume A6–A7. Then ψ_t^0 is unique in $L^2(P_{\mathbf{X}, \mathbf{Z}}; \overline{\mathcal{T}}_{g_o})$ and the map $P \mapsto H_P(t \mid \mathbf{X}_0, \mathbf{Z}_0)$ is pathwise differentiable at $\mathbb{P}_0$ as an element of $L^2(P_{\mathbf{X}, \mathbf{Z}})$; that is, for every regular submodel $\mathbb{P}_\varepsilon$ through $\mathbb{P}_0$ with score $S(\mathbf{W}; \mathbf{a}, b)$,

$$\left. \frac{d}{d\varepsilon} H_\varepsilon(t \mid \mathbf{X}_0, \mathbf{Z}_0) \right|_{\varepsilon=0} = \mathbb{E}_0 \left\{ \phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) S(\mathbf{W}; \mathbf{a}, b) \mid \mathbf{X}_0, \mathbf{Z}_0 \right\} \quad \text{in } L^2(P_{\mathbf{X}, \mathbf{Z}}).$$

Equivalently, (5.2) can be stated as a unique (up to $P_{\mathbf{X}, \mathbf{Z}}$ -null sets) random element $\psi_t^0(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0) \in L^2(P_{\mathbf{X}, \mathbf{Z}}; \overline{\mathcal{T}}_{g_o})$ such that for all $b \in \overline{\mathcal{T}}_{g_o}$,

$$(5.10) \quad \mathbb{E}_{(\mathbf{X}_0, \mathbf{Z}_0)} \left[\left\{ \langle \psi_t^0(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0), b \rangle_0 - L_t(b) \right\}^2 \right] = 0.$$

Finally, an EIF for $H_o(t \mid \mathbf{X}_0, \mathbf{Z}_0)$ is

$$(5.11) \quad \phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) = \int_0^\tau \psi_t^0(u, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) dM(u) + c_{0,t}(\mathbf{X}_0, \mathbf{Z}_0)^\top I(\boldsymbol{\theta}_o)^{-1} \ell_{\theta_o}^*(\mathbf{W}).$$

PROOF OF LEMMA 5. By A7, $Q : \overline{\mathcal{T}}_{g_o} \rightarrow L^2(P_{\mathbf{X}, \mathbf{Z}})$ is a bounded linear operator. Next, we invoke assumption A7, there exists $\psi_t^0(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0) \in L^2(P_{\mathbf{X}, \mathbf{Z}}; \overline{\mathcal{T}}_{g_o})$ such that for all $b \in \overline{\mathcal{T}}_{g_o}$,

$$(B.1) \quad (Qb)(\mathbf{X}_0, \mathbf{Z}_0) = \langle \psi_t^0(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0), b \rangle_0 \quad \text{in } L^2(P_{\mathbf{X}, \mathbf{Z}}).$$

Uniqueness follows from the Riesz representation theorem in the Hilbert space $\overline{\mathcal{T}}_{g_o}$: if $\psi, \tilde{\psi} \in L^2(P_{\mathbf{X}, \mathbf{Z}}; \overline{\mathcal{T}}_{g_o})$ both satisfy (B.1), then $\langle \psi - \tilde{\psi}, b \rangle_0 = 0$ in $L^2(P_{\mathbf{X}, \mathbf{Z}})$ for all b , hence $\psi = \tilde{\psi}$ in $L^2(P_{\mathbf{X}, \mathbf{Z}}; \overline{\mathcal{T}}_{g_o})$.

For each $b \in \overline{\mathcal{T}}_{g_o}$, the martingale isometry associated with (5.1) yields

$$\langle \psi_t^0(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0), b \rangle_0 = \mathbb{E}_0 \left[\left(\int_0^\tau \psi_t^0(u, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) dM(u) \right) \left(\int_0^\tau b(u, \mathbf{X}) dM(u) \right) \mid \mathbf{X}_0, \mathbf{Z}_0 \right],$$

and since $\langle \psi_t^0, b \rangle_0 = L_t(b)$ in $L^2(P_{\mathbf{X}, \mathbf{Z}})$, the nuisance contribution to the canonical gradient in the $L^2(P_{\mathbf{X}, \mathbf{Z}})$ sense is $\int_0^\tau \psi_t^0(u, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) dM(u)$.

The parametric derivative term in (5.9) equals $\mathbf{a}^\top c_{0,t}(\mathbf{X}_0, \mathbf{Z}_0)$. Since $\ell_{\theta_o}^*$ is the efficient score and $I(\boldsymbol{\theta}_o) = \mathbb{E}_0 \{ \ell_{\theta_o}^*(\mathbf{W})^{\otimes 2} \}$, the corresponding efficient influence function for $\boldsymbol{\theta}_o$ is $I(\boldsymbol{\theta}_o)^{-1} \ell_{\theta_o}^*(\mathbf{W})$. Thus the parametric contribution to the EIF for $H_o(t \mid \mathbf{X}_0, \mathbf{Z}_0)$ is

$$c_{0,t}(\mathbf{X}_0, \mathbf{Z}_0)^\top I(\boldsymbol{\theta}_o)^{-1} \ell_{\theta_o}^*(\mathbf{W}).$$

Combining nuisance and parametric pieces gives (5.11). Finally, using (5.9), (5.10), and the score decomposition (5.6), one checks that

$$\left. \frac{d}{d\varepsilon} H_\varepsilon(t \mid \mathbf{X}_0, \mathbf{Z}_0) \right|_{\varepsilon=0} - \mathbb{E}_0 [\phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) S(\mathbf{W}; \mathbf{a}, b) \mid \mathbf{X}_0, \mathbf{Z}_0] = 0 \quad \text{in } L^2(P_{\mathbf{X}, \mathbf{Z}}),$$

which proves pathwise differentiability of $H_P(t \mid \mathbf{X}_0, \mathbf{Z}_0)$ as an $L^2(P_{\mathbf{X}, \mathbf{Z}})$ -valued functional. $\square$

LEMMA 6. *Under Assumptions A6–A8, the Riesz representer $\psi_t^0(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0)$ satisfying (5.10) admits the closed form*

$$(5.12) \quad \psi_t^0(u, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) = \mathbf{1}(u \leq t) \mathbf{1}(\mathbf{X} = \mathbf{X}_0) \frac{h_o(u | \mathbf{X}_0, \mathbf{Z}_0)}{\Pr(\mathbf{X} = \mathbf{X}_0) A_{\mathbf{X}_0}(u)}.$$

PROOF OF LEMMA 6. Since $\mathbf{X}$ takes values in the finite set $\{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(J)}\}$, every $b \in \overline{\mathcal{T}}_{g_o}$ admits the representation

$$b(u, \mathbf{X}) = \sum_{j=1}^J b_j(u) \mathbf{1}\{\mathbf{X} = \mathbf{x}^{(j)}\}, \quad b_j \in L_2([0, \tau]).$$

Let j_0 be such that $\mathbf{X}_0 = \mathbf{x}^{(j_0)}$. Using the inner product (5.1), for any $\psi \in \overline{\mathcal{T}}_{g_o}$ we obtain

$$\begin{aligned} \langle \psi, b \rangle_0 &= \mathbb{E}_0 \left[\int_0^\tau \psi(u, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) b(u, \mathbf{X}) Y(u) h_o(u | \mathbf{X}, \mathbf{Z}) du \right] \\ &= \sum_{j=1}^J \mathbb{E}_0 \left[\mathbf{1}\{\mathbf{X} = \mathbf{x}^{(j)}\} \int_0^\tau \psi(u, \mathbf{x}^{(j)}; \mathbf{X}_0, \mathbf{Z}_0) b_j(u) Y(u) h_o(u | \mathbf{X}, \mathbf{Z}) du \right] \\ &= \sum_{j=1}^J \mathbb{P}_0(\mathbf{X} = \mathbf{x}^{(j)}) \int_0^\tau \psi(u, \mathbf{x}^{(j)}; \mathbf{X}_0, \mathbf{Z}_0) b_j(u) \mathbb{E}_0 \left[Y(u) h_o(u | \mathbf{X}, \mathbf{Z}) | \mathbf{X} = \mathbf{x}^{(j)} \right] du \\ &= \sum_{j=1}^J \mathbb{P}_0(\mathbf{X} = \mathbf{x}^{(j)}) \int_0^\tau \psi(u, \mathbf{x}^{(j)}; \mathbf{X}_0, \mathbf{Z}_0) b_j(u) A_j(u) du. \end{aligned}$$

On the other hand, by definition of the evaluation functional L_t (as in (5.7)),

$$L_t(b) = \int_0^t h_o(u | \mathbf{X}_0, \mathbf{Z}_0) b(u, \mathbf{X}_0) du = \int_0^t h_o(u | \mathbf{X}_0, \mathbf{Z}_0) b_{j_0}(u) du.$$

Therefore, the Riesz equation $\langle \psi_t^0, b \rangle_0 = L_t(b)$ for all $b \in \overline{\mathcal{T}}_{g_o}$ implies that $\psi_t^0(u, \mathbf{x}^{(j)}; \mathbf{X}_0, \mathbf{Z}_0) = 0$ for $j \neq j_0$ (a.e. u), and for $j = j_0$,

$$\Pr(\mathbf{X} = \mathbf{X}_0) \int_0^\tau \psi_t^0(u, \mathbf{X}_0; \mathbf{X}_0, \mathbf{Z}_0) b_{j_0}(u) A_{\mathbf{X}_0}(u) du = \int_0^t h_o(u | \mathbf{X}_0, \mathbf{Z}_0) b_{j_0}(u) du, \quad \forall b_{j_0}.$$

Since this holds for all $b_{j_0} \in L_2([0, \tau])$, it follows that for a.e. $u \in [0, \tau]$,

$$\Pr(\mathbf{X} = \mathbf{X}_0) A_{\mathbf{X}_0}(u) \psi_t^0(u, \mathbf{X}_0; \mathbf{X}_0, \mathbf{Z}_0) = \mathbf{1}(u \leq t) h_o(u | \mathbf{X}_0, \mathbf{Z}_0).$$

Thus,

$$\psi_t^0(u, \mathbf{X}_0; \mathbf{X}_0, \mathbf{Z}_0) = \mathbf{1}(u \leq t) \frac{h_o(u | \mathbf{X}_0, \mathbf{Z}_0)}{\Pr(\mathbf{X} = \mathbf{X}_0) A_{\mathbf{X}_0}(u)} \quad \text{for a.e. } u.$$

Combining this with the fact that $\psi_t^0(u, \mathbf{x}^{(j)}; \mathbf{X}_0, \mathbf{Z}_0) = 0$ for all $j \neq j_0$ (a.e. u), we obtain for general $\mathbf{X}$,

$$\psi_t^0(u, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) = \mathbf{1}(\mathbf{X} = \mathbf{X}_0) \psi_t^0(u, \mathbf{X}_0; \mathbf{X}_0, \mathbf{Z}_0) = \mathbf{1}(u \leq t) \mathbf{1}(\mathbf{X} = \mathbf{X}_0) \frac{h_o(u | \mathbf{X}_0, \mathbf{Z}_0)}{\Pr(\mathbf{X} = \mathbf{X}_0) A_{\mathbf{X}_0}(u)},$$

which is (5.12). $\square$

LEMMA 7. For $i \in \mathcal{I}_k$ define the plug-in martingale

$$\widehat{M}_i^{(-k)}(u) = N_i(u) - \int_0^u Y_i(s) \widehat{h}^{(-k)}(s | \mathbf{X}_i, \mathbf{Z}_i) ds,$$

and the plug-in efficient score

$$(5.13) \quad \widehat{\ell}_{\theta_o}^{*,(-k)}(\mathbf{W}_i) = \int_0^\tau \{\mathbf{Z}_i - \widehat{\mathbf{g}}^{*,(-k)}(u, \mathbf{X}_i)\} d\widehat{M}_i^{(-k)}(u).$$

Assume A1–A7 and A9. Then

$$(5.14) \quad \frac{1}{\sqrt{n}} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \left\{ \widehat{\ell}_{\theta_o}^{*,(-k)}(\mathbf{W}_i) - \ell_{\theta_o}^*(\mathbf{W}_i) \right\} = o_p(1).$$

Consequently, in any root- n expansion in which the oracle score $\ell_{\theta_o}^*$ appears inside an empirical sum, it may be replaced by the cross-fitted plug-in score $\widehat{\ell}_{\theta_o}^{*,(-k)}$ without affecting the first-order limit.

PROOF OF LEMMA 7. Fix $k \in \{1, \dots, K\}$ and write $\mathcal{F}_{-k}$ for the sigma-field generated by the training sample $\{W_j : j \in \mathcal{I}_k^c\}$. By Assumption A9, conditional on $\mathcal{F}_{-k}$ the functions $\widehat{g}^{(-k)}$, $\widehat{\theta}^{(-k)}$ and $\widehat{\mathbf{g}}^{*,(-k)}$ are fixed, and the observations $\{\mathbf{W}_i : i \in \mathcal{I}_k\}$ are i.i.d. from $\mathbb{P}_0$.

For $i \in \mathcal{I}_k$ set

$$R_{i,k} = \widehat{\ell}_{\theta_o}^{*,(-k)}(\mathbf{W}_i) - \ell_{\theta_o}^*(\mathbf{W}_i).$$

Insert and subtract the intermediate integral $\int_0^\tau \{\mathbf{Z}_i - \widehat{\mathbf{g}}^{*,(-k)}(u, \mathbf{X}_i)\} dM_i(u)$ to obtain

$$R_{i,k} = \int_0^\tau \{\widehat{\mathbf{g}}^{*,(-k)}(u, \mathbf{X}_i) - \mathbf{g}^*(u, \mathbf{X}_i)\} dM_i(u) + \int_0^\tau \{\mathbf{Z}_i - \widehat{\mathbf{g}}^{*,(-k)}(u, \mathbf{X}_i)\} d(\widehat{M}_i^{(-k)} - M_i)(u).$$

Denote the first and second terms by $R_{i,k}^{(1)}$ and $R_{i,k}^{(2)}$, respectively. Consider $R_{i,k}^{(1)}$. Conditional on $\mathcal{F}_{-k}$, the integrand $\widehat{\mathbf{g}}^{*,(-k)} - \mathbf{g}^*$ is predictable, and M_i is a square-integrable martingale. Therefore, the martingale isometry yields

$$\begin{aligned} E_0 \left[(R_{i,k}^{(1)})^2 \mid \mathcal{F}_{-k} \right] &= E_0 \left[\int_0^\tau \{\widehat{\mathbf{g}}^{*,(-k)}(u, \mathbf{X}) - \mathbf{g}^*(u, \mathbf{X})\}^2 Y(u) h_o(u | \mathbf{X}, \mathbf{Z}) du \mid \mathcal{F}_{-k} \right] \\ &= \|\widehat{\mathbf{g}}^{*,(-k)} - \mathbf{g}^*\|_0^2. \end{aligned}$$

By Assumption A9, the right-hand side is $O_p(\tau_n^2)$ uniformly in k . Since $\{R_{i,k}^{(1)} : i \in \mathcal{I}_k\}$ are i.i.d. conditional on $\mathcal{F}_{-k}$, we have

$$\mathbb{V}_0 \left(\frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_k} R_{i,k}^{(1)} \mid \mathcal{F}_{-k} \right) = \frac{|\mathcal{I}_k|}{n} \mathbb{V}_0 \left(R_{1,k}^{(1)} \mid \mathcal{F}_{-k} \right) \leq \frac{|\mathcal{I}_k|}{n} E_0 \left[(R_{1,k}^{(1)})^2 \mid \mathcal{F}_{-k} \right] = O_p(\tau_n^2).$$

Chebyshev's inequality implies

$$\frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_k} R_{i,k}^{(1)} = O_p(\tau_n) = o_p(1),$$

because $\tau_n \rightarrow 0$. Next consider $R_{i,k}^{(2)}$. Since

$$\widehat{M}_i^{(-k)}(u) - M_i(u) = - \int_0^u Y_i(s) \left\{ \widehat{h}^{(-k)}(s | \mathbf{X}_i, \mathbf{Z}_i) - h_o(s | \mathbf{X}_i, \mathbf{Z}_i) \right\} ds,$$

we have

$$(B.2) \quad R_{i,k}^{(2)} = - \int_0^\tau \{ \mathbf{Z}_i - \widehat{\mathbf{g}}^{*,(-k)}(u, \mathbf{X}_i) \} Y_i(u) \left\{ \widehat{h}^{(-k)}(u | \mathbf{X}_i, \mathbf{Z}_i) - h_o(u | \mathbf{X}_i, \mathbf{Z}_i) \right\} du.$$

Write

$$(B.3) \quad \widehat{h}^{(-k)}(u | \mathbf{X}, \mathbf{Z}) - h_o(u | \mathbf{X}, \mathbf{Z}) = h_o(u | \mathbf{X}, \mathbf{Z}) \times \left[\exp \left\{ \widetilde{g}^{(-k)}(u, \mathbf{X}) - g_o(u, \mathbf{X}) + \mathbf{Z}^\top (\widetilde{\boldsymbol{\theta}}^{(-k)} - \boldsymbol{\theta}_o) \right\} - 1 \right].$$

By the mean value theorem, for each $(u, \mathbf{X}, \mathbf{Z})$ there exists a random $\xi^{(-k)}(u, \mathbf{X}, \mathbf{Z})$ between 0 and $\delta^{(-k)}(u, \mathbf{X}, \mathbf{Z}) = \widetilde{g}^{(-k)}(u, \mathbf{X}) - g_o(u, \mathbf{X}) + \mathbf{Z}^\top (\widetilde{\boldsymbol{\theta}}^{(-k)} - \boldsymbol{\theta}_o)$ such that

$$\widehat{h}^{(-k)}(u | \mathbf{X}, \mathbf{Z}) - h_o(u | \mathbf{X}, \mathbf{Z}) = h_o(u | \mathbf{X}, \mathbf{Z}) \exp \{ \xi^{(-k)}(u, \mathbf{X}, \mathbf{Z}) \} \delta^{(-k)}(u, \mathbf{X}, \mathbf{Z}).$$

By the bounded-sieve condition for $\widehat{g}^{(-k)}$, boundedness of g_o , boundedness of $\mathbf{Z}$, and boundedness of the parameter space for $\boldsymbol{\theta}$, there exists a deterministic constant $C < \infty$ such that

$$\sup_{u \leq \tau, \mathbf{X}, \mathbf{Z}} \left| \delta^{(-k)}(u, \mathbf{X}, \mathbf{Z}) \right| \leq C$$

with probability tending to one. Therefore, uniformly on this event, the exponential map admits the second-order expansion

$$e^x = 1 + x + R_C(x), \quad |R_C(x)| \leq C_1 x^2, \quad |x| \leq C,$$

for some finite constant C_1 . Hence

$$\widehat{h}^{(-k)}(u | \mathbf{X}, \mathbf{Z}) - h_o(u | \mathbf{X}, \mathbf{Z}) = h_o(u | \mathbf{X}, \mathbf{Z}) \delta^{(-k)}(u, \mathbf{X}, \mathbf{Z}) + h_o(u | \mathbf{X}, \mathbf{Z}) R^{(-k)}(u, \mathbf{X}, \mathbf{Z}),$$

where $R^{(-k)}(u, \mathbf{X}, \mathbf{Z}) = e^{\delta^{(-k)}(u, \mathbf{X}, \mathbf{Z})} - 1 - \delta^{(-k)}(u, \mathbf{X}, \mathbf{Z})$ and

$$|R^{(-k)}(u, \mathbf{X}, \mathbf{Z})| \leq C_1 (\delta^{(-k)}(u, \mathbf{X}, \mathbf{Z}))^2.$$

Substituting into (B.2) gives

$$R_{i,k}^{(2)} = - \int_0^\tau \{ \mathbf{Z}_i - \widehat{\mathbf{g}}^{*,(-k)}(u, \mathbf{X}_i) \} Y_i(u) h_o(u | \mathbf{X}_i, \mathbf{Z}_i) \delta^{(-k)}(u, \mathbf{X}_i, \mathbf{Z}_i) du + \widetilde{R}_{i,k},$$

where the remainder term $\widetilde{R}_{i,k}$ satisfies

$$(B.4) \quad E_0 \left[|\widetilde{R}_{i,k}| \mid \mathcal{F}_{-k} \right] \lesssim \|\widetilde{g}^{(-k)} - g_o\|_0^2 + \|\widetilde{\boldsymbol{\theta}}^{(-k)} - \boldsymbol{\theta}_o\|^2.$$

To verify (B.4), note first that

$$|\widetilde{R}_{i,k}| \lesssim \int_0^\tau |\mathbf{Z}_i - \widehat{\mathbf{g}}^{*,(-k)}(u, \mathbf{X}_i)| Y_i(u) h_o(u | \mathbf{X}_i, \mathbf{Z}_i) |\delta^{(-k)}(u, \mathbf{X}_i, \mathbf{Z}_i)|^2 du.$$

By boundedness of $\mathbf{Z}_i - \widehat{\mathbf{g}}^{*,(-k)}(u, \mathbf{X}_i)$ and assumptions A2 and A6, the factor

$$|\mathbf{Z}_i - \widehat{\mathbf{g}}^{*,(-k)}(u, \mathbf{X}_i)| Y_i(u) h_o(u | \mathbf{X}_i, \mathbf{Z}_i)$$

is bounded by a random quantity with finite expectation, uniformly over the relevant indices. Therefore, conditional on $\mathcal{F}_{-k}$,

$$E_0 \left[|\widetilde{R}_{i,k}| \mid \mathcal{F}_{-k} \right] \lesssim E_0 \left[\int_0^\tau |\delta^{(-k)}(u, \mathbf{X}_i, \mathbf{Z}_i)|^2 du \mid \mathcal{F}_{-k} \right].$$

Expanding $\delta^{(-k)}$ and using $(a + b)^2 \leq 2a^2 + 2b^2$, we obtain (B.4).

Now treat the leading linear term in $R_{i,k}^{(2)}$. Conditional on $\mathcal{F}_{-k}$, Cauchy–Schwarz and Assumption A9 imply that the conditional second moment of the integrand is bounded by a constant multiple of $\|\tilde{g}^{(-k)} - g_o\|_0^2 + \|\tilde{\boldsymbol{\theta}}^{(-k)} - \boldsymbol{\theta}_o\|^2$, and hence

$$E_0 \left[(R_{i,k}^{(2)} - \tilde{R}_{i,k})^2 \mid \mathcal{F}_{-k} \right] \lesssim \|\tilde{g}^{(-k)} - g_o\|_0^2 + \|\tilde{\boldsymbol{\theta}}^{(-k)} - \boldsymbol{\theta}_o\|^2 = O_p(\tau_n^2 + n^{-1}).$$

Since $\{R_{i,k}^{(2)} : i \in \mathcal{I}_k\}$ are i.i.d. conditional on $\mathcal{F}_{-k}$, it follows that

$$\mathbb{V}_0 \left(\frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_k} (R_{i,k}^{(2)} - \tilde{R}_{i,k}) \mid \mathcal{F}_{-k} \right) = \frac{|\mathcal{I}_k|}{n} \mathbb{V}_0(R_{1,k}^{(2)} - \tilde{R}_{1,k} \mid \mathcal{F}_{-k}) \lesssim \tau_n^2 + n^{-1},$$

and therefore

$$\frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_k} (R_{i,k}^{(2)} - \tilde{R}_{i,k}) = O_p(\tau_n + n^{-1/2}) = o_p(1).$$

It remains to control the remainder contribution coming from $\tilde{R}_{i,k}$. By Assumption A9, $\|\tilde{g}^{(-k)} - g_o\|_0 = O_p(\tau_n)$ and $\|\tilde{\boldsymbol{\theta}}^{(-k)} - \boldsymbol{\theta}_o\| = O_p(n^{-1/2})$, uniformly in k . Plugging these into (B.4) yields

$$E_0 \left[|\tilde{R}_{i,k}| \mid \mathcal{F}_{-k} \right] = O_p(\tau_n^2 + n^{-1}),$$

so

$$E_0 \left[\left| \frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_k} \tilde{R}_{i,k} \right| \mid \mathcal{F}_{-k} \right] \leq \frac{|\mathcal{I}_k|}{\sqrt{n}} E_0 \left[|\tilde{R}_{1,k}| \mid \mathcal{F}_{-k} \right] = O_p(\sqrt{n} \tau_n^2) + O_p(n^{-1/2}) = o_p(1),$$

because $\sqrt{n} \tau_n^2 \rightarrow 0$. Combining the bounds for $R_{i,k}^{(1)}$ and $R_{i,k}^{(2)}$ yields

$$\frac{1}{\sqrt{n}} \sum_{i \in \mathcal{I}_k} R_{i,k} = o_p(1) \quad \text{for each } k.$$

Since K is fixed, summing over $k = 1, \dots, K$ proves (5.14) and completes the proof. $\square$

LEMMA 8. *Define the out-of-fold (efficient) information estimator*

$$\hat{I}^{(-k)} = \frac{1}{|\mathcal{I}_k^c|} \sum_{j \in \mathcal{I}_k^c} \Delta_j \left\{ \mathbf{Z}_j - \hat{\mathbf{g}}^{*,(-k)}(T_j, \mathbf{X}_j) \right\}^{\otimes 2}, \quad k = 1, \dots, K.$$

Assume A1–A7 and A9–A10. Then,

- (a) $\max_{1 \leq k \leq K} \|\hat{I}^{(-k)} - I(\boldsymbol{\theta}_o)\| = o_p(1)$.
- (b) With probability tending to one, each $\hat{I}^{(-k)}$ is invertible and

$$\max_{1 \leq k \leq K} \|\{\hat{I}^{(-k)}\}^{-1} - I(\boldsymbol{\theta}_o)^{-1}\| = o_p(1).$$

- (c) If, in addition, $E_0\{\|\ell_{\boldsymbol{\theta}_o}^*(\mathbf{W})\|^2\} < \infty$, then

$$\frac{1}{\sqrt{n}} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \left(\{\hat{I}^{(-k)}\}^{-1} - I(\boldsymbol{\theta}_o)^{-1} \right) \ell_{\boldsymbol{\theta}_o}^*(\mathbf{W}_i) = o_p(1).$$

PROOF OF LEMMA 8. Write $\mathbb{P}_{n,-k}$ for the empirical measure on the training subsample $\mathcal{I}_k^c$ and define

$$\Psi_h(\mathbf{W}) = \Delta \{ \mathbf{Z} - h(T, \mathbf{X}) \}^{\otimes 2}.$$

Then $\hat{I}^{(-k)} = \mathbb{P}_{n,-k} \Psi_{\hat{\mathbf{g}}^{*,(-k)}}$ and $I(\theta_o) = \mathbb{P}_0 \Psi_{\mathbf{g}^*}$, hence

$$\hat{I}^{(-k)} - I(\theta_o) = (\mathbb{P}_{n,-k} - \mathbb{P}_0) \Psi_{\hat{\mathbf{g}}^{*,(-k)}} + \mathbb{P}_0 (\Psi_{\hat{\mathbf{g}}^{*,(-k)}} - \Psi_{\mathbf{g}^*}).$$

We first control the second (bias) term. Let $r^*(T, \mathbf{X}) = \mathbf{Z} - g^*(T, \mathbf{X})$ and $\hat{r}^{(-k)}(T, \mathbf{X}) = \mathbf{Z} - \hat{g}^{*,(-k)}(T, \mathbf{X})$. Expanding,

$$\Psi_{\hat{\mathbf{g}}^{*,(-k)}}(\mathbf{W}) - \Psi_{\mathbf{g}^*}(\mathbf{W}) = \Delta \left[(\hat{r}^{(-k)} - r^*) \otimes (\hat{r}^{(-k)} - r^*) + (\hat{r}^{(-k)} - r^*) \otimes r^* + r^* \otimes (\hat{r}^{(-k)} - r^*) \right].$$

Under A2 and A6, $\|r^*\|$ is bounded almost surely and $\|\hat{r}^{(-k)} - r^*\| = \|\hat{\mathbf{g}}^{*,(-k)} - \mathbf{g}^*\|$. Therefore, for a constant $C_1 < \infty$,

$$\|\mathbb{P}_0(\Psi_{\hat{\mathbf{g}}^{*,(-k)}} - \Psi_{\mathbf{g}^*})\| \leq C_1 \mathbb{P}_0 \|\hat{\mathbf{g}}^{*,(-k)}(T, \mathbf{X}) - \mathbf{g}^*(T, \mathbf{X})\| + C_1 \mathbb{P}_0 \|\hat{\mathbf{g}}^{*,(-k)}(T, \mathbf{X}) - \mathbf{g}^*(T, \mathbf{X})\|^2.$$

By Cauchy–Schwarz and the definition of $\|\cdot\|_0$, each of the two expectations above is $O_p(\|\hat{\mathbf{g}}^{*,(-k)} - \mathbf{g}^*\|_0)$ and $O_p(\|\hat{\mathbf{g}}^{*,(-k)} - \mathbf{g}^*\|_0^2)$, respectively. Using A9, the bias term is $O_p(\tau_n) + O_p(\tau_n^2) = o_p(1)$, uniformly in k .

For the empirical term $(\mathbb{P}_{n,-k} - \mathbb{P}_0) \Psi_{\hat{\mathbf{g}}^{*,(-k)}}$, condition on the training sigma-field in A9. Conditional on it, $\hat{\mathbf{g}}^{*,(-k)}$ is fixed and the summands $\Psi_{\hat{\mathbf{g}}^{*,(-k)}}(\mathbf{W}_j)$, $j \in \mathcal{I}_k^c$, are i.i.d. Moreover, under A2, A6 and boundedness of $\hat{\mathbf{g}}^{*,(-k)}$ induced by the model setup, $\|\Psi_{\hat{\mathbf{g}}^{*,(-k)}}(\mathbf{W})\|$ is uniformly bounded by a constant. Hence each entry of $(\mathbb{P}_{n,-k} - \mathbb{P}_0) \Psi_{\hat{\mathbf{g}}^{*,(-k)}}$ has conditional variance of order $|\mathcal{I}_k^c|^{-1}$, and a conditional Chebyshev (or LLN) yields

$$(\mathbb{P}_{n,-k} - \mathbb{P}_0) \Psi_{\hat{\mathbf{g}}^{*,(-k)}} = o_p(1),$$

uniformly in k . Combining with the bias bound proves (a).

For (b), fix k and write $A = \hat{I}^{(-k)}$ and $B = I(\theta_o)$. By (a), $\|A - B\| = o_p(1)$ uniformly in k , and by A10, $\|B^{-1}\| \leq c_0^{-1}$. On the event $\|A - B\| \leq (2\|B^{-1}\|)^{-1}$, the matrix A is invertible and $\|A^{-1}\| \leq 2\|B^{-1}\|$. Using the resolvent identity,

$$A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1},$$

we obtain

$$\|A^{-1} - B^{-1}\| \leq \|A^{-1}\| \|A - B\| \|B^{-1}\| \leq 2\|B^{-1}\|^2 \|A - B\|.$$

Taking maxima over k and using $\max_k \|A - B\| = o_p(1)$ gives (b).

For (c), write

$$R_n = \frac{1}{\sqrt{n}} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \left(\{\hat{I}^{(-k)}\}^{-1} - I(\theta_o)^{-1} \right) \ell_{\theta_o}^*(\mathbf{W}_i).$$

Then

$$\|R_n\| \leq \left(\max_{1 \leq k \leq K} \|\{\hat{I}^{(-k)}\}^{-1} - I(\theta_o)^{-1}\| \right) \left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \ell_{\theta_o}^*(\mathbf{W}_i) \right\|.$$

The first factor is $o_p(1)$ by (b). The second factor is $O_p(1)$ by the multivariate CLT under $E_0 \|\ell_{\theta_o}^*(\mathbf{W})\|^2 < \infty$. Therefore $\|R_n\| = o_p(1)$, proving (c). $\square$

LEMMA 9. Assume [A1–A11](#). Let $\phi_{t,n}^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0)$ denote the grid version of (5.11) obtained by replacing $\int_0^\tau \psi_t^0(u, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) dM(u)$ with $\sum_{j=1}^{m_n} \psi_t^0(t_j, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) \Delta M_j$ and $c_{0,t}(\mathbf{X}_0, \mathbf{Z}_0)$ with its grid analogue. Then, uniformly in $k = 1, \dots, K$,

$$\mathbb{E}_0 \left[\left(\widehat{\phi}_t^{(-k)}(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) - \phi_{t,n}^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) \right)^2 \right] = o_p(1),$$

and therefore (since $\sqrt{n}\delta_n \rightarrow 0$ implies $\phi_{t,n}^0$ and ϕ_t^0 are asymptotically equivalent),

$$\mathbb{E}_0 \left[\left(\widehat{\phi}_t^{(-k)}(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) - \phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) \right)^2 \right] = o_p(1), \quad \text{uniformly in } k.$$

PROOF OF LEMMA 9. Write

$$\widehat{\phi}_t^{(-k)} - \phi_{t,n}^0 = \left(\widehat{\phi}_{t,g}^{(-k)} - \phi_{t,g,n}^0 \right) + \left(\widehat{\phi}_{t,\theta}^{(-k)} - \phi_{t,\theta,n}^0 \right),$$

and control each term in $L^2(P_0)$. We start with the nuisance Riesz part. For brevity write

$$\widehat{\psi}_j^{(-k)} = \widehat{\psi}_t^{(-k)}(t_j, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0), \quad \psi_j^0 = \psi_t^0(t_j, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0).$$

Also write $\Delta \widehat{M}_j^{(-k)} = \Delta N_j - \Delta A_j^{(-k)}$ and $\Delta M_j^0 = \Delta N_j - \Delta A_j^0$, where

$$\Delta A_j^{(-k)} = (t_j - t_{j-1}) Y_j \widehat{h}^{(-k)}(t_j | \mathbf{X}, \mathbf{Z})$$

and

$$\Delta A_j^0 = (t_j - t_{j-1}) Y_j h_o(t_j | \mathbf{X}, \mathbf{Z}).$$

Then

$$(B.5) \quad \widehat{\phi}_{t,g}^{(-k)} - \phi_{t,g,n}^0 = \sum_{j=1}^{m_n} (\widehat{\psi}_j^{(-k)} - \psi_j^0) \Delta M_j^0 + \sum_{j=1}^{m_n} \widehat{\psi}_j^{(-k)} (\Delta \widehat{M}_j^{(-k)} - \Delta M_j^0).$$

The first sum is a martingale transform with predictable integrand $\widehat{\psi}_j^{(-k)} - \psi_j^0$, hence by the discrete-time martingale isometry,

$$\mathbb{E}_0 \left[\left(\sum_{j=1}^{m_n} (\widehat{\psi}_j^{(-k)} - \psi_j^0) \Delta M_j^0 \right)^2 \middle| \mathcal{F}_{-k} \right] = \mathbb{E}_0 \left[\sum_{j=1}^{m_n} (\widehat{\psi}_j^{(-k)} - \psi_j^0)^2 \Delta A_j^0 \middle| \mathcal{F}_{-k} \right].$$

In the discrete finite-support case, ψ_t^0 has the closed form in (5.12), and $\widehat{\psi}_t^{(-k)}$ is obtained by replacing h_o and the nuisance quantities appearing in (5.12) by their foldwise plug-in versions. Write

$$\pi_0 = \Pr(\mathbf{X} = \mathbf{X}_0), \quad \widehat{\pi}^{(-k)} = \widehat{\mathbb{P}}^{(-k)}(\mathbf{X} = \mathbf{X}_0),$$

and

$$A_{\mathbf{X}_0}(u) = E_0[Y(u) h_o(u | \mathbf{X}, \mathbf{Z}) | \mathbf{X} = \mathbf{X}_0], \quad \widehat{A}_{\mathbf{X}_0}^{(-k)}(u) = \widehat{E}_{-k} \left[Y(u) \widehat{h}^{(-k)}(u | \mathbf{X}, \mathbf{Z}) | \mathbf{X} = \mathbf{X}_0 \right].$$

Because $\Pr(X = X_0) = \pi_0 > 0$ and X has finite support, we have

$$\inf_k \widehat{\pi}^{(-k)} \geq c_\pi > 0$$

with probability tending to one. Next, by the bounded-sieve restriction for $\widehat{g}^{(-k)}$, boundedness of $\mathbf{Z}$, and boundedness of the parameter space for $\boldsymbol{\theta}$, there exists a deterministic constant $c_\lambda > 0$ such that

$$\inf_k \inf_{u \leq t} \inf_{\mathbf{x}, \mathbf{z}} \widehat{h}^{(-k)}(u | \mathbf{x}, \mathbf{z}) \geq c_\lambda$$

with probability tending to one. Hence, for $u \leq t$,

$$\widehat{A}_{\mathbf{X}_0}^{(-k)}(u) = \widehat{E}_{-k} \left[Y(u) \widehat{h}^{(-k)}(u \mid \mathbf{X}, \mathbf{Z}) \mid \mathbf{X} = \mathbf{X}_0 \right] \geq c_\lambda \widehat{E}_{-k} [Y(u) \mid \mathbf{X} = \mathbf{X}_0].$$

By the positivity assumptions, there exists $c_Y > 0$ such that

$$\inf_{u \leq t} E_0[Y(u) \mid \mathbf{X} = \mathbf{X}_0] \geq c_Y.$$

Moreover, since $Y(u)$ is monotone, the class $\{Y(u) : 0 \leq u \leq t\}$ is Glivenko–Cantelli, and therefore

$$\sup_k \sup_{u \leq t} \left| \widehat{E}_{-k} [Y(u) \mid \mathbf{X} = \mathbf{X}_0] - E_0[Y(u) \mid \mathbf{X} = \mathbf{X}_0] \right| = o_p(1).$$

It follows that

$$\inf_k \min_{j: t_j \leq t} \widehat{A}_{\mathbf{X}_0}^{(-k)}(t_j) \geq c_0$$

with probability tending to one for some constant $c_0 > 0$.

Because $\Pr(\mathbf{X} = \mathbf{X}_0) \geq \pi_0 > 0$ and $\mathbf{X}$ has finite support, $\widehat{\pi}^{(-k)} \rightarrow \pi_0$ in probability uniformly in k . Define the event

$$\mathcal{E}_n = \left\{ \inf_k \widehat{\pi}^{(-k)} \geq c_0 \text{ and } \inf_k \min_{j: t_j \leq t} \widehat{A}_{\mathbf{X}_0}^{(-k)}(t_j) \geq c_0 \right\}.$$

By the above arguments, there exists $c_0 > 0$ such that

$$\mathbb{P}(\mathcal{E}_n) \rightarrow 1.$$

On $\mathcal{E}_n$, all denominator terms appearing in $\widehat{\psi}_t^{(-k)}$ are uniformly bounded away from zero.

Now

$$\widehat{\psi}_t^{(-k)}(u, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) = \mathbf{1}(u \leq t) \mathbf{1}(\mathbf{X} = \mathbf{X}_0) \frac{\widehat{h}^{(-k)}(u \mid \mathbf{X}_0, \mathbf{Z}_0)}{\widehat{\pi}^{(-k)} \widehat{A}_{\mathbf{X}_0}^{(-k)}(u)},$$

while

$$\psi_t^0(u, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) = \mathbf{1}(u \leq t) \mathbf{1}(\mathbf{X} = \mathbf{X}_0) \frac{h_o(u \mid \mathbf{X}_0, \mathbf{Z}_0)}{\pi_0 A_{\mathbf{X}_0}(u)}.$$

For readability, suppress the common arguments $(u \mid \mathbf{X}_0, \mathbf{Z}_0)$ and write

$$\Delta_{h,j}^{(-k)} = \widehat{h}^{(-k)}(t_j \mid \mathbf{X}_0, \mathbf{Z}_0) - h_o(t_j \mid \mathbf{X}_0, \mathbf{Z}_0),$$

$$\Delta_\pi^{(-k)} = \widehat{\pi}^{(-k)} - \pi_0, \quad \Delta_{A,j}^{(-k)} = \widehat{A}_{\mathbf{X}_0}^{(-k)}(t_j) - A_{\mathbf{X}_0}(t_j).$$

On $\mathcal{E}_n$, a first-order ratio expansion gives, for each j ,

$$\widehat{\psi}_j^{(-k)} - \psi_j^0 = \mathbf{1}(t_j \leq t) \mathbf{1}(\mathbf{X} = \mathbf{X}_0) \left\{ L_{h,j}^{(-k)} + L_{\pi,j}^{(-k)} + L_{A,j}^{(-k)} + R_{\psi,j}^{(-k)} \right\},$$

where

$$L_{h,j}^{(-k)} = \frac{\Delta_{h,j}^{(-k)}}{\pi_0 A_{\mathbf{X}_0}(t_j)}, \quad L_{\pi,j}^{(-k)} = \frac{h_o(t_j \mid \mathbf{X}_0, \mathbf{Z}_0)}{A_{\mathbf{X}_0}(t_j)} \left(\frac{1}{\widehat{\pi}^{(-k)}} - \frac{1}{\pi_0} \right),$$

$$L_{A,j}^{(-k)} = \frac{h_o(t_j \mid \mathbf{X}_0, \mathbf{Z}_0)}{\widehat{\pi}^{(-k)}} \left(\frac{1}{\widehat{A}_{\mathbf{X}_0}^{(-k)}(t_j)} - \frac{1}{A_{\mathbf{X}_0}(t_j)} \right),$$

and $R_{\psi,j}^{(-k)}$ is a finite sum of products of estimation errors, hence second order. Using $(a + b + c + d)^2 \lesssim a^2 + b^2 + c^2 + d^2$, we obtain on $\mathcal{E}_n$,

$$(\widehat{\psi}_j^{(-k)} - \psi_j^0)^2 \lesssim \mathbf{1}(\mathbf{X} = \mathbf{X}_0) \left\{ |L_{h,j}^{(-k)}|^2 + |L_{\pi,j}^{(-k)}|^2 + |L_{A,j}^{(-k)}|^2 + |R_{\psi,j}^{(-k)}|^2 \right\}.$$

Therefore

$$\mathbb{E}_0 \left[\sum_{j=1}^{m_n} (\widehat{\psi}_j^{(-k)} - \psi_j^0)^2 \Delta A_j^0 \right] \lesssim T_h^{(-k)} + T_\pi^{(-k)} + T_A^{(-k)} + T_R^{(-k)},$$

where

$$T_h^{(-k)} = \mathbb{E}_0 \left[\sum_{j=1}^{m_n} \mathbf{1}(\mathbf{X} = \mathbf{X}_0) |\Delta_{h,j}^{(-k)}|^2 \Delta A_j^0 \right],$$

$$T_\pi^{(-k)} = |\Delta_\pi^{(-k)}|^2 \mathbb{E}_0 \left[\sum_{j=1}^{m_n} \mathbf{1}(\mathbf{X} = \mathbf{X}_0) \Delta A_j^0 \right],$$

$$T_A^{(-k)} = \mathbb{E}_0 \left[\sum_{j=1}^{m_n} \mathbf{1}(\mathbf{X} = \mathbf{X}_0) |\Delta_{A,j}^{(-k)}|^2 \Delta A_j^0 \right],$$

and $T_R^{(-k)}$ is the analogous contribution of the second-order remainder $R_{\psi,j}^{(-k)}$.

We bound these four terms separately.

For $T_h^{(-k)}$, note that

$$\Delta A_j^0 = (t_j - t_{j-1}) Y(t_{j-1}) h_o(t_j | \mathbf{X}, \mathbf{Z}),$$

so

$$T_h^{(-k)} = \mathbb{E}_0 \left[\sum_{j=1}^{m_n} (t_j - t_{j-1}) \mathbf{1}(\mathbf{X} = \mathbf{X}_0) Y(t_{j-1}) h_o(t_j | \mathbf{X}, \mathbf{Z}) |\Delta_{h,j}^{(-k)}|^2 \right].$$

By the same exponential-link expansion used in Lemma 3, $\widehat{h}^{(-k)} - h_o$ is linear in $\widetilde{g}^{(-k)} - g_o$ and $\widetilde{\boldsymbol{\theta}}^{(-k)} - \boldsymbol{\theta}_o$ plus a quadratic remainder. Therefore, exactly as in Lemma 3,

$$T_h^{(-k)} \lesssim \|\widetilde{g}^{(-k)} - g_o\|_0^2 + \|\widetilde{\boldsymbol{\theta}}^{(-k)} - \boldsymbol{\theta}_o\|^2 + o_p(1) = O_p(\tau_n^2) + O_p(n^{-1}) + o_p(1) = o_p(1),$$

uniformly in k , by Assumption A9.

For $T_A^{(-k)}$, write

$$\Delta_{A,j}^{(-k)} = B_{A,j}^{(-k)} + D_{A,j}^{(-k)},$$

where

$$B_{A,j}^{(-k)} = \widehat{E}_{-k} \left\{ Y(t_{j-1}) \widehat{h}^{(-k)}(t_j | \mathbf{X}, \mathbf{Z}) | \mathbf{X} = \mathbf{X}_0 \right\} - E_0 \left\{ Y(t_{j-1}) \widehat{h}^{(-k)}(t_j | \mathbf{X}, \mathbf{Z}) | \mathbf{X} = \mathbf{X}_0 \right\},$$

and

$$D_{A,j}^{(-k)} = E_0 \left\{ Y(t_{j-1}) (\widehat{h}^{(-k)}(t_j | \mathbf{X}, \mathbf{Z}) - h_o(t_j | \mathbf{X}, \mathbf{Z})) | \mathbf{X} = \mathbf{X}_0 \right\}.$$

Hence,

$$T_A^{(-k)} \lesssim T_{A,1}^{(-k)} + T_{A,2}^{(-k)},$$

where

$$T_{A,1}^{(-k)} = E_0 \left[\sum_{j=1}^{m_n} \mathbf{1}(\mathbf{X} = \mathbf{X}_0) |B_{A,j}^{(-k)}|^2 \Delta A_j^0 \right],$$

and

$$T_{A,2}^{(-k)} = E_0 \left[\sum_{j=1}^{m_n} \mathbf{1}(\mathbf{X} = \mathbf{X}_0) |D_{A,j}^{(-k)}|^2 \Delta A_j^0 \right].$$

We first bound $T_{A,1}^{(-k)}$. Because $\mathbf{X}$ has finite support and $\widehat{h}^{(-k)}$ is uniformly bounded with probability tending to one, the variables

$$Y(t_{j-1})\widehat{h}^{(-k)}(t_j | \mathbf{X}, \mathbf{Z})\mathbf{1}(\mathbf{X} = \mathbf{X}_0)$$

are uniformly bounded over j and k . Therefore, conditional on $\mathcal{F}_{-k}$,

$$E_0 \left[(B_{A,j}^{(-k)})^2 | \mathcal{F}_{-k} \right] \lesssim \frac{1}{n} \quad \text{uniformly in } j, k.$$

Thus

$$\begin{aligned} E_0 \left[T_{A,1}^{(-k)} | \mathcal{F}_{-k} \right] &\lesssim \frac{1}{n} E_0 \left[\sum_{j=1}^{m_n} \mathbf{1}(\mathbf{X} = \mathbf{X}_0) \Delta A_j^0 \middle| \mathcal{F}_{-k} \right] \\ &\lesssim \frac{1}{n} E_0 \left[\sum_{j=1}^{m_n} \Delta A_j^0 \right] = O\left(\frac{1}{n}\right), \end{aligned}$$

since

$$\sum_{j=1}^{m_n} \Delta A_j^0 = \sum_{j=1}^{m_n} (t_j - t_{j-1}) Y(t_{j-1}) h_o(t_j | \mathbf{X}, \mathbf{Z}) = O_p(1).$$

Hence

$$T_{A,1}^{(-k)} = O_p(n^{-1}) = o_p(1)$$

uniformly in k .

Next, for $T_{A,2}^{(-k)}$, Jensen's inequality gives

$$|D_{A,j}^{(-k)}|^2 \leq E_0 \left[Y(t_{j-1}) (\widehat{h}^{(-k)}(t_j | \mathbf{X}, \mathbf{Z}) - h_o(t_j | \mathbf{X}, \mathbf{Z}))^2 | \mathbf{X} = \mathbf{X}_0 \right].$$

Therefore,

$$T_{A,2}^{(-k)} \lesssim E_0 \left[\sum_{j=1}^{m_n} \mathbf{1}(\mathbf{X} = \mathbf{X}_0) E_0 \left\{ Y(t_{j-1}) (\widehat{h}^{(-k)}(t_j | \mathbf{X}, \mathbf{Z}) - h_o(t_j | \mathbf{X}, \mathbf{Z}))^2 | \mathbf{X} = \mathbf{X}_0 \right\} \Delta A_j^0 \right].$$

Since $\mathbf{X}$ has finite support, this is bounded by a constant multiple of

$$E_0 \left[\sum_{j=1}^{m_n} (t_j - t_{j-1}) Y(t_{j-1}) h_o(t_j | \mathbf{X}, \mathbf{Z}) (\widehat{h}^{(-k)}(t_j | \mathbf{X}, \mathbf{Z}) - h_o(t_j | \mathbf{X}, \mathbf{Z}))^2 \right],$$

which is exactly the same integrated hazard-error quantity already controlled in the bound for $T_h^{(-k)}$. Hence, by the exponential-link expansion together with A9 and the same quadratic remainder control as in Lemma 3,

$$T_{A,2}^{(-k)} = o_p(1)$$

uniformly in k . Combining the two bounds yields

$$T_A^{(-k)} = o_p(1) \quad \text{uniformly in } k.$$

Finally, $T_R^{(-k)} = o_p(1)$ uniformly in k , because each term in $R_{\psi,j}^{(-k)}$ is a product of first-order estimation errors and hence contributes only second-order terms after summation with weight ΔA_j^0 .

Combining the four bounds yields

$$\mathbb{E}_0 \left[\sum_{j=1}^{m_n} (\hat{\psi}_j^{(-k)} - \psi_j^0)^2 \Delta A_j^0 \right] = o_p(1), \quad \text{uniformly in } k.$$

Consequently, the first sum is $o_p(1)$ in $L^2(P_0)$ uniformly in k .

For the second sum of (B.5), note that

$$\Delta \widehat{M}_j^{(-k)} - \Delta M_j^0 = -(\Delta A_j^{(-k)} - \Delta A_j^0),$$

so we set

$$S_{2,k} = \sum_{j=1}^{m_n} \hat{\psi}_j^{(-k)} (\Delta \widehat{M}_j^{(-k)} - \Delta M_j^0) = - \sum_{j=1}^{m_n} \hat{\psi}_j^{(-k)} (t_j - t_{j-1}) Y_j \{ \widehat{h}^{(-k)}(t_j | \mathbf{X}, \mathbf{Z}) - h_o(t_j | \mathbf{X}, \mathbf{Z}) \}.$$

On the event $\mathcal{E}_n$, the discrete closed form of $\hat{\psi}_j^{(-k)}$ and boundedness of the denominators imply

$$\sup_k \sup_{j: t_j \leq t} |\hat{\psi}_j^{(-k)}| = O_p(1).$$

Hence, by Cauchy–Schwarz,

$$\begin{aligned} |S_{2,k}|^2 &\lesssim \left(\sum_{j=1}^{m_n} (t_j - t_{j-1}) Y_j |\widehat{h}^{(-k)}(t_j | \mathbf{X}, \mathbf{Z}) - h_o(t_j | \mathbf{X}, \mathbf{Z})| \right)^2 \\ &\leq \left(\sum_{j=1}^{m_n} \Delta A_j^0 \right) \left(\sum_{j=1}^{m_n} (t_j - t_{j-1}) Y_j \frac{(\widehat{h}^{(-k)}(t_j | \mathbf{X}, \mathbf{Z}) - h_o(t_j | \mathbf{X}, \mathbf{Z}))^2}{h_o(t_j | \mathbf{X}, \mathbf{Z})} \right). \end{aligned}$$

Since h_o is bounded away from zero and infinity, and

$$\sum_{j=1}^{m_n} \Delta A_j^0 = \sum_{j=1}^{m_n} (t_j - t_{j-1}) Y_j h_o(t_j | \mathbf{X}, \mathbf{Z}) = O_{P_0}(1),$$

it follows that

$$\mathbb{E}_0[|S_{2,k}|^2] \lesssim \mathbb{E}_0 \left[\sum_{j=1}^{m_n} (t_j - t_{j-1}) Y_j (\widehat{h}^{(-k)}(t_j | \mathbf{X}, \mathbf{Z}) - h_o(t_j | \mathbf{X}, \mathbf{Z}))^2 \right].$$

The right-hand side is exactly the same weighted integrated hazard-error quantity already controlled above in the bound for $T_h^{(-k)}$. Therefore, by the exponential-link expansion, Assumption A9, and the same quadratic remainder control as in Lemma 3,

$$\mathbb{E}_0[|S_{2,k}|^2] = o_p(1) \quad \text{uniformly in } k.$$

Thus the second sum of (B.5) is $o_p(1)$ in $L^2(P_0)$, uniformly in k . Therefore,

$$\mathbb{E}_0 \left[(\widehat{\phi}_{t,g}^{(-k)} - \phi_{t,g,n}^0)^2 \right] = o_p(1) \quad \text{uniformly in } k.$$

For the parametric part, we write

$$\widehat{\phi}_{t,\theta}^{(-k)} - \phi_{t,\theta,n}^0 = \left(\widehat{c}_t^{(-k)} - c_{0,t,n} \right)^\top I(\boldsymbol{\theta}_o)^{-1} \ell_{\theta_o}^* + \widehat{c}_t^{(-k)\top} \left(\{\widehat{I}^{(-k)}\}^{-1} - I(\boldsymbol{\theta}_o)^{-1} \right) \ell_{\theta_o}^* + \widehat{c}_t^{(-k)\top} \{\widehat{I}^{(-k)}\}^{-1} \left(\widehat{\ell}_{\theta_o}^{*,(-k)} - \ell_{\theta_o}^* \right)$$

where all quantities are evaluated at $(\mathbf{X}_0, \mathbf{Z}_0)$ and $\mathbf{W}$ (we suppress these arguments). In the discrete case,

$$\widehat{c}_t^{(-k)}(\mathbf{X}_0, \mathbf{Z}_0) = \sum_{j:t_j \leq t} (t_j - t_{j-1}) \widehat{h}^{(-k)}(t_j | \mathbf{X}_0, \mathbf{Z}_0) \left\{ \mathbf{Z}_0 - \widehat{\mathbf{g}}^{*,(-k)}(t_j, \mathbf{X}_0) \right\},$$

so by boundedness and the rate conditions in Assumption A9 $\|\widehat{c}_t^{(-k)} - c_{0,t,n}\| = o_p(1)$ uniformly in k . Since $\mathbb{E}_0 \|\ell_{\theta_o}^*(\mathbf{W})\|^2 < \infty$ under the standing integrability assumptions, the first term is $o_p(1)$ in $L^2(P_0)$.

For the second term, Lemma 8 yields

$$\max_k \|(\widehat{I}^{(-k)})^{-1} - I(\boldsymbol{\theta}_o)^{-1}\| = o_p(1),$$

while $\widehat{c}_t^{(-k)}$ is bounded in probability and $\ell_{\theta_o}^* \in L^2(P_0)$, so the product is $o_p(1)$ in $L^2(P_0)$ uniformly in k .

By the decomposition in the proof of Lemma 3, we have

$$E_0 \left[(\widehat{\ell}_{\theta_o}^{*,(-k)} - \ell_{\theta_o}^*)^2 | F_{-k} \right] = O_p(\tau_n^2 + n^{-1}),$$

uniformly in k . Therefore

$$\widehat{\ell}_{\theta_o}^{*,(-k)} - \ell_{\theta_o}^* = O_p(\tau_n) \quad \text{in } L^2(\mathbb{P}_0),$$

and the third term is $o_p(1)$ in $L^2(\mathbb{P}_0)$.

Putting the three bounds together gives

$$\mathbb{E}_0 \left[(\widehat{\phi}_{t,\theta}^{(-k)} - \phi_{t,\theta,n}^0)^2 \right] = o_p(1) \quad \text{uniformly in } k.$$

Combining the nuisance and parametric parts proves

$$\mathbb{E}_0 \left[(\widehat{\phi}_t^{(-k)} - \phi_{t,n}^0)^2 \right] = o_p(1) \quad \text{uniformly in } k.$$

Finally, $\sqrt{n} \delta_n \rightarrow 0$ implies that the grid approximation error between $\phi_{t,n}^0$ and ϕ_t^0 is $o(1)$ in $L^2(P_0)$, so the same conclusion holds with ϕ_t^0 in place of $\phi_{t,n}^0$. $\square$

THEOREM 6. Fix $t \in [0, \tau]$ and let $(\mathbf{X}_0, \mathbf{Z}_0)$ be an independent draw from the covariate distribution. Under Assumptions A1–A11, and assuming moreover that for each fold k ,

$$\|\widetilde{g}^{(-k)} - g_o\|_0 = O_p(\gamma_n \log^2 n), \quad \gamma_n \log^2 n = o(n^{-1/4}),$$

uniformly in k , and that $\tilde{\boldsymbol{\theta}}^{(-k)}$ satisfies the same first-order expansion as established in Theorem 4, namely,

$$\sqrt{|\mathcal{I}_k^c|}(\tilde{\boldsymbol{\theta}}^{(-k)} - \boldsymbol{\theta}_o) = I(\boldsymbol{\theta}_o)^{-1} \frac{1}{\sqrt{|\mathcal{I}_k^c|}} \sum_{i \in \mathcal{I}_k^c} \ell_{\boldsymbol{\theta}_o}^*(\mathbf{W}_i) + o_p(1),$$

uniformly in k . Then, conditional on the independent draw $(\mathbf{X}_0, \mathbf{Z}_0)$,

$$\sqrt{n} \left\{ \hat{H}^{1, \text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0) - H_o(t | \mathbf{X}_0, \mathbf{Z}_0) \right\} | (\mathbf{X}_0, \mathbf{Z}_0) \rightarrow^{\mathcal{D}} \mathcal{N}(0, \sigma_H^2(t; \mathbf{X}_0, \mathbf{Z}_0)),$$

where $\sigma_H^2(t; \mathbf{X}_0, \mathbf{Z}_0) = \mathbb{V}_0(\phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) | \mathbf{X}_0, \mathbf{Z}_0)$. Moreover, with

$$\hat{\sigma}_H^2(t; \mathbf{X}_0, \mathbf{Z}_0) = \frac{1}{n} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \left\{ \hat{\phi}_t^{(-k)}(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0) \right\}^2,$$

we have $\hat{\sigma}_H^2(t; \mathbf{X}_0, \mathbf{Z}_0) \xrightarrow{p} \sigma_H^2(t; \mathbf{X}_0, \mathbf{Z}_0)$. Finally,

$$\sqrt{n} \left\{ \hat{S}^{1, \text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0) - S_o(t | \mathbf{X}_0, \mathbf{Z}_0) \right\} | (\mathbf{X}_0, \mathbf{Z}_0) \rightarrow^{\mathcal{D}} \mathcal{N}(0, \sigma_S^2(t; \mathbf{X}_0, \mathbf{Z}_0))$$

where $\sigma_S^2(t; \mathbf{X}_0, \mathbf{Z}_0) = S_o(t | \mathbf{X}_0, \mathbf{Z}_0)^2 \sigma_H^2(t; \mathbf{X}_0, \mathbf{Z}_0)$ and a consistent variance estimator is

$$\hat{\sigma}_S^2(t; \mathbf{X}_0, \mathbf{Z}_0) = \left\{ \hat{S}^{1, \text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0) \right\}^2 \hat{\sigma}_H^2(t; \mathbf{X}_0, \mathbf{Z}_0).$$

PROOF OF THEOREM 6. We suppress $(t, \mathbf{X}_0, \mathbf{Z}_0)$ from the notation when no confusion arises. Let $\{\mathcal{I}_k\}_{k=1}^K$ be the fold partition of $\{1, \dots, n\}$ and, for each k , let $\mathcal{F}_{-k}$ be the σ -field generated by the training subsample $\{\mathbf{W}_i : i \in \mathcal{I}_k^c\}$, the fold split, and $(\mathbf{X}_0, \mathbf{Z}_0)$. By construction, $\{\mathbf{W}_i : i \in \mathcal{I}_k\}$ is independent of $\mathcal{F}_{-k}$. For each fold k , write

$$\delta g^{(-k)}(u, \mathbf{X}) = \tilde{g}^{(-k)}(u, \mathbf{X}) - g_o(u, \mathbf{X}), \quad \delta \boldsymbol{\theta}^{(-k)} = \tilde{\boldsymbol{\theta}}^{(-k)} - \boldsymbol{\theta}_o,$$

and write $\Delta t_j = t_j - t_{j-1}$. The cross-fitted one-step estimator is

$$\hat{H}^{1, \text{cf}} = \hat{H}^{\text{cf}} + \frac{1}{n} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \hat{\phi}_t^{(-k)}(\mathbf{W}_i), \quad \hat{H}^{\text{cf}} = \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \hat{H}^{(-k)},$$

where the out-of-fold plug-in estimator uses the Riemann sum

$$\hat{H}^{(-k)}(t, \mathbf{X}_0, \mathbf{Z}_0) = \sum_{j: t_j \leq t} \hat{h}^{(-k)}(t_j | \mathbf{X}_0, \mathbf{Z}_0) \Delta t_j.$$

Introduce also the Riemann approximation of the *true* cumulative hazard,

$$\tilde{H}_o(t, \mathbf{X}_0, \mathbf{Z}_0) = \sum_{j: t_j \leq t} h_o(t_j | \mathbf{X}_0, \mathbf{Z}_0) \Delta t_j, \quad H_o(t, \mathbf{X}_0, \mathbf{Z}_0) = \int_0^t h_o(u | \mathbf{X}_0, \mathbf{Z}_0) du.$$

Under the time-discretization condition A11, we have

$$(B.6) \quad |\tilde{H}_o - H_o| \leq C \delta_n,$$

for a finite constant C (possibly random but tight under the joint law), and thus $\sqrt{n}(\tilde{H}_o - H_o) = o(1)$ whenever $\sqrt{n} \delta_n \rightarrow 0$.

Decompose

$$\hat{H}^{1, \text{cf}} - H_o = \left(\hat{H}^{\text{cf}} - \tilde{H}_o \right) + \left(\tilde{H}_o - H_o \right) + \frac{1}{n} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \hat{\phi}_t^{(-k)}(\mathbf{W}_i).$$

The second term is negligible by (B.6), so it suffices to analyze

$$A_n = \hat{H}^{\text{cf}} - \tilde{H}_o, \quad B_n = \frac{1}{n} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \hat{\phi}_t^{(-k)}(\mathbf{W}_i).$$

Write $A_n = \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} (\hat{H}^{(-k)} - \tilde{H}_o)$. For each k ,

$$\hat{H}^{(-k)} - \tilde{H}_o = \sum_{j:t_j \leq t} \left\{ \hat{h}^{(-k)}(t_j | \mathbf{X}_0, \mathbf{Z}_0) - h_o(t_j | \mathbf{X}_0, \mathbf{Z}_0) \right\} \Delta t_j.$$

Define the fold-wise log-perturbation on the grid

$$r_j^{(-k)} = \delta g^{(-k)}(t_j, \mathbf{X}_0) + (\delta \boldsymbol{\theta}^{(-k)})^\top \mathbf{Z}_0, \quad j = 1, \dots, m_n.$$

Then

$$\hat{h}^{(-k)}(t_j | \mathbf{X}_0, \mathbf{Z}_0) = h_o(t_j | \mathbf{X}_0, \mathbf{Z}_0) \exp\{r_j^{(-k)}\}.$$

Using $\exp(r) - 1 = r + \frac{1}{2}r^2 \exp(\tilde{r})$ for some $\tilde{r}$ between 0 and r yields

$$(B.7) \quad \hat{H}^{(-k)} - \tilde{H}_o = \sum_{j:t_j \leq t} h_o(t_j | \mathbf{X}_0, \mathbf{Z}_0) r_j^{(-k)} \Delta t_j + R_{n,1}^{(-k)},$$

with remainder bounded by

$$(B.8) \quad |R_{n,1}^{(-k)}| \leq \frac{1}{2} \sum_{j:t_j \leq t} h_o(t_j | \mathbf{X}_0, \mathbf{Z}_0) (r_j^{(-k)})^2 \exp\{|r_j^{(-k)}|\} \Delta t_j.$$

The linear term in (B.7) decomposes into a g -part and a $\boldsymbol{\theta}$ -part:

$$\sum_{j:t_j \leq t} h_o(t_j | \mathbf{X}_0, \mathbf{Z}_0) r_j^{(-k)} \Delta t_j = L_{t,n}(\delta g^{(-k)}) + (\mathbf{Z}_0 \tilde{H}_o)^\top \delta \boldsymbol{\theta}^{(-k)},$$

where the discrete functional is

$$L_{t,n}(b) = \sum_{j:t_j \leq t} h_o(t_j | \mathbf{X}_0, \mathbf{Z}_0) b(t_j, \mathbf{X}_0) \Delta t_j.$$

Since $\tilde{H}_o = H_o + O(\delta_n)$, we may replace $\mathbf{Z}_0 \tilde{H}_o$ by $\mathbf{Z}_0 H_o$ at cost $O(\|\mathbf{Z}_0\| \delta_n \|\delta \boldsymbol{\theta}^{(-k)}\|)$, which is $o_p(n^{-1/2})$ under $\sqrt{n} \delta_n \rightarrow 0$ and $\|\delta \boldsymbol{\theta}^{(-k)}\| = O_p(n^{-1/2})$.

Next decompose the one-step correction into conditional mean plus fluctuation. Define

$$B_{n,0} = \frac{1}{n} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \left\{ \hat{\phi}_t^{(-k)}(\mathbf{W}_i) - \mathbb{E}_0[\hat{\phi}_t^{(-k)}(\mathbf{W}) | \mathcal{F}_{-k}] \right\}, \quad B_{n,1} = \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \mathbb{E}_0[\hat{\phi}_t^{(-k)}(\mathbf{W}) | \mathcal{F}_{-k}],$$

so that $B_n = B_{n,0} + B_{n,1}$. In the discrete finite-support setting, the Riesz representer has a closed form and the foldwise estimator $\hat{\psi}_t^{(-k)}$ is $\mathcal{F}_{-k}$ -measurable. For each fold k , conditional on $\mathcal{F}_{-k}$ the estimators $\tilde{g}^{(-k)}$, $\tilde{\boldsymbol{\theta}}^{(-k)}$, and $\hat{\psi}_t^{(-k)}$ are fixed functions, and the observations $\{\mathbf{W}_i : i \in \mathcal{I}_k\}$ are independent of $\mathcal{F}_{-k}$. Since, for each i , the process

$$M_{0,i}(t) = N_i(t) - \int_0^t Y_i(u) h_o(u | \mathbf{X}_i, \mathbf{Z}_i) du$$

is a square-integrable martingale with respect to its natural filtration, it follows that for any $\mathcal{F}_{-k}$ -measurable predictable process $h(u, \mathbf{X}_i)$,

$$\mathbb{E}_0 \left[\int_0^\tau h(u, \mathbf{X}_i) dM_{0,i}(u) \mid \mathcal{F}_{-k} \right] = 0.$$

This identity justifies the foldwise conditional mean calculations below. In particular, the foldwise conditional mean satisfies

$$(B.9) \quad \mathbb{E}_0[\widehat{\phi}_t^{(-k)}(\mathbf{W}) \mid \mathcal{F}_{-k}] = -L_{t,n}(\delta g^{(-k)}) - (\mathbf{Z}_0 H_o(t \mid \mathbf{X}_0, \mathbf{Z}_0))^\top \delta \boldsymbol{\theta}^{(-k)} + R_{n,2}^{(-k)},$$

where $R_{n,2}^{(-k)}$ is second order and satisfies

$$(B.10) \quad |R_{n,2}^{(-k)}| \lesssim \|\widetilde{g}^{(-k)} - g_o\|_0^2 + \|\widetilde{\boldsymbol{\theta}}^{(-k)} - \boldsymbol{\theta}_o\|^2,$$

under the conditions from Lemma 5 (and the corresponding foldwise analogues), together with the uniform boundedness of h_o ensured by Assumptions A1–A4.

Combine (B.7) with (B.9) and the foldwise expansions and use the replacement $\widetilde{H}_o = H_o + O(\delta_n)$ discussed above to obtain

$$A_n + B_{n,1} = \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \left(R_{n,1}^{(-k)} + R_{n,2}^{(-k)} \right) + o_p(n^{-1/2})$$

under the joint law. It remains to bound the remainders. The bound (B.8) together with $\exp\{|r|\} \leq 1 + |r| \exp\{|r|\}$ implies that, on events where $\max_{j \leq m_n} |r_j^{(-k)}|$ is bounded (which holds with probability tending to one under $\|\delta g^{(-k)}\|_0 = o_p(1)$ and $\|\delta \boldsymbol{\theta}^{(-k)}\| = o_p(1)$ plus the uniform boundedness assumptions),

$$|R_{n,1}^{(-k)}| \lesssim \sum_{j:t_j \leq t} h_o(t_j \mid \mathbf{X}_0, \mathbf{Z}_0) (r_j^{(-k)})^2 \Delta t_j \lesssim (L_{t,n}(|\delta g^{(-k)}|))^2 + \|\delta \boldsymbol{\theta}^{(-k)}\|^2.$$

We now bound the discrete linear functional $L_{t,n}(b)$. By Cauchy–Schwarz and boundedness of h_o ,

$$|L_{t,n}(b)| \lesssim \left(\sum_{j:t_j \leq t} h_o(t_j \mid \mathbf{X}_0, \mathbf{Z}_0) b(t_j, \mathbf{X}_0)^2 \Delta t_j \right)^{1/2}.$$

The discrete sum is a Riemann approximation of $\int_0^t h_o(u \mid \mathbf{X}_0, \mathbf{Z}_0) b(u, \mathbf{X}_0)^2 du$ with error $O(\delta_n \|b\|_\infty^2)$. Under the boundedness of the sieve and $\sqrt{n} \delta_n \rightarrow 0$, this error is negligible. Because $\mathbf{X}$ takes values in a finite set and $\mathbb{P}(\mathbf{X} = x_0) \geq c_0 > 0$, we may write

$$\int_0^t h_o(u \mid \mathbf{X}_0, \mathbf{Z}_0) b(u, \mathbf{X}_0)^2 du = \frac{1}{\mathbb{P}(\mathbf{X} = x_0)} \mathbb{E}_0 \left[\int_0^t h_o(u \mid \mathbf{X}, \mathbf{Z}) b(u, \mathbf{X})^2 du \mathbf{1}\{\mathbf{X} = x_0\} \right].$$

Under the overlap condition $\mathbb{E}_0\{Y(u) \mid \mathbf{X}, \mathbf{Z}\} \geq \kappa > 0$ and boundedness of h_o , the right-hand side is bounded by a constant multiple of

$$\mathbb{E}_0 \left[\int_0^t Y(u) h_o(u \mid \mathbf{X}, \mathbf{Z}) b(u, \mathbf{X})^2 du \right] = \|b\|_0^2.$$

Therefore

$$|L_{t,n}(b)| \lesssim \|b\|_0 + o(n^{-1/2}).$$

Consequently, $|R_{n,1}^{(-k)}| \lesssim \|\delta g^{(-k)}\|_0^2 + \|\delta \boldsymbol{\theta}^{(-k)}\|^2$ in probability, uniformly in k . Together with (B.10) we obtain

$$(B.11) \quad |A_n + B_{n,1}| \lesssim \max_k \|\widetilde{g}^{(-k)} - g_o\|_0^2 + \max_k \|\widetilde{\boldsymbol{\theta}}^{(-k)} - \boldsymbol{\theta}_o\|^2 + o_p(n^{-1/2}).$$

Under the assumption $\max_k \|\tilde{g}^{(-k)} - g_o\|_0 = O_p(\gamma_n \log^2 n)$ with $\gamma_n \log^2 n = o(n^{-1/4})$, we have $\sqrt{n} \max_k \|\tilde{g}^{(-k)} - g_o\|_0^2 = o_p(1)$. Under $\max_k \|\tilde{\theta}^{(-k)} - \theta_o\| = O_p(n^{-1/2})$, we also have

$$\sqrt{n} \max_k \|\tilde{\theta}^{(-k)} - \theta_o\|^2 = o_p(1).$$

Therefore (B.11) implies

$$\sqrt{n} (A_n + B_{n,1}) = o_p(1).$$

We now analyze $B_{n,0}$ fold by fold. For each k , define

$$B_{n,0}^{(k)} = \frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \left\{ \hat{\phi}_t^{(-k)}(\mathbf{W}_i) - \mathbb{E}_0[\hat{\phi}_t^{(-k)}(\mathbf{W}) \mid \mathcal{F}_{-k}] \right\},$$

so that $B_{n,0} = \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} B_{n,0}^{(k)}$. Fix k . Conditional on $\mathcal{F}_{-k}$, the variables $\{\mathbf{W}_i : i \in \mathcal{I}_k\}$ are i.i.d., and $\hat{\phi}_t^{(-k)}$ is fixed. Hence the summands in $B_{n,0}^{(k)}$ are conditionally i.i.d. and mean zero. Now insert and subtract the true EIF. For $i \in \mathcal{I}_k$, write

$$\Delta_{i,k} = \hat{\phi}_t^{(-k)}(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0) - \phi_t^0(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0), \quad \Delta_k(\mathbf{W}) = \hat{\phi}_t^{(-k)}(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) - \phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0).$$

Then

$$\begin{aligned} B_{n,0}^{(k)} &= \frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \left\{ \phi_t^0(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0) - \mathbb{E}_0[\phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) \mid \mathcal{F}_{-k}] \right\} \\ &\quad + \frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \left\{ \Delta_{i,k} - \mathbb{E}_0[\Delta_k(\mathbf{W}) \mid \mathcal{F}_{-k}] \right\}. \end{aligned}$$

Because $\mathbf{W}$ is independent of the training subsample generating $\mathcal{F}_{-k}$, and $(\mathbf{X}_0, \mathbf{Z}_0) \in \mathcal{F}_{-k}$, we have

$$\mathbb{E}_0[\phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) \mid \mathcal{F}_{-k}] = \mathbb{E}_0[\phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) \mid \mathbf{X}_0, \mathbf{Z}_0].$$

Since ϕ_t^0 is the EIF,

$$\mathbb{E}_0[\phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) \mid \mathbf{X}_0, \mathbf{Z}_0] = 0,$$

and

$$\mathbb{E}_0[\phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) \mid \mathcal{F}_{-k}] = 0.$$

Therefore

$$B_{n,0}^{(k)} = \frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \phi_t^0(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0) + R_{n,3}^{(k)},$$

where

$$R_{n,3}^{(k)} = \frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \left\{ \Delta_{i,k} - \mathbb{E}_0[\Delta_k(\mathbf{W}) \mid \mathcal{F}_{-k}] \right\}.$$

We next show that the remainder is negligible. Conditional on $\mathcal{F}_{-k}$, the summands in $R_{n,3}^{(k)}$ are centered and independent, so

$$\mathbb{E}_0 \left[\left(R_{n,3}^{(k)} \right)^2 \mid \mathcal{F}_{-k} \right] \leq \frac{1}{|\mathcal{I}_k|} \mathbb{E}_0 [\Delta_k(\mathbf{W})^2 \mid \mathcal{F}_{-k}].$$

Based on Lemma 9,

$$\max_{1 \leq k \leq K} \mathbb{E}_0 [\Delta_k(\mathbf{W})^2] = o_p(1),$$

and therefore, by taking expectations,

$$\mathbb{E}_0 \left[\left(R_{n,3}^{(k)} \right)^2 \right] \leq \frac{1}{|\mathcal{I}_k|} \mathbb{E}_0 [\Delta_k(\mathbf{W})^2] = o\left(\frac{1}{n}\right),$$

uniformly in k , because $|\mathcal{I}_k| \asymp n$ and K is fixed. Hence

$$\sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} R_{n,3}^{(k)} = o_p(n^{-1/2}).$$

Consequently,

$$B_{n,0} = \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \left\{ \frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \phi_t^0(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0) + R_{n,3}^{(k)} \right\} = \frac{1}{n} \sum_{i=1}^n \phi_t^0(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0) + o_p(n^{-1/2}),$$

and hence

$$\sqrt{n} B_{n,0} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \phi_t^0(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0) + o_p(1).$$

Combining this with $\sqrt{n}(A_n + B_{n,1}) = o_p(1)$ and $\sqrt{n}(\tilde{H}_o - H_o) = o(1)$ yields

$$\sqrt{n}(\hat{H}^{1,\text{cf}} - H_o) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \phi_t^0(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0) + o_p(1).$$

Conditional on $(\mathbf{X}_0, \mathbf{Z}_0)$, the summands $\phi_t^0(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0)$ are i.i.d. with mean 0 and variance $\sigma_H^2(t; \mathbf{X}_0, \mathbf{Z}_0) = \mathbb{V}(\phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0) \mid \mathbf{X}_0, \mathbf{Z}_0)$, and have finite second moment by Lemma 5. The (conditional) Lindeberg–Feller CLT therefore gives

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \phi_t^0(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0) \Rightarrow \mathcal{N}(0, \sigma_H^2(t; \mathbf{X}_0, \mathbf{Z}_0)),$$

proving conditional asymptotic normality of $\hat{H}^{1,\text{cf}}(t \mid \mathbf{X}_0, \mathbf{Z}_0)$ given $(\mathbf{X}_0, \mathbf{Z}_0)$.

For $\hat{S}^{1,\text{cf}} = \exp\{-\hat{H}^{1,\text{cf}}\}$, the delta method applies since $x \mapsto e^{-x}$ is continuously differentiable, giving

$$\sqrt{n}(\hat{S}^{1,\text{cf}} - S_o) = -S_o \sqrt{n}(\hat{H}^{1,\text{cf}} - H_o) + o_p(1),$$

and thus

$$\sqrt{n}(\hat{S}^{1,\text{cf}} - S_o) \rightarrow^{\mathcal{D}} \mathcal{N}(0, S_o(t \mid \mathbf{X}_0, \mathbf{Z}_0)^2 \sigma_H^2(t; \mathbf{X}_0, \mathbf{Z}_0)).$$

Finally, consider the variance estimator

$$\hat{\sigma}_H^2(t; \mathbf{X}_0, \mathbf{Z}_0) = \frac{1}{n} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \left\{ \hat{\phi}_t^{(-k)}(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0) \right\}^2.$$

For each i , let $k(i)$ denote the fold index such that $i \in \mathcal{I}_{k(i)}$, and write

$$\hat{\phi}_i = \hat{\phi}_t^{(-k(i))}(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0), \quad \phi_i = \phi_t^0(\mathbf{W}_i; \mathbf{X}_0, \mathbf{Z}_0), \quad \mathbb{P}_n f = \frac{1}{n} \sum_{i=1}^n f(\mathbf{W}_i).$$

Then

$$\hat{\sigma}_H^2(t; \mathbf{X}_0, \mathbf{Z}_0) - \sigma_H^2(t; \mathbf{X}_0, \mathbf{Z}_0) = (\mathbb{P}_n - \mathbb{P}_0)\phi^2 + \mathbb{P}_n(\hat{\phi}^2 - \phi^2).$$

Conditionally on $(\mathbf{X}_0, \mathbf{Z}_0)$, $\{\phi_i^2\}_{i=1}^n$ are i.i.d. with $\mathbb{E}_0(\phi^2 | \mathbf{X}_0, \mathbf{Z}_0) < \infty$, hence the law of large numbers yields $(\mathbb{P}_n - \mathbb{P}_0)\phi^2 = o_p(1)$. For the second term, use $\hat{\phi}^2 - \phi^2 = (\hat{\phi} - \phi)(\hat{\phi} + \phi)$ and Cauchy–Schwarz, and get

$$\left| \mathbb{P}_n(\hat{\phi}^2 - \phi^2) \right| \leq \left\{ \mathbb{P}_n(\hat{\phi} - \phi)^2 \right\}^{1/2} \left\{ \mathbb{P}_n(\hat{\phi} + \phi)^2 \right\}^{1/2}.$$

By the $L^2(P_0)$ -consistency established above, $\mathbb{E}_0[(\hat{\phi} - \phi)^2] \rightarrow 0$ (uniformly in folds), which implies $\mathbb{P}_n(\hat{\phi} - \phi)^2 = o_p(1)$. Moreover,

$$\mathbb{P}_n(\hat{\phi} + \phi)^2 \leq 2\mathbb{P}_n\hat{\phi}^2 + 2\mathbb{P}_n\phi^2 = O_p(1),$$

since $\sup_n \mathbb{E}_0(\hat{\phi}^2) < \infty$ and $\mathbb{E}_0(\phi^2) < \infty$. Therefore $\mathbb{P}_n(\hat{\phi}^2 - \phi^2) = o_p(1)$, and hence

$$\hat{\sigma}_H^2(t; \mathbf{X}_0, \mathbf{Z}_0) \xrightarrow{p} \sigma_H^2(t; \mathbf{X}_0, \mathbf{Z}_0).$$

Finally, by the delta method,

$$\sqrt{n} \left\{ \hat{S}^{1, \text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0) - S_o(t | \mathbf{X}_0, \mathbf{Z}_0) \right\} \Rightarrow \mathcal{N}(0, \sigma_S^2(t; \mathbf{X}_0, \mathbf{Z}_0)),$$

and consistency of $\hat{\sigma}_S^2(t; \mathbf{X}_0, \mathbf{Z}_0) = \{\hat{S}^{1, \text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0)\}^2 \hat{\sigma}_H^2(t; \mathbf{X}_0, \mathbf{Z}_0)$ follows by Slutsky's theorem together with $\hat{S}^{1, \text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0) \xrightarrow{p} S_o(t | \mathbf{X}_0, \mathbf{Z}_0)$. $\square$

APPENDIX C: GENERAL COVARIATE DISTRIBUTIONS OF $\mathbf{X}$: SIEVE APPROXIMATION OF THE RIESZ REPRESENTER

When $\mathbf{X}$ has a general distribution, no closed-form expression is available and $\psi_t^0(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0)$ must be approximated in a finite-dimensional sieve. In this section we construct a spline-based approximation $\hat{\psi}_t^{(-k)}(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0)$ and define a cross-fitted one-step estimator based on the same Riemann grid $\{t_j\}_{j=0}^{m_n}$ used throughout (with $\Delta t_j = t_j - t_{j-1}$), and we state the corresponding $\sqrt{n}$ asymptotic normality under additional sieve conditions.

Fix $t \in [0, \tau]$ and a random evaluation point $(\mathbf{X}_0, \mathbf{Z}_0)$ drawn independently from the target covariate distribution. By conditions, L_t is bounded on $(\mathcal{H}, \|\cdot\|_0)$, $\mathcal{H} = \overline{\mathcal{T}}_{g_o}$, and there exists a unique $\psi_t^0(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0) \in \mathcal{H}$ solving the Riesz equation

$$\langle \psi_t^0(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0), b \rangle_0 = L_t(b), \quad \forall b \in \mathcal{H}.$$

Let $\{\mu_r(u, \mathbf{X}) : r = 1, \dots, d_n\}$ be a spline basis on $[0, \tau] \times \mathcal{X}$ (e.g., tensor-product B -splines in u and each coordinate of $\mathbf{X}$ after rescaling $\mathbf{X}$ to $[0, 1]^r$), and define the sieve space

$$\mathcal{H}_{d_n} = \text{span}\{\mu_1, \dots, \mu_{d_n}\} \subset \mathcal{H}.$$

Define the population sieve-Riesz projection $\psi_t^{0, (d_n)}(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0) \in \mathcal{H}_{d_n}$ as the unique solution of

$$\langle \psi_t^{0, (d_n)}(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0), b \rangle_0 = L_t(b), \quad \forall b \in \mathcal{H}_{d_n}.$$

The key additional requirement for $\sqrt{n}$ inference is that the sieve approximation bias is negligible, namely,

$$(C.1) \quad \left\| \psi_t^{0, (d_n)}(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0) - \psi_t^0(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0) \right\|_0 = o(n^{-1/2}).$$

It might be tempting to assume that for each $t \in [0, \tau]$, the Riesz representer $\psi_t^0(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0)$ (viewed as a function of $(u, \mathbf{X}) \in [0, \tau] \times \mathcal{X}$) belongs to a Sobolev (or Hölder) ball of smoothness s , uniformly over $t \in [0, \tau]$ (and uniformly over $(\mathbf{X}_0, \mathbf{Z}_0)$ in its support), so that the spline sieve $\mathcal{H}_{d_n}$ satisfies the approximation property (C.1). Clearly, this condition is separate from the composite Hölder assumption on g_o , which is used to guarantee $\|\hat{g} - g_o\|_0 = O_p(\gamma_n \log^2 n)$.

In the absence of additional structural restrictions, approximation rates in the norm $\|\cdot\|_0$ deteriorate rapidly with the dimension of $\mathbf{X}$ (curse of dimensionality), so that achieving $o(n^{-1/2})$ accuracy typically requires sieve dimension d_n that grows so quickly with n that the required n for a given accuracy becomes impractically large. Section 1.3 provides two alternative assumption sets that ensure (C.1). Throughout the remainder of this section we assume that (C.1) holds.

Now, we provide an estimator of $\psi_t^{0, (d_n)}(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0)$ under the cross-fitted setting. For each fold k and functions $f, h \in \mathcal{H}$, define the fold-wise empirical inner product (computed on $\mathcal{I}_k^c$) by the Riemann sum

$$(C.2) \quad \langle f, h \rangle_{n, -k} = \frac{1}{|\mathcal{I}_k^c|} \sum_{i \in \mathcal{I}_k^c} \sum_{j=1}^{m_n} f(t_j, \mathbf{X}_i) h(t_j, \mathbf{X}_i) Y_i(t_{j-1}) \hat{h}^{(-k)}(t_j | \mathbf{X}_i, \mathbf{Z}_i) \Delta t_j.$$

Similarly, define the fold-wise empirical linear functional by

$$(C.3) \quad \hat{L}_t^{(-k)}(b) = \sum_{j: t_j \leq t} \hat{h}^{(-k)}(t_j | \mathbf{X}_0, \mathbf{Z}_0) b(t_j, \mathbf{X}_0) \Delta t_j.$$

We estimate the Riesz representer in the sieve space $\mathcal{H}_{d_n}$ by solving the ridge-stabilized empirical Riesz equation. Namely, we find $\hat{\psi}_t^{(-k)}(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0) \in \mathcal{H}_{d_n}$ such that for all $b \in \mathcal{H}_{d_n}$,

$$(C.4) \quad \langle \hat{\psi}_t^{(-k)}(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0), b \rangle_{n, -k} + \rho_n \langle \hat{\psi}_t^{(-k)}(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0), b \rangle_{\text{coef}} = \hat{L}_t^{(-k)}(b),$$

where $\rho_n \downarrow 0$, $\sqrt{n}\rho_n \rightarrow 0$, and $\langle \cdot, \cdot \rangle_{\text{coef}}$ is the Euclidean inner product of spline coefficients. Here $\rho_n > 0$ is a vanishing ridge parameter used only to stabilize the inversion of the empirical Gram matrix. Specifically, write $\hat{\psi}_t^{(-k)}(u, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) = \sum_{r=1}^{d_n} \hat{\alpha}_r^{(-k)} \mu_r(u, \mathbf{X})$ and $b(u, \mathbf{X}) = \sum_{s=1}^{d_n} \beta_s \mu_s(u, \mathbf{X})$. Let $\hat{G}_{-k} \in \mathbb{R}^{d_n \times d_n}$ and $\hat{\ell}_{-k} \in \mathbb{R}^{d_n}$ be defined such that their components are given by

$$(\hat{G}_{-k})_{rs} = \langle \mu_r, \mu_s \rangle_{n, -k}, \quad r, s = 1, \dots, d_n.$$

$$(\hat{\ell}_{-k})_s = \hat{L}_t^{(-k)}(\mu_s) = \sum_{j: t_j \leq t} \hat{h}^{(-k)}(t_j | \mathbf{X}_0, \mathbf{Z}_0) \mu_s(t_j, \mathbf{X}_0) \Delta t_j, \quad s = 1, \dots, d_n.$$

Let $\hat{\alpha}^{(-k)} = (\hat{\alpha}_1^{(-k)}, \dots, \hat{\alpha}_{d_n}^{(-k)})^\top \in \mathbb{R}^{d_n}$. Then (C.4) is equivalent to

$$(\hat{G}_{-k} + \rho_n I_{d_n}) \hat{\alpha}^{(-k)} = \hat{\ell}_{-k},$$

so that $\hat{\alpha}^{(-k)} = (\hat{G}_{-k} + \rho_n I_{d_n})^{-1} \hat{\ell}_{-k}$, where I_{d_n} denotes the $d_n \times d_n$ identity matrix.

Finally, we are ready to define the cross-fitted one-step estimator based on the above spline approximation. Using the fold-wise spline weight $\hat{\psi}_t^{(-k)}(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0)$, define the fold-wise nuisance influence contribution in discrete (Riemann-sum) form

$$\hat{\phi}_{t,g}^{(-k)}(W_i; \mathbf{X}_0, \mathbf{Z}_0) = \sum_{j=1}^{m_n} \hat{\psi}_t^{(-k)}(t_j, \mathbf{X}_i; \mathbf{X}_0, \mathbf{Z}_0) \Delta \hat{M}_i^{(-k)}(t_j).$$

Let $\widehat{\phi}_{t,\theta}^{(-k)}(W_i; \mathbf{X}_0, \mathbf{Z}_0)$ denote the parametric EIF piece, constructed exactly as in the discrete-support section. Set $\widehat{\phi}_t^{(-k)} = \widehat{\phi}_{t,g}^{(-k)} + \widehat{\phi}_{t,\theta}^{(-k)}$. The cross-fitted one-step estimator is then

$$\widehat{H}^{1,\text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0) = \widehat{H}^{\text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0) + \frac{1}{n} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \widehat{\phi}_t^{(-k)}(W_i; \mathbf{X}_0, \mathbf{Z}_0),$$

and $\widehat{S}^{1,\text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0) = \exp\{-\widehat{H}^{1,\text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0)\}$.

Relative to the discrete case, the general- $\mathbf{X}$ setting introduces three additional sources of error that must be controlled to obtain a $\sqrt{n}$ expansion:

1. *Sieve approximation (bias)*: the population sieve-Riesz representer $\psi_t^{0,(d_n)} \in \mathcal{H}_{d_n}$ must approximate ψ_t^0 in the $\|\cdot\|_0$ norm, i.e. $\|\psi_t^{0,(d_n)} - \psi_t^0\|_0$ (cf. (C.1)).
2. *Estimation of the sieve-Riesz weights*: the data-driven solution $\widehat{\psi}_t$ must be close to $\psi_t^{0,(d_n)}$, i.e. $\|\widehat{\psi}_t - \psi_t^{0,(d_n)}\|_0 = o_p(1)$.
3. *Stability*: the (regularized) Gram matrix defining $\widehat{\psi}_t$ must be well-conditioned, and the effective sieve complexity must grow slowly enough with n so that the associated empirical-process remainder terms are $o_p(n^{-1/2})$.

The general- $\mathbf{X}$ setting is substantially more delicate than the finite-support case. In particular, a fully rigorous analysis of the sieve-Riesz step would require additional assumptions on the covariate distribution and on the regularity and approximability of the population Riesz representer, together with quantitative control of the sieve approximation bias, the ridge-stabilized Gram system, and the empirical-process terms arising from estimation of the Riesz coefficients. A complete investigation of these issues for the specific sieve-Riesz implementation considered here would take us beyond the intended scope of the present paper. Accordingly, the following theorem is best viewed as a high-level conditional asymptotic result for the general- $\mathbf{X}$ setting. Even in this form, however, the result remains valuable, as it helps clarify the ingredients that would be needed to extend the one-step asymptotic theory beyond the finite-support case. From a practical point of view, Theorem 7 is already highly relevant, since when $\mathbf{X}$ is handled through the DNN nuisance-estimation step, discretization or grouping of covariates is often not expected to materially compromise prediction accuracy.

THEOREM 7. *Fix $t \in [0, \tau]$ and let $(\mathbf{X}_0, \mathbf{Z}_0)$ be an independent draw from the covariate distribution, independent of the training sample. Assume Assumptions A1–A4 hold, so that the hazard is uniformly bounded. Let $\{t_j\}_{j=0}^{m_n}$ be a deterministic grid with $0 = t_0 < \dots < t_{m_n} = t$ and mesh $\delta_n = \max_j(t_j - t_{j-1})$. Assume $\sqrt{n}\delta_n \rightarrow 0$ and the time-grid Riemann approximation error is $O(\delta_n)$ uniformly over a neighborhood of $\eta_o = (\theta_o, g_o)$. Suppose the conditions of Lemma 5 hold and $\mathbb{E}_0\{\phi_t^0(\mathbf{W}; \mathbf{X}_0, \mathbf{Z}_0)^2\} < \infty$. Suppose moreover that for each fold k ,*

- (i) $\|\widehat{g}^{(-k)} - g_o\|_0 = o_p(n^{-1/4})$ (up to polylogarithmic factors) and, uniformly in k ,

$$\sqrt{|\mathcal{I}_k^c|}(\widehat{\theta}^{(-k)} - \theta_o) = I(\theta_o)^{-1} \frac{1}{\sqrt{|\mathcal{I}_k^c|}} \sum_{i \in \mathcal{I}_k^c} \ell_{\theta_o}^*(\mathbf{W}_i) + o_p(1);$$

- (ii) the sieve bias condition (C.1) holds;

- (iii) $\|\widehat{\psi}_t^{(-k)}(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0) - \psi_t^{0,(d_n)}(\cdot, \cdot; \mathbf{X}_0, \mathbf{Z}_0)\|_0 = o_p(1)$;

- (iv) the ridge-stabilized Gram matrix $\widehat{G}_{-k} + \rho_n I_{d_n}$ is well-conditioned and d_n grows slowly enough so that the additional empirical-process terms due to estimating $\widehat{\psi}$ are $o_p(n^{-1/2})$.

Then, conditional on the independent draw $(\mathbf{X}_0, \mathbf{Z}_0)$,

$$\sqrt{n} \left\{ \hat{H}^{1,\text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0) - H_0(t | \mathbf{X}_0, \mathbf{Z}_0) \right\} | (\mathbf{X}_0, \mathbf{Z}_0) \Rightarrow N(0, \sigma_H^2(t; \mathbf{X}_0, \mathbf{Z}_0)),$$

where $\sigma_H^2(t; \mathbf{X}_0, \mathbf{Z}_0) = \mathbb{V}_0(\phi_t^0(W; \mathbf{X}_0, \mathbf{Z}_0) | \mathbf{X}_0, \mathbf{Z}_0)$. Consequently,

$$\sqrt{n} \left\{ \hat{S}^{1,\text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0) - S_0(t | \mathbf{X}_0, \mathbf{Z}_0) \right\} | (\mathbf{X}_0, \mathbf{Z}_0) \Rightarrow N(0, \sigma_S^2(t; \mathbf{X}_0, \mathbf{Z}_0)),$$

and $\sigma_S^2(t; \mathbf{X}_0, \mathbf{Z}_0) = S_0(t | \mathbf{X}_0, \mathbf{Z}_0)^2 \sigma_H^2(t; \mathbf{X}_0, \mathbf{Z}_0)$.

Theorem 7 should be read as a conditional asymptotic-distribution result under assumptions (i)–(iv), where assumption (iv) collects the high-level control needed for the sieve-Riesz estimation step, including the associated empirical-process and projection-approximation errors.

PROOF OF THEOREM 7. We suppress $(t, \mathbf{X}_0, \mathbf{Z}_0)$ from the notation when no confusion arises. Let $\{\mathcal{I}_k\}_{k=1}^K$ be the fixed K -fold partition and let $\mathcal{F}_{-k}$ denote the σ -field generated by the training subsample $\{\mathbf{W}_j : j \in \mathcal{I}_k^c\}$, the fold split, and $(\mathbf{X}_0, \mathbf{Z}_0)$. By construction, conditional on $\mathcal{F}_{-k}$, the observations $\{\mathbf{W}_i : i \in \mathcal{I}_k\}$ are i.i.d. from $\mathbb{P}_0$.

Throughout this proof, $L_{t,n}$ and $c_{0,t,n}$ denote the grid-based analogues of L_t and $c_{0,t}$, respectively. Under $\sqrt{n} \delta_n \rightarrow 0$, the discrepancy between the continuous and grid-based versions is $o(n^{-1/2})$ and is absorbed into the remainder notation.

Let $\tilde{H}_0$ be the Riemann approximation of H_0 on the grid, so that $\|\tilde{H}_0 - H_0\| \lesssim \delta_n$ and hence $\sqrt{n}(\tilde{H}_0 - H_0) = o(1)$ whenever $\sqrt{n} \delta_n \rightarrow 0$. Write

$$\hat{H}^{1,\text{cf}} - H_0 = \underbrace{(\hat{H}^{\text{cf}} - \tilde{H}_0)}_{A_n} + \underbrace{(\tilde{H}_0 - H_0)}_{o(n^{-1/2})} + \underbrace{\mathbb{P}_n \hat{\phi}_t}_{B_n},$$

where $\hat{H}^{\text{cf}} = \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \hat{H}^{(-k)}$ and $\mathbb{P}_n \hat{\phi}_t = n^{-1} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \hat{\phi}_t^{(-k)}(\mathbf{W}_i)$.

As in Theorem 6, a first-order exponential expansion yields

$$A_n = \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \sum_{j:t_j \leq t} \Delta t_j h_o(t_j | \mathbf{X}_0, \mathbf{Z}_0) \left\{ \delta g^{(-k)}(t_j, \mathbf{X}_0) + (\delta \boldsymbol{\theta}^{(-k)})^\top \mathbf{Z}_0 \right\} + R_{n,1},$$

with $\sqrt{n} R_{n,1} = o_p(1)$ under assumption (i), the same second-order remainder control as in Theorem 6.

Let $\phi_t^0(\mathbf{W})$ denote the population EIF from Lemma 5. Since ϕ_t^0 is an influence function, $\mathbb{P}_0(\phi_t^0(\mathbf{W}) | \mathbf{X}_0, \mathbf{Z}_0) = 0$, hence $\mathbb{P}_n \phi_t^0 = (\mathbb{P}_n - \mathbb{P}_0) \phi_t^0$. Insert and subtract $\mathbb{P}_n \phi_t^0$ to obtain

$$\sqrt{n}(\hat{H}^{1,\text{cf}} - H_0) = \underbrace{\sqrt{n}(\mathbb{P}_n - \mathbb{P}_0) \phi_t^0}_{\text{CLT term}} + \sqrt{n} \left\{ A_n + \mathbb{P}_n(\hat{\phi}_t - \phi_t^0) \right\} + o_p(1).$$

By conditional Lindeberg–Feller (given $(\mathbf{X}_0, \mathbf{Z}_0)$) and $\mathbb{E}_0[(\phi_t^0)^2 | \mathbf{X}_0, \mathbf{Z}_0] < \infty$,

$$\sqrt{n}(\mathbb{P}_n - \mathbb{P}_0) \phi_t^0 \xrightarrow{d} N(0, \sigma_H^2(t; \mathbf{X}_0, \mathbf{Z}_0)), \quad \sigma_H^2(t; \mathbf{X}_0, \mathbf{Z}_0) = \mathbb{V}_0(\phi_t^0(\mathbf{W}) | \mathbf{X}_0, \mathbf{Z}_0).$$

Thus it remains to show

$$\sqrt{n} \left\{ A_n + \mathbb{P}_n(\hat{\phi}_t - \phi_t^0) \right\} = o_p(1).$$

Decomposing $\hat{\phi}_t - \phi_t^0$ into the nuisance-Riesz part and the $\boldsymbol{\theta}$ -part gives

$$\hat{\phi}_t^{(-k)} - \phi_t^0 = (\hat{\phi}_{t,g}^{(-k)} - \phi_{t,g}^0) + (\hat{\phi}_{t,\theta}^{(-k)} - \phi_{t,\theta}^0).$$

By the same first-order expansion used in the discrete setting for the parametric EIF component,

$$\sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \left\{ c_{0,t,n}(\mathbf{X}_0, \mathbf{Z}_0)^\top \delta \boldsymbol{\theta}^{(-k)} - T_{n,1,\theta}^{(-k)} \right\} + \mathbb{P}_n(\hat{\phi}_{t,\theta} - \phi_{t,\theta}^0) = o_p(n^{-1/2}),$$

by Lemmas 7–8 and assumption (i).

Now we focus on the nuisance-Riesz part, where the new ingredient is the sieve estimation of ψ_t^0 . Write

$$(C.5) \quad \hat{\phi}_{t,g}^{(-k)} - \phi_{t,g}^0 = \sum_{j=1}^{m_n} (\hat{\psi}_j^{(-k)} - \psi_j^0) \Delta M_{0,j} + \sum_{j=1}^{m_n} \hat{\psi}_j^{(-k)} (\Delta \widehat{M}_j^{(-k)} - \Delta M_{0,j}).$$

We start with the second sum of the right-hand side of (C.5). Set

$$B_{n,1}^{(g)} = \frac{1}{n} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \sum_{j=1}^{m_n} \hat{\psi}_t^{(-k)}(t_j, \mathbf{X}_i; \mathbf{X}_0, \mathbf{Z}_0) (\Delta \widehat{M}_{ij}^{(-k)} - \Delta M_{0,ij}),$$

so that

$$\mathbb{P}_n(\hat{\phi}_{t,g} - \phi_{t,g}^0) = B_{n,0}^{(g)} + B_{n,1}^{(g)},$$

where

$$B_{n,0}^{(g)} = \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \left\{ \frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \sum_{j=1}^{m_n} (\hat{\psi}_t^{(-k)}(t_j, \mathbf{X}_i; \mathbf{X}_0, \mathbf{Z}_0) - \psi_t^0(t_j, \mathbf{X}_i; \mathbf{X}_0, \mathbf{Z}_0)) \Delta M_{0,ij} \right\}.$$

We first analyze $B_{n,1}^{(g)}$. Since $\Delta \widehat{M}_{ij}^{(-k)} - \Delta M_{0,ij} = -(\Delta \widehat{A}_{ij}^{(-k)} - \Delta A_{0,ij})$, we have

$$B_{n,1}^{(g)} = -\frac{1}{n} \sum_{k=1}^K \sum_{i \in \mathcal{I}_k} \sum_{j=1}^{m_n} \hat{\psi}_t^{(-k)}(t_j, \mathbf{X}_i; \mathbf{X}_0, \mathbf{Z}_0) Y_i(t_{j-1}) \{ \widehat{h}^{(-k)}(t_j | \mathbf{X}_i, \mathbf{Z}_i) - h_o(t_j | \mathbf{X}_i, \mathbf{Z}_i) \} \Delta t_j,$$

and $B_{n,1}^{(g)} = \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} B_{n,1}^{(g,k)}$ where

$$B_{n,1}^{(g,k)} = \frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \sum_{j=1}^{m_n} \hat{\psi}_t^{(-k)}(t_j, \mathbf{X}_i; \mathbf{X}_0, \mathbf{Z}_0) (\Delta \widehat{M}_{ij}^{(-k)} - \Delta M_{0,ij}).$$

Conditional on $\mathcal{F}_{-k}$, the observations $\{\mathbf{W}_i : i \in \mathcal{I}_k\}$ are i.i.d., hence

$$\mathbb{E}_0(B_{n,1}^{(g,k)} | \mathcal{F}_{-k}) = -\sum_{j=1}^{m_n} \mathbb{E}_0 \left[\hat{\psi}_t^{(-k)}(t_j, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) Y(t_{j-1}) \{ \widehat{\lambda}^{(-k)}(t_j | \mathbf{X}, \mathbf{Z}) - h_o(t_j | \mathbf{X}, \mathbf{Z}) \} \mid \mathcal{F}_{-k} \right] \Delta t_j.$$

Now write

$$\widehat{h}^{(-k)}(t_j | \mathbf{X}, \mathbf{Z}) - h_o(t_j | \mathbf{X}, \mathbf{Z}) = h_o(t_j | \mathbf{X}, \mathbf{Z}) r_j^{(-k)}(\mathbf{X}, \mathbf{Z}) + R_{\lambda,j}^{(-k)}(\mathbf{X}, \mathbf{Z}),$$

where

$$r_j^{(-k)}(\mathbf{X}, \mathbf{Z}) = \delta g^{(-k)}(t_j, \mathbf{X}) + (\delta \boldsymbol{\theta}^{(-k)})^\top \mathbf{Z},$$

and the exponential-link remainder satisfies

$$|R_{\lambda,j}^{(-k)}(\mathbf{X}, \mathbf{Z})| \lesssim h_o(t_j | \mathbf{X}, \mathbf{Z}) (r_j^{(-k)}(\mathbf{X}, \mathbf{Z}))^2$$

with probability tending to one, uniformly in j and k . Let $\mathbf{g}^*(u, \mathbf{X})$ denote the L_0^2 -projection of $\mathbf{Z}$ onto $\mathcal{H}$, as in Lemma 5, and define

$$b_j^{(-k)}(\mathbf{X}) = \delta g^{(-k)}(t_j, \mathbf{X}) + (\delta \boldsymbol{\theta}^{(-k)})^\top \mathbf{g}^*(t_j, \mathbf{X}).$$

Then

$$r_j^{(-k)}(\mathbf{X}, \mathbf{Z}) = b_j^{(-k)}(\mathbf{X}) + (\delta \boldsymbol{\theta}^{(-k)})^\top \{\mathbf{Z} - \mathbf{g}^*(t_j, \mathbf{X})\}.$$

Accordingly,

$$\mathbb{E}_0(B_{n,1}^{(g,k)} | \mathcal{F}_{-k}) = -\left\{T_{n,1,g}^{(-k)} + T_{n,1,\theta}^{(-k)} + R_{n,2}^{(-k)}\right\},$$

where

$$T_{n,1,g}^{(-k)} = \sum_{j=1}^{m_n} \mathbb{E}_0 \left[\widehat{\psi}_t^{(-k)}(t_j, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) Y(t_{j-1}) h_o(t_j | \mathbf{X}, \mathbf{Z}) b_j^{(-k)}(\mathbf{X}) \mid \mathcal{F}_{-k} \right] \Delta t_j,$$

and

$$T_{n,1,\theta}^{(-k)} = \sum_{j=1}^{m_n} \mathbb{E}_0 \left[\widehat{\psi}_t^{(-k)}(t_j, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) Y(t_{j-1}) h_o(t_j | \mathbf{X}, \mathbf{Z}) (\delta \boldsymbol{\theta}^{(-k)})^\top \{\mathbf{Z} - \mathbf{g}^*(t_j, \mathbf{X})\} \mid \mathcal{F}_{-k} \right] \Delta t_j,$$

and

$$|R_{n,2}^{(-k)}| \lesssim \sum_{j=1}^{m_n} \mathbb{E}_0 \left[\left| \widehat{\psi}_t^{(-k)}(t_j, \mathbf{X}; \mathbf{X}_0, \mathbf{Z}_0) \right| Y(t_{j-1}) h_o(t_j | \mathbf{X}, \mathbf{Z}) (r_j^{(-k)}(\mathbf{X}, \mathbf{Z}))^2 \mid \mathcal{F}_{-k} \right] \Delta t_j.$$

The term $T_{n,1,g}^{(-k)}$ is the nuisance-space part, since $b^{(-k)}$ is a function of $(u, \mathbf{X})$ only. The term $T_{n,1,\theta}^{(-k)}$ corresponds to the residual parametric direction generated by $\mathbf{Z} - \mathbf{g}^*(u, \mathbf{X})$; this will match the first-order parametric contribution in $\mathbb{P}_n(\widehat{\phi}_{t,\theta} - \phi_{t,\theta}^0)$. Write $\Pi_{d_n} b^{(-k)}$ for its sieve projection onto $\mathcal{H}_{d_n}$ on the grid. We next identify the leading nuisance term $T_{n,1,g}^{(-k)}$ with the evaluation functional applied to $b^{(-k)}$. Write

$$T_{n,1,g}^{(-k)} = \left\langle \widehat{\psi}_t^{(-k)}, \Pi_{d_n} b^{(-k)} \right\rangle_{0,n} + R_{n,3}^{(-k)},$$

where $R_{n,3}^{(-k)} = o_p(n^{-1/2})$ uniformly in k by the projection approximation and the same Riemann discretization error bound used throughout. Next,

$$\left\langle \widehat{\psi}_t^{(-k)}, \Pi_{d_n} b^{(-k)} \right\rangle_{0,n} = \left\langle \psi_t^{0,(d_n)}, \Pi_{d_n} b^{(-k)} \right\rangle_0 + R_{n,4}^{(-k)},$$

where $R_{n,4}^{(-k)} = o_p(n^{-1/2})$ uniformly in k by assumption (iv), which absorbs the additional empirical-process terms arising from estimating $\widehat{\psi}_t^{(-k)}$ and replacing the empirical inner product by its population version. Finally, by the defining property of the population sieve-Riesz representer,

$$\left\langle \psi_t^{0,(d_n)}, \Pi_{d_n} b^{(-k)} \right\rangle_0 = L_{t,n}(\Pi_{d_n} b^{(-k)}).$$

Hence

$$T_{n,1,g}^{(-k)} = L_{t,n}(\Pi_{d_n} b^{(-k)}) + R_{n,3}^{(-k)} + R_{n,4}^{(-k)}.$$

It remains to remove the sieve projection. Assumption (iv) also absorbs the projection-removal error inside the evaluation functional, hence

$$L_{t,n}(\Pi_{d_n} b^{(-k)}) - L_{t,n}(b^{(-k)}) = o_p(n^{-1/2}),$$

uniformly in k . Therefore

$$T_{n,1,g}^{(-k)} = L_{t,n}(b^{(-k)}) + o_p(n^{-1/2}),$$

uniformly in k .

On the other hand, by the definition of

$$c_{0,t}(\mathbf{X}_0, \mathbf{Z}_0) = \mathbf{Z}_0 H_0(t | \mathbf{X}_0, \mathbf{Z}_0) - L_t(\mathbf{g}^*),$$

and its Riemann approximation $c_{0,t,n}(\mathbf{X}_0, \mathbf{Z}_0)$, we have

$$\sum_{j:t_j \leq t} h_o(t_j | \mathbf{X}_0, \mathbf{Z}_0) (\delta \boldsymbol{\theta}^{(-k)})^\top \{ \mathbf{Z}_0 - \mathbf{g}^*(t_j, \mathbf{X}_0) \} \Delta t_j = (c_{0,t,n}(\mathbf{X}_0, \mathbf{Z}_0))^\top \delta \boldsymbol{\theta}^{(-k)},$$

with $c_{0,t,n}(\mathbf{X}_0, \mathbf{Z}_0) - c_{0,t}(\mathbf{X}_0, \mathbf{Z}_0) = O(\delta_n) = o(n^{-1/2})$. Thus the first-order expansion of A_n can be written as a nuisance component $L_{t,n}(b^{(-k)})$ plus the parametric component $c_{0,t,n}(\mathbf{X}_0, \mathbf{Z}_0)^\top \delta \boldsymbol{\theta}^{(-k)}$. The latter is canceled by the corresponding first-order contribution from $\mathbb{P}_n(\hat{\phi}_{t,\theta} - \phi_{t,\theta}^0)$, exactly as in the discrete setting.

Now rewrite the first-order expansion of A_n as

$$A_n = \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \left[L_{t,n}(b^{(-k)}) + c_{0,t,n}(\mathbf{X}_0, \mathbf{Z}_0)^\top \delta \boldsymbol{\theta}^{(-k)} \right] + R_{n,1},$$

where

$$b_j^{(-k)}(\mathbf{X}) = \delta g^{(-k)}(t_j, \mathbf{X}) + (\delta \boldsymbol{\theta}^{(-k)})^\top \mathbf{g}^*(t_j, \mathbf{X}),$$

and

$$c_{0,t,n}(\mathbf{X}_0, \mathbf{Z}_0) = \sum_{j:t_j \leq t} h_o(t_j | \mathbf{X}_0, \mathbf{Z}_0) \{ \mathbf{Z}_0 - \mathbf{g}^*(t_j, \mathbf{X}_0) \} \Delta t_j, \quad c_{0,t,n}(\mathbf{X}_0, \mathbf{Z}_0) = c_{0,t}(\mathbf{X}_0, \mathbf{Z}_0) + o(1).$$

Combining the previous display with

$$T_{n,1,g}^{(-k)} = L_{t,n}(b^{(-k)}) + o_p(n^{-1/2}),$$

we see that the nuisance-space contribution in

$$A_n = \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \left\{ L_{t,n}(b^{(-k)}) + c_{0,t,n}(\mathbf{X}_0, \mathbf{Z}_0)^\top \delta \boldsymbol{\theta}^{(-k)} \right\} + R_{n,1}$$

cancels fold-by-fold with the term

$$- \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} T_{n,1,g}^{(-k)}$$

appearing in $\sum_k \frac{|\mathcal{I}_k|}{n} \mathbb{E}_0(B_{n,1}^{(g,k)} | \mathcal{F}_{-k})$. Therefore,

$$A_n + \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \mathbb{E}_0(B_{n,1}^{(g,k)} | \mathcal{F}_{-k}) = \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \left\{ c_{0,t,n}(\mathbf{X}_0, \mathbf{Z}_0)^\top \delta \boldsymbol{\theta}^{(-k)} - T_{n,1,\theta}^{(-k)} \right\} + R_{n,1} + \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \tilde{R}_{n,2}^{(-k)},$$

where $\tilde{R}_{n,2}^{(-k)} = o_p(n^{-1/2})$ uniformly in k .

By the same first-order expansion used earlier for the parametric EIF component, the weighted sum

$$\sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \left\{ c_{0,t,n}(\mathbf{X}_0, \mathbf{Z}_0)^\top \delta \boldsymbol{\theta}^{(-k)} - T_{n,1,\theta}^{(-k)} \right\} + \mathbb{P}_n(\hat{\phi}_{t,\theta} - \phi_{t,\theta}^0) = o_p(n^{-1/2}).$$

Consequently,

$$A_n + \sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \mathbb{E}_0(B_{n,1}^{(g,k)} | \mathcal{F}_{-k}) + \mathbb{P}_n(\hat{\phi}_{t,\theta} - \phi_{t,\theta}^0) = o_p(n^{-1/2}),$$

because assumption (i) also implies

$$\sqrt{n} \max_k \|\delta g^{(-k)}\|_0^2 = o_p(1), \quad \sqrt{n} \max_k \|\delta \theta^{(-k)}\|^2 = o_p(1).$$

It remains to control the centered empirical fluctuation

$$\sum_{k=1}^K \frac{|\mathcal{I}_k|}{n} \left\{ B_{n,1}^{(g,k)} - \mathbb{E}_0(B_{n,1}^{(g,k)} | \mathcal{F}_{-k}) \right\},$$

which is $o_p(n^{-1/2})$ by conditional independence across $i \in \mathcal{I}_k$, bounded conditional second moments, and assumption (iv). Here the dependence on $\hat{\psi}_t^{(-k)}$ is handled entirely through assumption (iv), conditional on $\mathcal{F}_{-k}$. Hence

$$A_n + B_{n,1}^{(g)} = -\mathbb{P}_n(\hat{\phi}_{t,\theta} - \phi_{t,\theta}^0) + o_p(n^{-1/2}).$$

It remains to control $B_{n,0}^{(g)}$, which reflects sieve-Riesz approximation and estimation error. Let $\psi_t^{0,(d_n)} \in \mathcal{H}_{d_n}$ be the population sieve-Riesz representer and write

$$\hat{\psi}_t^{(-k)} - \psi_t^0 = \underbrace{(\psi_t^{0,(d_n)} - \psi_t^0)}_{\text{sieve bias}} + \underbrace{(\hat{\psi}_t^{(-k)} - \psi_t^{0,(d_n)})}_{\text{estimation error}}.$$

Using the triangle inequality,

$$\|\hat{\psi}_t^{(-k)} - \psi_t^0\|_0 \leq \|\psi_t^{0,(d_n)} - \psi_t^0\|_0 + \|\hat{\psi}_t^{(-k)} - \psi_t^{0,(d_n)}\|_0.$$

By assumptions (ii) and (iii),

$$\max_{1 \leq k \leq K} \|\hat{\psi}_t^{(-k)} - \psi_t^0\|_0 = o_p(1).$$

Now, conditional on $\mathcal{F}_{-k}$ we can apply the discrete-time martingale isometry to obtain

$$\begin{aligned} & \mathbb{E}_0 \left[\left(\sum_{j=1}^{m_n} (\hat{\psi}_t^{(-k)}(t_j, \mathbf{X}) - \psi_t^0(t_j, \mathbf{X})) \Delta M_{0,j} \right)^2 \middle| \mathcal{F}_{-k} \right] \\ &= \mathbb{E}_0 \left[\sum_{j=1}^{m_n} (\hat{\psi}_t^{(-k)}(t_j, \mathbf{X}) - \psi_t^0(t_j, \mathbf{X}))^2 \Delta A_{0,j} \middle| \mathcal{F}_{-k} \right]. \end{aligned}$$

Conditional on $\mathcal{F}_{-k}$, the summands

$$\sum_{j=1}^{m_n} (\hat{\psi}_t^{(-k)}(t_j, \mathbf{X}_i) - \psi_t^0(t_j, \mathbf{X}_i)) \Delta M_{0,ij}, \quad i \in \mathcal{I}_k,$$

are independent and centered. Hence, by the discrete-time martingale isometry,

$$\mathbb{V}_0 \left(\frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \sum_{j=1}^{m_n} (\hat{\psi}_t^{(-k)}(t_j, \mathbf{X}_i) - \psi_t^0(t_j, \mathbf{X}_i)) \Delta M_{0,ij} \middle| \mathcal{F}_{-k} \right) \lesssim \frac{1}{|\mathcal{I}_k|} \|\hat{\psi}_t^{(-k)} - \psi_t^0\|_0^2.$$

Therefore,

$$\frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \sum_{j=1}^{m_n} (\widehat{\psi}_t^{(-k)}(t_j, \mathbf{X}_i) - \psi_t^0(t_j, \mathbf{X}_i)) \Delta M_{0,ij} = O_p\left(|\mathcal{I}_k|^{-1/2} \|\widehat{\psi}_t^{(-k)} - \psi_t^0\|_0\right).$$

Since K is fixed and $|\mathcal{I}_k| \asymp n$,

$$B_{n,0}^{(g)} = O_p\left(n^{-1/2} \max_{1 \leq k \leq K} \|\widehat{\psi}_t^{(-k)} - \psi_t^0\|_0\right)$$

and hence

$$B_{n,0}^{(g)} = o_p(n^{-1/2}).$$

Combining this with the previously established bound

$$A_n + B_{n,1}^{(g)} + \mathbb{P}_n(\widehat{\phi}_{t,\theta} - \phi_{t,\theta}^0) = o_p(n^{-1/2}),$$

and with

$$B_{n,0}^{(g)} = o_p(n^{-1/2}),$$

we obtain

$$A_n + \mathbb{P}_n(\widehat{\phi}_t - \phi_t^0) = o_p(n^{-1/2}).$$

and therefore

$$\sqrt{n}(\widehat{H}^{1,\text{cf}}(t | \mathbf{X}_0, \mathbf{Z}_0) - H_0(t | \mathbf{X}_0, \mathbf{Z}_0)) \xrightarrow{d} N(0, \sigma_H^2(t; \mathbf{X}_0, \mathbf{Z}_0)).$$

The result for $\widehat{S}^{1,\text{cf}} = \exp(-\widehat{H}^{1,\text{cf}})$ follows by the delta method with derivative $-S_0$. $\square$

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